

Unified Discrete Diffusion for Categorical Data

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Abstract

Discrete diffusion models have attracted significant attention for their application to naturally discrete data, such as language and graphs. While discrete-time discrete diffusion has been established for some time, it was only recently that [Campbell et al. \[2022\]](#) introduced the first framework for continuous-time discrete diffusion. However, their training and backward sampling processes significantly differ from those of the discrete-time version, requiring nontrivial approximations for tractability. In this paper, we first introduce a series of generalizations and simplifications of the variational lower bound (VLB) that facilitate more accurate and easier optimization both discrete- and continuous-time discrete diffusion. We further establish a unification of discrete- and continuous-time discrete diffusion through shared forward process and backward parameterization. Thanks to this unification, the continuous-time diffusion can now utilize the exact and efficient backward process developed for the discrete-time case, avoiding the need for costly and inexact approximations. Similarly, the discrete-time diffusion now also employ the MCMC corrector, which was previously exclusive to the continuous-time case. Extensive experiments and ablations demonstrate the significant improvement.

1 Introduction

Deep generative models have taken the world by storm, capturing complex data distributions and producing realistic data, from human-like text [\[Brown et al., 2020, Li et al., 2022, OpenAI, 2023\]](#) and natural looking images [\[Dhariwal and Nichol, 2021, Ramesh et al., 2022, Zhang et al., 2023\]](#) to novel compounds like molecules and drugs [\[Kang and Cho, 2018, Li et al., 2021\]](#) and video synthesis [\[Ho et al., 2022\]](#). Denoising diffusion models [\[Ho et al., 2020b\]](#), a powerful class of generative models, are trained through a forward diffusion process that gradually adds noise to the training samples, and a backward process that denoises these diffusion trajectories. New data are then generated by sampling from the noise distribution and employing the trained model for recursive denoising.

Discrete diffusion for categorical data has two modeling paradigms: discrete-time and continuous-time. The former discretizes time such that backward denoising is learned only at pre-specified time points. This limits generation, which can only “jump back” through fixed points. In contrast, continuous-time case allows a path through any point in range, and often yields higher sample quality. Current literature on discrete-time discrete diffusion is relatively established, while only recently [Campbell et al. \[2022\]](#) introduced the first continuous-time discrete diffusion framework. While groundbreaking, their their loss requires multiple network evaluations during training. Moreover, the exact sampling through their learned backward process is extremely tedious for multi-dimensional variables. Due to mathematically complicated and computationally demanding formulations, [Campbell et al. \[2022\]](#) propose nontrivial approximations for tractability with unknown potential errors.

In this paper, we introduce a series of improvements to two critical aspects of discrete diffusion: loss computation and backward sampling, for both discrete- and continuous-time scenarios. What is more,

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用于分类数据的统一离散扩散

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摘要

离散扩散模型因其对自然离散数据的适用性（如语言和图）而备受关注。尽管离散时间离散扩散模型已存在一段时间，但坎贝尔等人 [\[2022\]](#)直到最近才首次提出了连续时间离散扩散的框架。然而，他们的训练和反向采样过程与离散时间版本存在显著差异，需要非平凡的近似方法才能保证可行性。在本文中，我们首先介绍了一系列变分下界（VLB）的推广和简化，这些方法有助于离散时间和连续时间离散扩散的更准确和更易于优化的实现。我们进一步通过共享的前向过程和反向参数化，建立了离散时间和连续时间离散扩散的统一。得益于这一统一，连续时间扩散现在可以利用为离散时间情况开发的精确且高效的反向过程，避免了昂贵且不精确的近似需求。类似地，离散时间扩散现在也采用了MCMC校正器，这一方法此前仅限于连续时间情况。大量的实验和消融研究证明了显著的改进。

1 引言

深度生成模型风靡全球，能够捕捉复杂的数据分布并生成逼真的数据，从类似人类的文本 [\[Brown等人, 2020年, 李等人, 2022年, OpenAI, 2023\]](#)到自然图像 [\[Dhariwal和Nichol, 2021年, Ramesh等人, 2022年, 张等人, 2023\]](#)，再到新型化合物如分子和药物 [\[Kang和Cho, 2018年, 李等人, 2021\]](#)以及视频合成 [\[Ho等人, 2022\]](#)。去噪扩散模型 [\[Ho等人, 2020b\]](#)，是一类强大的生成模型，通过正向扩散过程逐渐向训练样本添加噪声，以及一个反向去噪过程来处理这些扩散轨迹。然后通过从噪声分布中采样，并利用训练好的模型进行递归去噪来生成新数据。

用于分类数据的离散扩散有两种建模范式：离散时间和连续时间。前者将时间离散化，使得反向去噪仅在预指定的时间点学习。这限制了生成，只能“跳跃”通过固定点。相比之下，连续时间的情况允许通过范围内的任何点，并且通常能产生更高的样本质量。当前关于离散时间离散扩散的文献相对成熟，而坎贝尔等人 [\[2022\]](#)最近首次引入了第一个连续时间离散扩散框架。虽然具有开创性，但他们的损失在训练期间需要多次网络评估。此外，通过他们学习的反向过程进行精确采样对于多维变量来说极其繁琐。由于数学上复杂且计算需求高，坎贝尔等人 [\[2022\]](#)提出了非平凡的近似方法以实现可处理性，但存在潜在的错误。

在本文中，我们介绍了一系列针对离散扩散两个关键方面的改进：损失计算和反向采样，适用于离散时间和连续时间场景。此外，

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for the first time, we show that the discrete-time and continuous-time discrete diffusion can be unified together such that they share exactly the same forward diffusion and backward sampling process. We elaborate on the details and their benefits as follows:

- **Loss simplification and generalization:** We derive significantly simplified yet exact VLB calculations for both discrete&continuous-time discrete diffusion, taking into account the properties of categorical data. Our simplified formulations allow both forward and backward probabilities to accommodate any noise distribution element-wisely, which is particularly attractive for multi-element objects where each element can exhibit a different noise distribution.
- **Efficient backward sampling:** We present a closed-form backward probability $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ for all $s < t$ with reconstruction-based model parameterization, paving the way for VLB simplification and accelerated backward sampling, initially derived in the discrete-time case. Furthermore, we demonstrate that this improved backward process can be applied to continuous-time case, eliminating the need for the costly and inexact sampling approximations in Campbell et al. [2022].
- **Unification:** We demonstrate that continuous-time diffusion can share the exact same forward process as discrete-time diffusion for *any* noise distribution. This unified forward process, along with shared backward parameterization, leads to a unification of both forward and backward processes for discrete-& continuous-time cases. This offers mutual benefits: discrete-time diffusion can utilize continuous time’s MCMC corrector to enhance performance, while continuous-time diffusion can employ fast and exact backward sampling derived in discrete-time case.

We present both discrete-time (§2) and continuous-time (§3) discrete diffusion in a self-contained manner, accompanied by easy-to-use open-source code at <https://github.com/LingxiaoShawn/USD3>. We explicitly mark our novel contributions in (sub)sections with summarization.

2 Discrete-time Discrete Diffusion

Notation: Let $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x}_0)$ be the random variable of observed data with underlying distribution $p_{\text{data}}(\mathbf{x}_0)$. Let $\mathbf{x}_t \sim q(\mathbf{x}_t)$ be the latent variable at time t of a single-element object, like a pixel of an image or a node/edge of a graph, with maximum time T . Let $\mathbf{x}_{t|s} \sim q(\mathbf{x}_t|\mathbf{x}_s)$ be the conditional random variable. We model the *forward diffusion* process independently for each element of the object, while the *backward denoising* process is modeled jointly for all elements of the object. For simplicity and clarity of presentation, we first assume that the object only has 1 element and extend to multi-element object later. Let $\mathbf{x}_0^{1:D}$ denote the object with D elements, and $\mathbf{x}_t^i$ be the i -th element of latent object at time t . We assume all random variables take categorical values from $\{1, 2, \dots, K\}$. Let $\mathbf{e}_k \in \{0, 1\}^K$ be the one-hot encoding of category k . For a random variable $\mathbf{x}$, we use $\mathbf{x}$ denoting its one-hot encoded sample where $\mathbf{x} \in \{\mathbf{e}_1, \dots, \mathbf{e}_K\}$. Also, we interchangeably use $q(\mathbf{x}_t|\mathbf{x}_s)$, $q(\mathbf{x}_t = \mathbf{x}_t|\mathbf{x}_s = \mathbf{x}_s)$, and $q_{t|s}(\mathbf{x}_t|\mathbf{x}_s)$ when no ambiguity is introduced. Let $\langle \cdot, \cdot \rangle$ denote inner product. All vectors are column-wise vectors.

2.1 Graphical Model View of Diffusion Models

Diffusion models can be represented by latent variable graphical models (see Appx. Fig. 1). We can write the joint probability as $p_\theta(\mathbf{x}_{0:T}) := p_\theta(\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_T) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ using the Markov condition. Parameters θ are learned by maximizing the loglikelihood of the observed variable $\mathbf{x}_0$: $\log p_\theta(\mathbf{x}_0) = \log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T}$. However the marginalization is intractable, and instead the following variational lower bound (VLB) is used.

$$\log p_\theta(\mathbf{x}_0) = \log \int q(\mathbf{x}_{1:T}|\mathbf{x}_0) \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} d\mathbf{x}_{1:T} \geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_{0:T}) - \log q(\mathbf{x}_{1:T}|\mathbf{x}_0)]. \quad (1)$$

The above inequality holds for any conditional probability $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ and finding the best $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ to tighten the bound is the inference problem in graphical models (i.e. E step in EM algorithm). Exact inference is intractable, thus $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ in diffusion models is fixed or chosen specifically to simplify the learning objective. To simplify Eq. (1), it is important to assume $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ is *decomposable*. The typical assumption is $q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$ [Ho et al., 2020a], which we also adopt. Others that have been explored include $q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)$ [Song et al., 2021a].

Assuming $q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$, Eq. (1) can be simplified as the following

$$\underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]}_{-\mathcal{L}_1(\theta)} - \underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)||p_\theta(\mathbf{x}_T))}_{\mathcal{L}_{\text{prior}}} - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\mathcal{L}_t(\theta)} \quad (2)$$

我们首次展示了离散时间和连续时间离散扩散可以统一在一起，使得它们共享完全相同的正向扩散和反向采样过程。我们详细阐述其细节和优势如下：

- **损失简化与泛化：**我们推导出针对离散时间与连续时间离散扩散的显著简化且精确的 VLB 计算，同时考虑了分类数据的特性。我们的简化公式允许正向与反向概率以逐元素的方式适应任何噪声分布，这对于多元素对象特别有吸引力，因为每个元素可以表现出不同的噪声分布。
- **高效的逆向采样：**我们提出了基于重建模型参数化的方法，适用于所有 $s < t$ 的闭式反向概率 $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ ，为 VLB 简化与加速逆向采样铺平了道路，最初是在离散时间情况下推导的。此外，我们证明了这种改进的逆向过程可以应用于连续时间情况，从而消除了 Campbell 等人 [2022] 中昂贵且不精确的采样近似。
- **统一性：**我们证明了连续时间扩散可以与离散时间扩散共享完全相同的正向过程，适用于任何噪声分布。这种统一的正向过程，加上共享的逆向参数化，导致了离散时间与连续时间情况的正向与逆向过程的统一。这带来了互惠的好处：离散时间扩散可以利用连续时间的 MCMC 校正器来提升性能，而连续时间扩散可以采用在离散时间情况下推导出的快速且精确的逆向采样。

我们以自包含的方式呈现了离散时间 (§2) 和连续时间 (§3) 的离散扩散，并提供了易于使用的开源代码，可在 <https://github.com/LingxiaoShawn/USD3> 获取。我们在 (子) 节中明确标注了我们的创新贡献，并附有总结。

2 离散时间离散扩散

符号说明：令 $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x}_0)$ 为观测数据的随机变量，其底层分布为 $p_{\text{data}}(\mathbf{x}_0)$ 。令 $\mathbf{x}_t \sim q(\mathbf{x}_t)$ 为单个元素对象在时间 t 的潜变量，例如图像的像素或图的节点/边，其最大时间为 T 。令 $\mathbf{x}_{t|s} \sim q(\mathbf{x}_t|\mathbf{x}_s)$ 为条件随机变量。我们对对象的每个元素独立建模正向扩散过程，而逆向去噪过程则对所有对象的元素联合建模。为简洁和清晰起见，我们首先假设对象只有一个元素，然后扩展到多元素对象。令 $\mathbf{x}_0^{1:D}$ 表示具有 D 个元素的对象， $\mathbf{x}_t^i$ 为潜对象在时间 t 的第 i 个元素。我们假设所有随机变量从 $\{1, 2, \dots, K\}$ 中取分类值。令 $\mathbf{e}_k \in \{0, 1\}^K$ 为类别 k 的独热编码。对于随机变量 $\mathbf{x}$ ，我们使用 $\mathbf{x}$ 表示其独热编码样本，其中 $\mathbf{x} \in \{\mathbf{e}_1, \dots, \mathbf{e}_K\}$ 。当不引入歧义时，我们可互换使用 $q(\mathbf{x}_t|\mathbf{x}_s)$ $q(\mathbf{x}_t = \mathbf{x}_t|\mathbf{x}_s = \mathbf{x}_s)$ 和 $q_{t|s}(\mathbf{x}_t|\mathbf{x}_s)$ 。令 $\langle \cdot, \cdot \rangle$ 表示内积。所有向量都是列向量。

2.1 扩散模型的图模型视图

扩散模型可以通过潜变量图模型表示 (参见附录图1)。我们可以使用马尔可夫条件将联合概率表示为 $p_\theta(\mathbf{x}_{0:T}) := p_\theta(\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_T) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ 。参数 θ 通过最大化观测变量 $\mathbf{x}_0$ 的对数似然性： $\log p_\theta(\mathbf{x}_0) = \log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T}$ 来学习。然而，边缘化是不可行的，因此使用以下变分下界 (VLB)。

$$\log p_\theta(\mathbf{x}_0) = \log \int q(\mathbf{x}_{1:T}|\mathbf{x}_0) \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} d\mathbf{x}_{1:T} \geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_{0:T}) - \log q(\mathbf{x}_{1:T}|\mathbf{x}_0)]. \quad (1)$$

上述不等式对任何条件概率 $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ 都成立，找到最佳的 $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ 来收紧界限是图模型中的推理问题 (即 EM 算法中的 E 步)。精确推理是不可行的，因此在扩散模型中， $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ 是固定的或特意选择的以简化学习目标。为了简化公式(1)，假设 $q(\mathbf{x}_{1:T}|\mathbf{x}_0)$ 是可分解的非常重要。典型假设是 $q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$ [Ho 等人, 2020a]，我们同样采用。其他已探索的假设包括 $q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)$ [Song 等人, 2021a]。

假设 $q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$ ，式(1) 可以简化为以下形式

$$\underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]}_{-\mathcal{L}_1(\theta)} - \underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)||p_\theta(\mathbf{x}_T))}_{\mathcal{L}_{\text{prior}}} - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\mathcal{L}_t(\theta)} \quad (2)$$

Where $\mathcal{L}_{\text{prior}} \approx 0$, since $p_\theta(\mathbf{x}_T) \approx q(\mathbf{x}_T|\mathbf{x}_0)$ is designed as a fixed noise distribution that is easy to sample from. (See Appx. §A.1 for derivation.) To compute Eq. (2), we need to formalize distributions (i) $q(\mathbf{x}_t|\mathbf{x}_0)$ and (ii) $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$, as well as (iii) the parameterization of $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$. We specify these respectively in §2.2.1, §2.2.2, and §2.2.3.

2.2 Components of Discrete-time Discrete Diffusion

After reviewing the forward process for discrete diffusion (§2.2.1), we contribute a series of analytical simplifications for various components (§2.2.2, §2.2.3) of the VLB, providing exact closed-form formulation in §2.2.4, as well as an approximated loss for easier optimization in §2.2.5. We present fast backward sampling in Appx. §A.8, and give the extension to multi-element case in Appx. §A.9.

2.2.1 The Forward Diffusion Process

We assume each discrete random variable $\mathbf{x}_t$ has a categorical distribution, i.e. $\mathbf{x}_t \sim \text{Cat}(\mathbf{x}_t; \mathbf{p})$ with $\mathbf{p} \in [0, 1]^K$ and $\mathbf{1}^\top \mathbf{p} = 1$. One can verify that $p(\mathbf{x}_t = \mathbf{x}_t) = \mathbf{x}_t^\top \mathbf{p}$, or simply $p(\mathbf{x}_t) = \mathbf{x}_t^\top \mathbf{p}$. As shown in [Hoogeboom et al., 2021, Austin et al., 2021], the forward process with discrete variables $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ at t step can be represented as a transition matrix $Q_t \in [0, 1]^{K \times K}$ such that $[Q_t]_{ij} = q(\mathbf{x}_t = \mathbf{e}_j|\mathbf{x}_{t-1} = \mathbf{e}_i)$. Then, we can write the distribution explicitly as

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \text{Cat}(\mathbf{x}_t; Q_t^\top \mathbf{x}_{t-1}). \quad (3)$$

Given transition matrices $Q_1, \dots, Q_T$, the t -step marginal distribution conditioning on $s, \forall t > s$ is

$$q(\mathbf{x}_t|\mathbf{x}_s) = \text{Cat}(\mathbf{x}_t; \overline{Q}_{t|s}^\top \mathbf{x}_s), \text{ with } \overline{Q}_{t|s} = Q_{s+1} \dots Q_t. \quad (4)$$

The s -step posterior distribution conditioning on $\mathbf{x}_0$ and t -step can be derived as

$$q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_s)q(\mathbf{x}_s|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} = \text{Cat}(\mathbf{x}_s; \frac{\overline{Q}_{t|s} \mathbf{x}_t \odot \overline{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \overline{Q}_t^\top \mathbf{x}_0}), \forall t > s. \quad (5)$$

The above formulations are valid (derivation in Appx. §A.3) for any transition matrices $Q_1, \dots, Q_T$. However, to achieve uninformative noise $\mathbf{x}_T$, $\{Q_i\}_{i=1}^T$ should be chosen such that every row of $\overline{Q}_t = \overline{Q}_{t|0}$ converge to the same known stationary distribution when t becomes large enough (at T). Let the known stationary distribution be $\mathbf{m}_0 \sim \text{Cat}(\mathbf{m}_0; \mathbf{m})$. Then, the constraint can be stated as

$$\lim_{t \rightarrow T} \overline{Q}_t = \mathbf{1} \mathbf{m}^\top. \quad (6)$$

In addition, this paper focuses on *nominal data* (see Appx. §A.2) where categories are unordered and only equality comparison is defined. Hence, no ordering prior *except checking equality* should be used to define Q_t ². To achieve the desired convergence on nominal data while keeping the flexibility of choosing any categorical stationary distribution $\mathbf{m}_0 \sim \text{Cat}(\mathbf{m}_0; \mathbf{m})$, we define Q_t as

$$Q_t = \alpha_t I + (1 - \alpha_t) \mathbf{1} \mathbf{m}^\top, \quad (7)$$

where $\alpha_t \in [0, 1]$. This results in the accumulated transition matrix $\overline{Q}_{t|s}$ being equal to

$$\overline{Q}_{t|s} = \overline{\alpha}_{t|s} I + (1 - \overline{\alpha}_{t|s}) \mathbf{1} \mathbf{m}^\top, \forall t > s, \quad (8)$$

where $\overline{\alpha}_{t|s} = \prod_{i=s+1}^t \alpha_i$. Note that $\overline{\alpha}_t = \overline{\alpha}_{t|0} = \overline{\alpha}_{t|s} \overline{\alpha}_s$. We can achieve the uninformative noise with satisfying the constraint Eq. (6) by picking α_t such that $\lim_{t \rightarrow T} \overline{\alpha}_t = 0$.

2.2.2 Analytical Form of $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$

The formulation in Eq. (7) can be used to simplify $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$. We provide a general formulation of $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ for any s, t with $0 < s < t \leq T$, which will be useful for unifying with continuous-time diffusion. One can recover $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ by setting s as $t-1$.

Proposition 2.1 (Zheng et al. [2023]). *For both discrete- and continuous-time discrete diffusion, we can write the conditional distribution as*

$$q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) = \begin{cases} \text{Cat}(\mathbf{x}_s; (1 - \lambda_{t|s}) \cdot \mathbf{x}_t + \lambda_{t|s} \cdot \mathbf{m}) & \text{when } \mathbf{x}_t = \mathbf{x}_0 \\ \text{Cat}(\mathbf{x}_s; (1 - \mu_{t|s}) \cdot \mathbf{x}_0 + \mu_{t|s} \overline{\alpha}_{t|s} \cdot \mathbf{x}_t + \mu_{t|s} (1 - \overline{\alpha}_{t|s}) \cdot \mathbf{m}) & \text{when } \mathbf{x}_t \neq \mathbf{x}_0 \end{cases} \quad (9)$$

where $\lambda_{t|s} := \frac{(1 - \overline{\alpha}_s)(1 - \overline{\alpha}_{t|s}) \langle \mathbf{m}, \mathbf{x}_t \rangle}{\overline{\alpha}_t + (1 - \overline{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle}$, $\mu_{t|s} := \frac{1 - \overline{\alpha}_s}{1 - \overline{\alpha}_t}$.

We remark that the above formulation is originally presented in Zheng et al. [2023] with $s = t - 1$. Appx. §A.3 gives the detailed derivation of the probability's formulation.

²Mathematically, this means $\forall k, j, q(\mathbf{x}_t|\mathbf{x}_{t-1} = \mathbf{e}_j, \mathbf{x}_t \neq \{\mathbf{e}_j, \mathbf{e}_k\}) = q(\mathbf{x}_t|\mathbf{x}_{t-1} = \mathbf{e}_k, \mathbf{x}_t \neq \{\mathbf{e}_j, \mathbf{e}_k\})$.

其中 $\mathcal{L}_{\text{prior}} \approx 0$, 因为 $p_\theta(\mathbf{x}_T) \approx q(\mathbf{x}_T|\mathbf{x}_0)$ 被设计为一种易于采样的固定噪声分布。(参见附录 §A.1 的推导。)要计算式(2), 我们需要将分布 (i) $q(\mathbf{x}_t|\mathbf{x}_0)$ 和 (ii) $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ 以及 (iii) $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ 的参数化形式化, 我们分别在 §2.2.1、§2.2.2 和 §2.2.3 中具体说明这些内容。

2.2 离散时间离散扩散的组成部分

在回顾了离散扩散的前向过程 (§2.2.1) 之后, 我们为 VLB 的各个组成部分 (§2.2.2, §2.2.3) 提出了一系列分析简化方法, 并在 §2.2.4 中提供了精确的闭式公式, 以及在 §2.2.5 中提供了近似损失以便于优化。我们在 Appx. §A.8 中展示了快速反向采样, 并在 Appx. §A.9 中给出了多元素情况的扩展。

2.2.1 前向扩散过程

我们假设每个离散随机变量 $\mathbf{x}_t$ 服从分类分布, 即 $\mathbf{x}_t \sim \text{Cat}(\mathbf{x}_t; \mathbf{p})$ 具有 $\mathbf{p} \in [0, 1]^K$ 和 $\mathbf{1}^\top \mathbf{p} = 1$ 。可以验证 $p(\mathbf{x}_t = \mathbf{x}_t) = \mathbf{x}_t^\top \mathbf{p}$, 或者简单地说 $p(\mathbf{x}_t) = \mathbf{x}_t^\top \mathbf{p}$ 。正如 [Hoogeboom 等人, 2021 年, Austin 等人, 2021], 所示, 具有离散变量的前向过程 $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ 在 t 步骤可以被表示为一个转移矩阵 $Q_t \in [0, 1]^{K \times K}$, 使得 $[Q_t]_{ij} = q(\mathbf{x}_t = \mathbf{e}_j|\mathbf{x}_{t-1} = \mathbf{e}_i)$ 。然后, 我们可以显式地写出分布为

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \text{Cat}(\mathbf{x}_t; Q_t^\top \mathbf{x}_{t-1}). \quad (3)$$

给定转移矩阵 $Q_1 \dots Q_T$, t -步边缘分布在 $s, \forall t > s$ 条件下为

$$q(\mathbf{x}_t|\mathbf{x}_s) = \text{Cat}(\mathbf{x}_t; \overline{Q}_{t|s}^\top \mathbf{x}_s), \text{ with } \overline{Q}_{t|s} = Q_{s+1} \dots Q_t. \quad (4)$$

在 $\mathbf{x}_0$ 条件下 s -步后验分布和 t -步可以推导为

$$q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_s)q(\mathbf{x}_s|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} = \text{Cat}(\mathbf{x}_s; \frac{\overline{Q}_{t|s} \mathbf{x}_t \odot \overline{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \overline{Q}_t^\top \mathbf{x}_0}), \forall t > s. \quad (5)$$

上述公式对任何转移矩阵 $Q_1 \dots Q_T$ 都有效 (推导见附录 §A.3)。然而, 为了实现无信息的噪声 $\mathbf{x}_T$ 、 $\{Q_i\}_{i=1}^T$, 应选择使得当 t 足够大时 (在 T 处), $Q_t = Q_{t|0}$ 的每一行都收敛到相同的已知稳态分布。设已知稳态分布为 $\mathbf{m}_0 \sim \text{Cat}(\mathbf{m}_0; \mathbf{m})$ 。那么, 约束条件可以表述为

$$\lim_{t \rightarrow T} \overline{Q}_t = \mathbf{1} \mathbf{m}^\top. \quad (6)$$

此外, 本文关注名义数据 (参见附录 §A.2), 其中类别是无序的, 仅定义了相等比较。因此, 在定义 Q_t ² 时不应使用除检查相等性之外的任何排序先验。为了在名义数据上实现所需的收敛, 同时保持选择任何分类稳态分布 $\mathbf{m}_0 \sim \text{Cat}(\mathbf{m}_0; \mathbf{m})$ 的灵活性, 我们定义 Q_t 为

$$Q_t = \alpha_t I + (1 - \alpha_t) \mathbf{1} \mathbf{m}^\top, \quad (7)$$

在 $\alpha_t \in [0, 1]$ 处。这导致累积转移矩阵 $\overline{Q}_{t|s}$ 等于

$$\overline{Q}_{t|s} = \overline{\alpha}_{t|s} I + (1 - \overline{\alpha}_{t|s}) \mathbf{1} \mathbf{m}^\top, \forall t > s, \quad (8)$$

在 $\alpha_{t|s} = \prod_{i=s+1}^t \alpha_i$ 处。请注意 $\alpha_t = \alpha_{t|0} = \alpha_{t|s} \alpha_s$ 。我们可以通过选择 α_t 使得 $\lim_{t \rightarrow T} \alpha_t = 0$ 满足约束 Eq. (6) 来实现无信息噪声。

2.2.2 $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ 的解析形式

可以使用 Eq. (7) 中的公式来简化 $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ 。我们为任何 s, t (满足 $0 < s < t \leq T$) 提供了 $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ 的通用公式, 这将对与连续时间扩散的统一有用。通过将 s 设置为 $t-1$, 可以恢复 $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ 。

命题 2.1 (Zheng 等人 [2023])。对于离散时间和连续时间的离散扩散, 我们可以将条件分布写成

$$q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) = \begin{cases} \text{Cat}(\mathbf{x}_s; (1 - \lambda_{t|s}) \cdot \mathbf{x}_t + \lambda_{t|s} \cdot \mathbf{m}) & \text{when } \mathbf{x}_t = \mathbf{x}_0 \\ \text{Cat}(\mathbf{x}_s; (1 - \mu_{t|s}) \cdot \mathbf{x}_0 + \mu_{t|s} \overline{\alpha}_{t|s} \cdot \mathbf{x}_t + \mu_{t|s} (1 - \overline{\alpha}_{t|s}) \cdot \mathbf{m}) & \text{when } \mathbf{x}_t \neq \mathbf{x}_0 \end{cases} \quad (9)$$

where $\lambda_{t|s} := \frac{(1 - \overline{\alpha}_s)(1 - \overline{\alpha}_{t|s}) \langle \mathbf{m}, \mathbf{x}_t \rangle}{\overline{\alpha}_t + (1 - \overline{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle}$, $\mu_{t|s} := \frac{1 - \overline{\alpha}_s}{1 - \overline{\alpha}_t}$ 。

我们指出, 上述公式最初由郑等人 [2023] 与 $s = t - 1$ 提出。附录 §A.3 给出了概率公式的详细推导过程。

²从数学上讲, 这意味着 $\forall k, j, q(\mathbf{x}_t|\mathbf{x}_{t-1} = \mathbf{e}_j, \mathbf{x}_t \neq \{\mathbf{e}_j, \mathbf{e}_k\}) = q(\mathbf{x}_t|\mathbf{x}_{t-1} = \mathbf{e}_k, \mathbf{x}_t \neq \{\mathbf{e}_j, \mathbf{e}_k\})$

2.2.3 Contrib. 1: Analytical Form of $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$

The literature has explored three different parameterizations of $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$: (1) parameterizing $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ directly; (2) parameterizing $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ with f_t^θ such that $p_\theta(\mathbf{x}_0|\mathbf{x}_t) = \text{Cat}(\mathbf{x}_0; f_t^\theta(\mathbf{x}_t))$ and letting $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = q(\mathbf{x}_{t-1}|\mathbf{x}_t, f_t^\theta(\mathbf{x}_t))$; and (3) parameterizing $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ with f_t^θ and then marginalizing $q(\mathbf{x}_{t-1}, \mathbf{x}_0|\mathbf{x}_t)$ such that $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \sum_{\mathbf{x}_0} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)p_\theta(\mathbf{x}_0|\mathbf{x}_t)$.

Method (1) does not reuse any distribution in forward process, and hence is less effective. Method (2) has been widely used for continuous diffusion models as in Ho et al. [2020a] and Song et al. [2021a], and some discrete diffusion models like in Hoogeboom et al. [2021] and Zheng et al. [2023]. It avoids marginalization and works effectively for continuous diffusion. However for discrete diffusion, as shown in Eq. (9), sample $\mathbf{x}_0$ determines which categorical distribution should be used, which cannot be determined without the true $\mathbf{x}_0$. Some heuristics have been proposed in Zheng et al. [2023], however those can have a large gap to the true $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$, leading to an inaccurate sampling process.

Method (3) has been used in Austin et al. [2021] by directly marginalizing out $\mathbf{x}_0$, which introduces additional computational cost in both loss function computation and sampling process as the formulation of $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ has not been simplified to a closed-form distribution. In this paper, we show that method (3) parameterization can be simplified to a clean formulation of categorical distribution.

Proposition 2.2. *The parameterization of $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ can be simplified for any $0 < s < t \leq T$ as*

$$p_\theta(\mathbf{x}_s|\mathbf{x}_t) = \text{Cat}\left(\mathbf{x}_s; (1 - \mu_{t|s}) \cdot f_t^\theta(\mathbf{x}_t) + (\mu_{t|s}\bar{\alpha}_{t|s} + \gamma_{t|s}^\theta) \cdot \mathbf{x}_t + (\mu_{t|s}(1 - \bar{\alpha}_{t|s}) - \gamma_{t|s}^\theta) \cdot \mathbf{m}\right) \quad (10)$$

where $\gamma_{t|s}^\theta$ is affected by $f_t^\theta(\mathbf{x}_t)$, $\gamma_{t|s}^\theta := (\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s})\langle f_t^\theta(\mathbf{x}_t), \mathbf{x}_t \rangle$.

The detailed proof is in Appx. §A.4. Notice that $f_t^\theta(\mathbf{x}_t)$ is a parameterized neural network with $\mathbf{1}^T f_t^\theta(\mathbf{x}_t) = 1$. As we show next, the above formulation simplifies the negative VLB loss computation greatly (§2.2.4), further motivates an approximated loss that is much easier to optimize (§2.2.5), and accelerates the sampling process through reparameterization (§A.8).

2.2.4 Loss Function

With $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ in Eq. (9) and $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ in Eq. (10), the $\mathcal{L}_t(\theta)$ in Eq. (2) can be written as

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\delta_{\mathbf{x}_t, \mathbf{x}_0} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t = \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) + (1 - \delta_{\mathbf{x}_t, \mathbf{x}_0}) D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t \neq \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) \right], \quad (11)$$

where $\delta_{\mathbf{x}_t, \mathbf{x}_0}$ denotes the Kronecker delta of $\mathbf{x}_t$ and $\mathbf{x}_0$. $q(\mathbf{x}_{t-1}|\mathbf{x}_t = \mathbf{x}_0)$ and $q(\mathbf{x}_{t-1}|\mathbf{x}_t \neq \mathbf{x}_0)$ represent the first and second categorical distribution in Eq. (9), respectively.

Apart from negative VLB, another commonly employed auxiliary loss is the cross-entropy (CE) loss between $q(\mathbf{x}_t|\mathbf{x}_0)$ and $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$, which measures the reconstruction quality.

$$\mathcal{L}_t^{CE}(\theta) := \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [-\log p_\theta(\mathbf{x}_0|\mathbf{x}_t)] \quad (12)$$

Eq. (12) and Eq. (11) share the same global minima with $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ being the true posterior $q(\mathbf{x}_0|\mathbf{x}_t)$. However they have different optimization landscape; and thus under limited data and network capacity, which loss is easier to minimize is unknown [Tewari and Bartlett, 2007, Demirkaya et al., 2020].

2.2.5 Contrib. 2: Further Loss Simplification for Easier Optimization

While Eq. (11) is the *exact* negative VLB loss, in practice we find it harder to minimize than $\mathcal{L}_t^{CE}$. In this section, we first derive a much simpler, approximated loss by observing a relation between $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ and $p_\theta(\mathbf{x}_t|\mathbf{x}_0)$. We then show that coefficient simplification and $\mathcal{L}_t^{CE}$ are *both* valuable for optimizing a general negative VLB where only partial time steps are observed. We combine all designs to derive the final approximated loss, denoted as $\tilde{\mathcal{L}}_t$.

Proposition 2.3. *For any $0 < s < t \leq T$, with $\mathbf{x}_0$ known, $\Delta p_\theta(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ is defined as*

$$p_\theta(\mathbf{x}_s|\mathbf{x}_t) - q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) = (1 - \mu_{t|s})[f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0 + \phi_{t|s}\langle f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0, \mathbf{x}_t \rangle(\mathbf{x}_t - \mathbf{m})], \quad (13)$$

where $\phi_{t|s} := \frac{(1 - \bar{\alpha}_s)\bar{\alpha}_{t|s}}{\bar{\alpha}_t + (1 - \bar{\alpha}_t)\langle \mathbf{x}_t, \mathbf{m} \rangle}$.

Proposition 2.3 shows that the distribution difference between $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ and $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ has a closed-form formulation. (See Appx. §A.5 for the proof.) With it, we can apply the Taylor expansion (up to second order) to approximate the KL divergence directly (See Appx. §A.6.)

$$D_{\text{KL}}(q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_s|\mathbf{x}_t)) \approx \sum_{\mathbf{x}_s} \frac{|\Delta p_\theta(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)|^2}{q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)} \quad (14)$$

2.2.3 贡献1: $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ 的解析形式

文献探讨了 $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$: (1) 直接参数化; (2) 以 $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ 为参数, 使用 f_t^θ 进行参数化, 使得 $p_\theta(\mathbf{x}_0|\mathbf{x}_t) = \text{Cat}(\mathbf{x}_0; f_t^\theta(\mathbf{x}_t))$ 并让 $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = q(\mathbf{x}_{t-1}|\mathbf{x}_t, f_t^\theta(\mathbf{x}_t))$; 以及 (3) 以 $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ 为参数, 使用 f_t^θ 然后边缘化 $q(\mathbf{x}_{t-1}, \mathbf{x}_0|\mathbf{x}_t)$ 使得 $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \sum_{\mathbf{x}_0} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ 。

方法 (1) 在前向过程中不重用任何分布, 因此效率较低。方法 (2) 已被广泛用于连续扩散模型, 如 Ho 等人 [2020a] 和 Song 等人 [2021a], 的研究, 以及一些离散扩散模型, 如 Hoogeboom 等人 [2021] 和 Zheng 等人 [2023] 的研究。它避免了边缘化, 并且对于连续扩散非常有效。然而对于离散扩散, 如式 (9) 所示, 样本 $\mathbf{x}_0$ 决定了应使用哪个分类分布, 而如果没有真实的 $\mathbf{x}_0$ 则无法确定。Zheng 等人 [2023], 提出了一些启发式方法, 但这些方法与真实值 $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ 可能存在较大差距, 导致采样过程不准确。

方法 (3) 已被 Austin 等人 [2021] 采用, 通过直接边缘化 $\mathbf{x}_0$, , 这给损失函数计算和采样过程引入了额外的计算成本, 因为 $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ 的公式尚未简化为封闭形式的分布。在本文中, 我们证明了方法 (3) 的参数化可以简化为分类分布的简洁公式。

命题 2.2。 $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ 的参数化可以简化为任意 $0 < s < t \leq T$ 的

$$p_\theta(\mathbf{x}_s|\mathbf{x}_t) = \text{Cat}\left(\mathbf{x}_s; (1 - \mu_{t|s}) \cdot f_t^\theta(\mathbf{x}_t) + (\mu_{t|s}\bar{\alpha}_{t|s} + \gamma_{t|s}^\theta) \cdot \mathbf{x}_t + (\mu_{t|s}(1 - \bar{\alpha}_{t|s}) - \gamma_{t|s}^\theta) \cdot \mathbf{m}\right) \quad (10)$$

其中 $\gamma_{t|s}^\theta$ 受 $f_t^\theta(\mathbf{x}_t)$ 、 $\gamma_{t|s}^\theta := (\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s})\langle f_t^\theta(\mathbf{x}_t), \mathbf{x}_t \rangle$ 影响

详细证明在附录 §A.4 中。注意 $f_t^\theta(\mathbf{x}_t)$ 是一个带 $\mathbf{1}^T f_t^\theta(\mathbf{x}_t) = 1$ 的参数化神经网络。正如我们接下来所展示的, 上述公式极大地简化了负 VLB 损失计算 (§2.2.4), 进一步促使我们采用更容易优化的近似损失 (§2.2.5), 并通过重新参数化加速采样过程 (§A.8)。

2.2.4 损失函数

根据公式 (9) 中的 $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ 和公式 (10) 中的 $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$, 公式 (2) 中的 $\mathcal{L}_t(\theta)$ 可以写成

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\delta_{\mathbf{x}_t, \mathbf{x}_0} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t = \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) + (1 - \delta_{\mathbf{x}_t, \mathbf{x}_0}) D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t \neq \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) \right], \quad (11)$$

$\delta_{\mathbf{x}_t, \mathbf{x}_0}$ 处, t 和 $\{x3\}$ 的 Kronecker delta。 $q(\mathbf{x}_{t-1}|\mathbf{x}_t = \mathbf{x}_0)$ 和 $\{x5\}$ 分别表示公式(9)中的第一类和第二类分类分布。

除了负 VLB 之外, 另一种常用的辅助损失是 $\{x1\}$ 和 $\{x2\}$ 之间的交叉熵(CE)损失, 它用于衡量重建质量。

$$\mathcal{L}_t^{CE}(\theta) := \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [-\log p_\theta(\mathbf{x}_0|\mathbf{x}_t)] \quad (12)$$

公式(12)和公式(11)与 $\{x1\}$ 作为真实后验 $\{x2\}$ 具有相同的全局最小值。然而, 它们的优化景观不同; 因此, 在有限数据和网络容量的情况下, 哪种损失更容易最小化尚不清楚[Tewari and Bartlett, 2007, Demirkaya et al., 2020]。

2.2.5 贡献2: 进一步简化损失以实现更易于优化

虽然公式(11)是精确的负 VLB 损失, 但在实践中我们发现它比 $\mathcal{L}_t^{CE}$ 更难最小化。在本节中, 我们首先通过观察 $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ 和 $p_\theta(\mathbf{x}_t|\mathbf{x}_0)$ 之间的关系, 推导出一个更简单、近似化的损失。然后, 我们证明系数简化和 $\mathcal{L}_t^{CE}$ 对于优化仅观察到部分时间步的通用负 VLB 都是有价值的。我们将所有设计结合起来, 推导出最终的近似化损失, 记为 $\mathcal{L}_{to}$ 。

命题 2.3。对于任何 $0 < s < t \leq T$, 当 $\mathbf{x}_0$ 已知时, $\Delta p_\theta(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ 定义为

$$p_\theta(\mathbf{x}_s|\mathbf{x}_t) - q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) = (1 - \mu_{t|s})[f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0 + \phi_{t|s}\langle f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0, \mathbf{x}_t \rangle(\mathbf{x}_t - \mathbf{m})], \quad (13)$$

where $\phi_{t|s} := \frac{(1 - \bar{\alpha}_s)\bar{\alpha}_{t|s}}{\bar{\alpha}_t + (1 - \bar{\alpha}_t)\langle \mathbf{x}_t, \mathbf{m} \rangle}$ 。

命题 2.3 表明, $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ 与 $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ 之间的分布差异具有封闭形式解。(参见附录 §A.5 证明。)有了它, 我们可以直接应用泰勒展开(二阶以内)来近似 KL 散度(参见附录 §A.6)。

$$D_{\text{KL}}(q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_s|\mathbf{x}_t)) \approx \sum_{\mathbf{x}_s} \frac{|\Delta p_\theta(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)|^2}{q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)} \quad (14)$$

The above formulation along with prop.2.3 are valid for *any* $0 < s < t \leq T$. We next show that minimizing divergence between q and p_θ at any s and t is also valid, as it is inside a general negative VLB with partial time steps. The initial version of the VLB is derived under the assumption that observations are made at every time step. Its backward denoising process is designed to advance by a single time step during each generation step for best generation quality. Let us consider a more general case where only partial time steps are observed in the forward process, then, minimizing its negative VLB can help improve generation quality with fewer steps. Assuming only $\mathbf{x}_s$ and $\mathbf{x}_t$ are observed, where $0 < s < t \leq T$, a derivation analogous to that of Eq. (2) can show that

$$\log p_\theta(\mathbf{x}_0) \geq -D_{\text{KL}}[q(\mathbf{x}_t|\mathbf{x}_0)\|p_\theta(\mathbf{x}_t)] + \mathbb{E}_{q(\mathbf{x}_s|\mathbf{x}_0)}[\log p_\theta(\mathbf{x}_0|\mathbf{x}_s)] - D_{\text{KL}}[q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)\|p_\theta(\mathbf{x}_s|\mathbf{x}_t)], \quad (15)$$

where the first divergence term between the prior and posterior quantifies the quality of the backward denoising process from T to t . The second term represents the CE loss, $\mathcal{L}_s^{\text{CE}}$, which influences the generation quality from time s to time 0. The final term is given by Eq. (14), and contributes to the generation process from t to s . (See Appx. §A.7 for proof.)

This generalized formulation of the VLB highlights the significance of the CE loss and the alignment between $q(\mathbf{x}_s|\mathbf{x}_s, \mathbf{x}_0)$ and $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ at any observed times s and t . While CE loss does not have a coefficient that depends on noise schedules and time, changing s and t or noise schedule (which determines $\bar{\alpha}_t$) during training will greatly impact the scale of the term in Eq. (14). By rendering the loss scaling term independent of time and the noise schedule, the minimization of this adjusted loss concurrently leads to the minimization of the original loss in Eq. (14) for any given s and t . Hence, by removing the sensitive scale $\frac{(1-\mu_{t|s})^2}{q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)}$ in Eq. (14), we reformulate the loss as

$$\mathcal{L}_t^2 := \|f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0 + \phi_{t|s}\langle f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0, \mathbf{x}_t \rangle(\mathbf{x}_t - \mathbf{m})\|_2^2, \quad (16)$$

where we can further clip $\phi_{t|s}$ to $\min(1, \phi_{t|s})$ for minimal scaling influence. It is important to note that while we have modified the coefficient to be invariant to the noise schedule and time s to effectively minimize Eq. (14) at any t and s , a similar approach to coefficient revision has been previously explored in Ho et al. [2020a], Karras et al. [2022], primarily to facilitate an easier optimization by achieving a balance of terms in the loss function.

Overall, we have the final approximated loss as $\tilde{\mathcal{L}}_t(\theta) = \mathcal{L}_t^2(\theta) + \mathcal{L}_t^{CE}(\theta)$. We find that in practice this loss is much easier to optimize than the original exact negative VLB for more complicated distributions, leading to faster convergence and improved generation quality.

3 Continuous-time Discrete Diffusion

Despite being simple, discrete-time diffusion limits the generation process as we can only “jump back” through fixed time points. Recent works generalize continuous-state diffusion models to continuous-time [Song et al., 2021b]. This generalization enables great flexibility in backward generation as one can “jump back” through any time in $[0, T]$ with improved sample quality.

Nevertheless, generalizing discrete-state diffusion model from discrete-time to continuous-time is nontrivial, as the score-matching based technique [Song et al., 2021b] in continuous-state models requires the score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ to be available. This function, however, is evidently non-existent for discrete distributions. Recently, Campbell et al. [2022] presented the first continuous-time diffusion model for discrete data. It formulates the forward process through a Continuous Time Markov Chain (CTMC) and aims at learning a reverse CTMC that matches the marginal distribution with the forward CTMC at any time t . While being theoretically solid, the formulation in Campbell et al. [2022] has two problems: (1) the negative VLB loss of matching the forward and backward CTMCs is analytically complicated and hard to implement; and (2) the exact sampling through the learned backward CTMC is unrealistic. Campbell et al. [2022] propose approximate solutions, which however, trade off computational tractability with unknown errors and sample quality. SDDM Sun et al. [2023] takes a different approach with ratio matching Hyvärinen [2007], Lyu [2009], which is the generalization of score matching to discrete data. However, it needs a specific model architecture. Lou et al. [2024] advances ratio matching, yielding a loss that requires only a single model pass. Interestingly, this loss is similar to ours in formulation, despite originating from a different angle.

In this paper, we build on the CTMC formulation in Campbell et al. [2022], and show that the loss can be analytically simplified for nominal data. This simplification also inspires an improved MCMC corrector with closed-form formulation. For problem (2), we argue that the difficulty arises from directly using the learned backward CTMC’s transition rate. Instead, realizing that the reverse transition rate is computed based on the learned $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$, we propose to compute $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ through

上述公式连同命题2.3对任何 $0 < s < t \leq T$ 都成立。我们接下来证明，在任何 s 和 t 处最小化 q 与 p_θ 之间的散度也是有效的，因为它包含在一个具有部分时间步长的通用负VLB内。VLB的初始版本是在假设每个时间步都进行观测的条件下推导的。其反向去噪过程设计为在每次生成步骤中前进一个时间步，以获得最佳生成质量。让我们考虑一个更一般的情况，其中正向过程中只有部分时间步长被观测到，那么，最小化其负VLB可以帮助用更少的步骤提高生成质量。假设仅观测到 $\mathbf{x}_s$ 和 $\mathbf{x}_t$ ，其中 $0 < s < t \leq T$ ，一个类似于等式(2)的推导可以表明

$$\log p_\theta(\mathbf{x}_0) \geq -D_{\text{KL}}[q(\mathbf{x}_t|\mathbf{x}_0)\|p_\theta(\mathbf{x}_t)] + \mathbb{E}_{q(\mathbf{x}_s|\mathbf{x}_0)}[\log p_\theta(\mathbf{x}_0|\mathbf{x}_s)] - D_{\text{KL}}[q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)\|p_\theta(\mathbf{x}_s|\mathbf{x}_t)], \quad (15)$$

先验与后验之间的第一个发散项量化了从 T 到 t 的逆向去噪过程的质量。第二个项表示CE损失， $\mathcal{L}_s^{\text{CE}}$ ，它影响了从时间 s 到时间0的生成质量。最后一项由公式(14)给出，它对从 t 到 s 的生成过程做出贡献。（参见附录 § A.7的证明。）

这种VLB的广义公式突出了CE损失的重要性以及 $q(\mathbf{x}_s|\mathbf{x}_s, \mathbf{x}_0)$ 与 $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ 在任何观测时间 s 和 t 之间的对齐。虽然CE损失没有依赖于噪声调度和时间的系数，但在训练过程中改变 s 和 t 或噪声调度（它决定了 $\bar{\alpha}_t$ ）会极大地影响公式(14)中该项的规模。通过使损失缩放项与时间和噪声调度无关，调整后损失的最小化同时导致在任何给定的 s 和 t 下公式(14)中原始损失的最小化。因此，通过移除公式(14)中的敏感尺度 $\frac{(1-\mu_{t|s})^2}{q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)}$ ，我们将损失重新表述为

$$\mathcal{L}_t^2 := \|f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0 + \phi_{t|s}\langle f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0, \mathbf{x}_t \rangle(\mathbf{x}_t - \mathbf{m})\|_2^2, \quad (16)$$

在何处我们可以进一步将 $\phi_{t|s}$ 剪辑到 $\min(1, \phi_{t|s})$ 以实现最小的缩放影响。重要的是要注意，虽然我们已经修改了系数使其对噪声调度和时间 s 不变，以有效地在任何 t 和 s 处最小化公式 (14)，但类似系数修订的方法在 Ho 等人 [2020a], Karras 等人 [2022], 中已被先前探索，主要目的是通过在损失函数的项之间取得平衡来简化优化过程。

总体而言，我们得到了最终的近似损失 $\tilde{\mathcal{L}}_t(\theta) = \mathcal{L}_t^2(\theta) + \mathcal{L}_t^{CE}(\theta)$ 。我们发现，在实践中，这个损失比原始的精确负 VLB 对于更复杂的分布更容易优化，从而实现了更快的收敛和更好的生成质量。

3 连续时间离散扩散

尽管简单，离散时间扩散限制了生成过程，因为我们只能通过固定的时间点“跳跃”回过去。最近的研究将连续状态扩散模型推广到连续时间 [Song 等人, 2021b]。这种推广在反向生成中提供了极大的灵活性，因为人们可以在 $[0, T]$ 中通过任何时间点“跳跃”回过去，并提高了样本质量。

然而，将离散状态扩散模型从离散时间推广到连续时间并非易事，因为基于分数匹配的技术 [Song 等人, 2021b] 在连续状态模型中需要分数函数 $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ 的可用性。然而，该函数对离散分布显然是不存在的。最近，Campbell 等人 [2022] 提出了第一个用于离散数据的连续时间扩散模型。它通过连续时间马尔可夫链 (CTMC) 构建前向过程，并旨在学习一个与任何时间 t 的前向 CTMC 匹配边际分布的逆向 CTMC。虽然理论上很扎实，但 Campbell 等人 [2022] 的公式存在两个问题：(1) 匹配前向和逆向 CTMC 的负 VLB 损失在解析上很复杂且难以实现；(2) 通过学习的逆向 CTMC 进行精确采样是不现实的。Campbell 等人 [2022] 提出了近似解决方案，但这些方案在计算可伸缩性与未知误差和样本质量之间进行了权衡。SDDM Sun 等人 [2023] 采用了不同的方法，即比例匹配 Hyvärinen [2007], Lyu [2009]，这是分数匹配到离散数据的推广。然而，它需要特定的模型架构。Lou 等人 [2024] 推进了比例匹配，产生一个只需要单次模型遍历的损失。有趣的是，尽管来自不同的角度，这个损失在公式上与我们的损失相似。

在本文中，我们基于Campbell等人 [2022], 中的CTMC公式，并证明了对于名义数据，损失函数可以解析简化。这种简化也启发了一种具有闭式公式的改进MCMC校正器。对于问题(2)，我们论证困难在于直接使用学习到的反向CTMC的转移率。相反，认识到反向转移率是基于学习到的 $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ 计算的，我们提出通过计算 $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ 来

$p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ without using the transition rate matrix. This avoids the approximation error of sampling directly from the backward CTMC and greatly simplifies the generation process.

Remarkably, with the new formulation of generation, we show that the continuous-time and discrete-time diffusion models can be unified together, with exactly the same forward diffusion and now also backward generation process. Moreover, we demonstrate that this unification offers mutual benefits: the continuous-time diffusion can leverage the swift and precise sampling formulation derived from the discrete-time case (as detailed in §A.8), while the discrete-time diffusion can utilize the MCMC corrector from the continuous-time scenario (see Appx. §A.17 and §A.19.6).

3.1 Background: Continuous-Time Markov Chain

CTMC generalizes Markov chain from discrete- to continuous-time via the Markov property: $\mathbf{x}_{t_1} \perp\!\!\!\perp \mathbf{x}_{t_3} | \mathbf{x}_{t_2}, \forall t_1 < t_2 < t_3$. Anderson [2012] provides an introduction to time-homogeneous CTMC. It can be derived from discrete-time Markov chain by increasing the number of time stamps N to infinite while keeping the total time T fixed. Specifically, we can define $\Delta t = \frac{T}{N}$, $t_i = i\Delta t$, and a discrete-time Markov chain characterized by transition probability $q(\mathbf{x}_{t_i}|\mathbf{x}_{t_{i-1}})$ and transition matrix Q_{t_i} with $[Q_{t_i}]_{jk} = q(\mathbf{x}_{t_i} = \mathbf{e}_k | \mathbf{x}_{t_{i-1}} = \mathbf{e}_j)$. By setting N to infinite, the transition probability $q(\mathbf{x}_{t_i}|\mathbf{x}_{t_{i-1}})$ converges to 0, hence is not suitable for describing CTMC. Instead, CTMC is fully characterized by its *transition rate* $r_t(\mathbf{y}|\mathbf{x})$, s.t.

$$r_t(\mathbf{y}|\mathbf{x}) = \lim_{N \rightarrow \infty} \frac{q(\mathbf{x}_{t_i} = \mathbf{y} | \mathbf{x}_{t_{i-1}} = \mathbf{x}) - \delta_{\mathbf{x}, \mathbf{y}}}{t_i - t_{i-1}} = \lim_{\Delta t \rightarrow 0} \frac{q_{t|t-\Delta t}(\mathbf{y}|\mathbf{x}) - \delta_{\mathbf{x}, \mathbf{y}}}{\Delta t}. \quad (17)$$

As the name suggests, $r_t(\mathbf{y}|\mathbf{x})$ measures the change rate of the transition probability of moving from state $\mathbf{x}$ to state $\mathbf{y}$ at time t in the direction of the process. The corresponding *transition rate matrix* R_t with $[R_t]_{ij} = r_t(\mathbf{e}_j|\mathbf{e}_i)$ fully determines the underlying stochastic process. A CTMC's transition probabilities satisfy the Kolmogorov equations [Kolmogoroff, 1931] (see Appx. §A.10), which have unique solution. As Rindos et al. [1995] stated, when R_{t_1} and R_{t_2} commute (i.e. $R_{t_1}R_{t_2} = R_{t_2}R_{t_1}$) for any t_1, t_2 , the transition probability matrix can be written as $\bar{Q}_{t|s} = \exp\left(\int_s^t R_a da\right)$, where $\exp(M) := \sum_{k=0}^{\infty} \frac{M^k}{k!}$. The commutative property of R_t can be achieved by choosing $R_t = \beta(t)R_b$ where $R_b \in \mathbb{R}^{K \times K}$ is a time-independent base rate matrix.

3.2 Contrib. 3: Simplified Forward & Backward CTMCs that Connect to Discrete-Time Case

Forward CTMC. Two properties are needed for modeling the forward process: P1) the process can converge to an easy-to-sample stationary distribution at final time T ; and P2) the conditional marginal distribution $q(\mathbf{x}_t|\mathbf{x}_0)$ can be obtained analytically for efficient training. As given above, P2) can be achieved by choosing commutative transition rate matrices with $R_t = \beta(t)R_b$.

We next show that property P1), i.e. $\lim_{t \rightarrow T} \bar{Q}_{t|0} = \bar{Q}_{T|0} = \mathbf{1}\mathbf{m}^\top$ for some stationary distribution $\mathbf{m}$, can be achieved by choosing $R_b = \mathbf{1}\mathbf{m}^\top - I$, which is a valid transition rate matrix with the property $(-R_b)^2 = (-R_b)$ (see derivation in Appx. §A.11). Then we have

$$\bar{Q}_{t|s} = \exp(\bar{\beta}_{t|s}R_b) = e^{-\bar{\beta}_{t|s}}I + (1 - e^{-\bar{\beta}_{t|s}})\mathbf{1}\mathbf{m}^\top. \quad (18)$$

★ **Unified forward process.** Eq. (18) has the same formulation as the transition matrix of the discrete-time case in Eq. (8), if we set $\bar{\alpha}_{t|s} = \exp(-\bar{\beta}_{t|s}) = \exp(-\int_s^t \beta(a)da)$. Thus, *this formulation unifies the forward processes of adding noise for both discrete- and continuous-time discrete diffusion*. With $\lim_{t \rightarrow T} \bar{\alpha}_{t|0} = 0$, or equivalently $\lim_{t \rightarrow T} \int_0^T \beta(a)da = \infty$, we achieve the goal $\bar{Q}_{T|0} = \mathbf{1}\mathbf{m}^\top$. We use $\bar{\alpha}_{t|s}$ directly in the following sections, i.e. Eq. (8). To summarize, for the forward CTMC

$$R_t = \beta(t)(\mathbf{1}\mathbf{m}^\top - I), \text{ and } \bar{Q}_{t|s} = \bar{\alpha}_{t|s}I + (1 - \bar{\alpha}_{t|s})\mathbf{1}\mathbf{m}^\top. \quad (19)$$

We can further get vector-form forward rate

$$r_t(\mathbf{x}|\cdot) = R_t\mathbf{x} = \beta(t)(\langle \mathbf{x}, \mathbf{m} \rangle \mathbf{1} - \mathbf{x}), \quad r_t(\cdot|\mathbf{x}) = R_t^\top \mathbf{x} = \beta(t)(\mathbf{m} - \mathbf{x}). \quad (20)$$

Backward CTMC. To generate samples from the target distribution, we have to reverse the process of the forward CTMC. Let $\hat{r}_t$ be the transition rate of the backward CTMC with corresponding matrix $\hat{R}_t$. When the forward and backward CTMCs are matched exactly, theoretically the forward and

$p_\theta(\mathbf{x}_0|\mathbf{x}_t)$ 不使用转移率矩阵。这避免了直接从反向CTMC采样而产生的近似误差，并极大地简化了生成过程。

值得注意的是，通过新的生成公式，我们证明了连续时间扩散模型和离散时间扩散模型可以统一在一起，并且具有完全相同的正向扩散过程，现在也包括逆向生成过程。此外，我们展示了这种统一具有互惠性：连续时间扩散可以利用从离散时间情况中推导出的快速精确采样公式（如 §A.8 所述），而离散时间扩散则可以利用来自连续时间场景的 MCMC 校正器（参见 Appx. §A.17 和 §A.19.6）。

3.1 背景：连续时间马尔可夫链

CTMC 通过马尔可夫特性将马尔可夫链从离散时间推广到连续时间： $\mathbf{x}_{t_1} \perp\!\!\!\perp \mathbf{x}_{t_3} | \mathbf{x}_{t_2}, \forall t_1 < t_2 < t_3$. Anderson [2012] 介绍了齐次连续时间 CTMC。它可以通过将离散时间马尔可夫链的时间戳数量 N 增加到无限，同时保持总时间 T 固定来推导。具体来说，我们可以定义 $\Delta t = \frac{T}{N}$, $t_i = i\Delta t$, 以及一个由转移概率 $q(\mathbf{x}_{t_i}|\mathbf{x}_{t_{i-1}})$ 和转移矩阵 Q_{t_i} （具有 $[Q_{t_i}]_{jk} = q(\mathbf{x}_{t_i} = \mathbf{e}_k | \mathbf{x}_{t_{i-1}} = \mathbf{e}_j)$ ）表征的离散时间马尔可夫链。通过将 N 设置为无限，转移概率 $q(\mathbf{x}_{t_i}|\mathbf{x}_{t_{i-1}})$ 收敛到 0，因此不适用于描述 CTMC。相反，CTMC 完全由其转移率 $r_t(\mathbf{y}|\mathbf{x})$ 表征，即

$$r_t(\mathbf{y}|\mathbf{x}) = \lim_{N \rightarrow \infty} \frac{q(\mathbf{x}_{t_i} = \mathbf{y} | \mathbf{x}_{t_{i-1}} = \mathbf{x}) - \delta_{\mathbf{x}, \mathbf{y}}}{t_i - t_{i-1}} = \lim_{\Delta t \rightarrow 0} \frac{q_{t|t-\Delta t}(\mathbf{y}|\mathbf{x}) - \delta_{\mathbf{x}, \mathbf{y}}}{\Delta t}. \quad (17)$$

顾名思义， $r_t(\mathbf{y}|\mathbf{x})$ 衡量了在时间 t 时从状态 $\mathbf{x}$ 到状态 $\mathbf{y}$ 的转移概率随时间的变化率，在过程的方向上。相应的转移率矩阵 R_t 与 $[R_t]_{ij} = r_t(\mathbf{e}_j|\mathbf{e}_i)$ 完全决定了底层随机过程。CTMC 的转移概率满足 Kolmogorov 方程 [Kolmogoroff, 1931]（参见附录 §A.10），这些方程有唯一解。正如 Rindos 等人 [1995] 所述，当 R_{t_1} 和 R_{t_2} 交换（即 $R_{t_1}R_{t_2} = R_{t_2}R_{t_1}$ ）对任何 t_1, t_2 时，转移概率矩阵可以表示为 $\bar{Q}_{t|s} = \exp\left(\int_s^t R_a da\right)$ ，其中 $\exp(M) := \sum_{k=0}^{\infty} \frac{M^k}{k!}$ 。通过选择 $R_t = \beta(t)R_b$ ，其中 $R_b \in \mathbb{R}^{K \times K}$ 是一个与时间无关的基础率矩阵，可以实现 R_t 的交换性质。

3.2 贡献3：简化连接到离散时间案例的前向&后向CTMC

前向CTMC。建模前向过程需要两个性质：P1) 该过程在最终时间 T 可以收敛到一个易于采样的稳态分布；以及P2) 条件边缘分布 $q(\mathbf{x}_t|\mathbf{x}_0)$ 可以解析地获得以进行高效训练。如上所述，P2) 可以通过选择具有 $R_t = \beta(t)R_b$ 的交换过渡率矩阵来实现。

我们接下来证明性质P1)，即对于某个稳态分布 $\mathbf{m}$ 的 $\lim_{t \rightarrow T} \bar{Q}_{t|0} = \bar{Q}_{T|0} = \mathbf{1}\mathbf{m}^\top$ ，可以通过选择 $R_b = \mathbf{1}\mathbf{m}^\top - I$ 来实现， $R_b = \mathbf{1}\mathbf{m}^\top - I$ 是一个具有性质 $(-R_b)^2 = (-R_b)$ 的有效过渡率矩阵（参见附录 §A.11 的推导）。然后我们有

$$\bar{Q}_{t|s} = \exp(\bar{\beta}_{t|s}R_b) = e^{-\bar{\beta}_{t|s}}I + (1 - e^{-\bar{\beta}_{t|s}})\mathbf{1}\mathbf{m}^\top, \quad (18)$$

★ 统一的前向过程。如果我们将 $\alpha_{t|s} = \exp(-\bar{\beta}_{t|s}) = \exp(-\int_s^t \beta(a)da)$ 代入公式(18)，它就与公式(8)中离散时间案例的转移矩阵具有相同的表述形式。因此，这种表述形式统一了离散时间和连续时间离散扩散添加噪声的前向过程。通过 $\lim_{t \rightarrow T} \alpha_{t|0} = 0$ ，或等价地通过 $\lim_{t \rightarrow T} \int_0^T \beta(a)da = \infty$ ，我们实现了目标 $\bar{Q}_{T|0} = \mathbf{1}\mathbf{m}^\top$ 。我们在以下各节中直接使用 $\alpha_{t|s}$ ，即公式(8)。总结来说，对于前向CTMC

$$R_t = \beta(t)(\mathbf{1}\mathbf{m}^\top - I), \text{ and } \bar{Q}_{t|s} = \alpha_{t|s}I + (1 - \alpha_{t|s})\mathbf{1}\mathbf{m}^\top. \quad (19)$$

我们可以进一步获取向量形式的远期利率

$$r_t(\mathbf{x}|\cdot) = R_t\mathbf{x} = \beta(t)(\langle \mathbf{x}, \mathbf{m} \rangle \mathbf{1} - \mathbf{x}), \quad r_t(\cdot|\mathbf{x}) = R_t^\top \mathbf{x} = \beta(t)(\mathbf{m} - \mathbf{x}). \quad (20)$$

反向CTMC。要从目标分布中生成样本，我们必须逆转正向CTMC的过程。设 $\hat{r}_t$ 为反向CTMC的转移率，对应矩阵 $\hat{R}_t$ 。当正向和反向CTMC完全匹配时，理论上正向和

backward CTMCs have the following relationship (see Campbell et al. [2022]’s Proposition 1):

$$\hat{r}_t(\mathbf{x}|\mathbf{y}) = r_t(\mathbf{y}|\mathbf{x}) \frac{q_t(\mathbf{x})}{q_t(\mathbf{y})}, \quad \forall \mathbf{x} \neq \mathbf{y}. \quad (21)$$

However, the marginal distributions $q_t(\mathbf{x})$ and $q_t(\mathbf{y})$ are intractable analytically, hence we cannot derive the backward CTMC directly from Eq. (21). Instead, Campbell et al. [2022] parameterize the transition rate $\hat{r}_t^\theta$, by observing that $\frac{q_t(\mathbf{x})}{q_t(\mathbf{y})} = \sum_{\mathbf{x}_0} \frac{q_{t|0}(\mathbf{x}|\mathbf{x}_0)}{q_{t|0}(\mathbf{y}|\mathbf{x}_0)} q_{0|t}(\mathbf{x}_0|\mathbf{y})^3$, as follows

$$\hat{r}_t^\theta(\mathbf{x}|\mathbf{y}) = r_t(\mathbf{y}|\mathbf{x}) \sum_{\mathbf{x}_0} \frac{q_{t|0}(\mathbf{x}|\mathbf{x}_0)}{q_{t|0}(\mathbf{y}|\mathbf{x}_0)} p_{0|t}^\theta(\mathbf{x}_0|\mathbf{y}). \quad (22)$$

Then, $\hat{r}_t^\theta$ is learned with θ to minimize the continuous-time negative VLB introduced next.

Negative VLB. Similar to the discrete-time case, the backward CTMC can be learned by maximizing the VLB for data log-likelihood. Computing VLB for CTMC is nontrivial, and fortunately Campbell et al. [2022] has derived (see their Proposition 2) that the negative VLB can be formulated as

$$T \mathbb{E}_{\substack{t \sim \text{Uni}(0,T) \\ \mathbf{x} \sim q(\mathbf{x}_t|\mathbf{x}_0)}} \left[\sum_{\mathbf{z} \neq \mathbf{x}} \hat{r}_t^\theta(\mathbf{z}|\mathbf{x}) - \sum_{\mathbf{z} \neq \mathbf{x}} r_t(\mathbf{z}|\mathbf{x}) \log \hat{r}_t^\theta(\mathbf{x}|\mathbf{z}) \right]. \quad (23)$$

However, Campbell et al. [2022]’s original design did not simplify the negative VLB with the parameterization of $\hat{r}_t^\theta$ in Eq. (22), making the implementation nontrivial and inefficient. In this section, we show that their formulation can be greatly simplified to a closed-form evaluation.

To simplify Eq. (23), we introduce $g_t^\theta(\mathbf{x}|\mathbf{y})$ such that $\hat{r}_t^\theta(\mathbf{x}|\mathbf{y}) = r_t(\mathbf{y}|\mathbf{x}) g_t^\theta(\mathbf{x}|\mathbf{y})$, with

$$g_t^\theta(\mathbf{x}|\mathbf{y}) := \sum_{\mathbf{x}_0} \frac{q_{t|0}(\mathbf{x}|\mathbf{x}_0)}{q_{t|0}(\mathbf{y}|\mathbf{x}_0)} p_{0|t}^\theta(\mathbf{x}_0|\mathbf{y}) \approx \frac{q_t(\mathbf{x})}{q_t(\mathbf{y})}, \quad (24)$$

which is the estimator of the marginal probability ratio.

Proposition 3.1. The vector form parameterization of $g_t^\theta(\mathbf{x}|\mathbf{y})$ can be simplified analytically as:

$$g_t^\theta(\cdot|\mathbf{y}) = \left[\left(1 - \frac{\bar{\alpha}_{t|0} \langle f_t^\theta(\mathbf{y}), \mathbf{y} \rangle}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle \mathbf{y}, \mathbf{m} \rangle} \right) \mathbf{m} + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} f_t^\theta(\mathbf{y}) \right] \odot \frac{\mathbf{1} - \mathbf{y}}{\langle \mathbf{y}, \mathbf{m} \rangle} + \mathbf{y} \quad (25)$$

The proof is in Appx. §A.12, with its extension to multi-element $g_t^{\theta,d}(\cdot|\mathbf{y}^{1:D})$ in Eq. (75).

3.3 Contrib. 4: Simplification of Continuous-time Negative VLB

As the derivation is much harder in multi-element case, we work on it directly and single-element can be induced as a special case. Given $\mathbf{x}^{1:D}$, let $\mathbf{x}^{\setminus d}$ represent $\mathbf{x}^{1:D \setminus d}$, i.e. the object without d -th element. As we assume the forward processes are independent for different elements, r_t^d represents the transition rate of the forward CTMC process at the d -th element.

Proposition 3.2. The negative VLB in Eq. (23) in multi-element case can be simplified as

$$T \mathbb{E}_{\substack{t \sim \text{Uni}(0,T) \\ \mathbf{x}^{1:D} \sim q_{t|0}(\cdot|\mathbf{x}_0^{1:D}) \\ \mathbf{z}^{1:D} \sim S_t(\cdot|\mathbf{x}^{1:D})}} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D}) - \frac{1}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D}|\mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot|\mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d|\mathbf{x}_0^d)} \right] \quad (26)$$

where $S_t(\mathbf{z}^{1:D}|\mathbf{x}^{1:D})$ is any unnormalized distribution (see Eq. (82) in Appx.) of sampling the auxiliary variable $\mathbf{z}^{1:D}$ from $\mathbf{x}^{1:D}$. The auxiliary variable is introduced to avoid multiple passes of the model for computing the second term of Eq. (23). $\mathcal{M}_{S_t}$ is a normalization scalar (see Eq. (86) in Appx.) that only depends on S_t , $\mathbf{z}^{1:D}$, and $\mathbf{x}_0^{1:D}$.

The detailed proof is in Appx. §A.14. This formulation simplifies and generalizes the result in Campbell et al. [2022], such that any S_t can be used for introducing the auxiliary variable $\mathbf{z}^{1:D}$. Importantly, Eq. (26) shows that changing S_t only affects a scalar weight. Computing the loss requires two passes of the model for $\mathbf{z}^{1:D}$ and $\mathbf{x}^{1:D}$ separately. Now we show that it can be further simplified to only a single pass of the model.

Proposition 3.3. The loss in Eq. (26) can be further simplified as

$$T \mathbb{E}_{\substack{t \sim \text{Uni}(0,T) \\ \mathbf{x}^{1:D} \sim q_{t|0}(\cdot|\mathbf{x}_0^{1:D})}} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d|\cdot)^\top \left(g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D}) - \frac{q_{t|0}(\cdot|\mathbf{x}_0^d) \odot \log g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D})}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \right) \right] \quad (27)$$

³As $q_t(\mathbf{x}) = \sum_{\mathbf{x}_0} q_{t|0}(\mathbf{x}|\mathbf{x}_0) q_{\text{data}}(\mathbf{x}_0)$ and $q_t(\mathbf{x}) = \frac{q_{t|0}(\mathbf{x}|\tilde{\mathbf{x}}_0) q_{\text{data}}(\tilde{\mathbf{x}}_0)}{q_{0|t}(\tilde{\mathbf{x}}_0|\mathbf{x})}$.

反向CTMC之间具有如下关系（参见Campbell等人 [2022]的命题1）：

$$\hat{r}_t(\mathbf{x}|\mathbf{y}) = r_t(\mathbf{y}|\mathbf{x}) \frac{q_t(\mathbf{x})}{q_t(\mathbf{y})}, \quad \forall \mathbf{x} \neq \mathbf{y}. \quad (21)$$

然而，边际分布 $q_t(\mathbf{x})$ 和 $q_t(\mathbf{y})$ 在解析上难以处理，因此我们无法直接从公式(21)推导出反向CTMC。相反，Campbell等人 [2022] 通过观察 $\frac{q_t(\mathbf{x})}{q_t(\mathbf{y})} = \sum_{\mathbf{x}_0} \frac{q_{t|0}(\mathbf{x}|\mathbf{x}_0)}{q_{t|0}(\mathbf{y}|\mathbf{x}_0)} q_{0|t}(\mathbf{x}_0|\mathbf{y})^3$ ，对转移率 $\hat{r}_t^\theta$ 进行参数化，如下

$$\hat{r}_t^\theta(\mathbf{x}|\mathbf{y}) = r_t(\mathbf{y}|\mathbf{x}) \sum_{\mathbf{x}_0} \frac{q_{t|0}(\mathbf{x}|\mathbf{x}_0)}{q_{t|0}(\mathbf{y}|\mathbf{x}_0)} p_{0|t}^\theta(\mathbf{x}_0|\mathbf{y}). \quad (22)$$

然后， $\hat{r}_t^\theta$ 与 θ 一起学习，以最小化接下来引入的连续时间负VLB。

负VLB。类似于离散时间情况，反向CTMC可以通过最大化数据对数似然的VLB来学习。计算CTMC的VLB并不简单，幸运的是，Campbell等人 [2022] 已经推导出（见他们的命题2），负VLB可以表示为

$$T \mathbb{E}_{\substack{t \sim \text{Uni}(0,T) \\ \mathbf{x} \sim q(\mathbf{x}_t|\mathbf{x}_0)}} \left[\sum_{\mathbf{z} \neq \mathbf{x}} \hat{r}_t^\theta(\mathbf{z}|\mathbf{x}) - \sum_{\mathbf{z} \neq \mathbf{x}} r_t(\mathbf{z}|\mathbf{x}) \log \hat{r}_t^\theta(\mathbf{x}|\mathbf{z}) \right]. \quad (23)$$

然而，Campbell等人 [2022]的原始设计没有通过公式(22)中的 $\hat{r}_t^\theta$ 参数化来简化负VLB，使得实现既复杂又低效。在本节中，我们表明他们的公式可以大大简化为封闭形式评估。

为简化公式(23)，我们引入 $g_t^\theta(\mathbf{x}|\mathbf{y})$ ，使得 $\hat{r}_t^\theta(\mathbf{x}|\mathbf{y}) = r_t(\mathbf{y}|\mathbf{x}) g_t^\theta(\mathbf{x}|\mathbf{y})$ ，其中

$$g_t^\theta(\mathbf{x}|\mathbf{y}) := \sum_{\mathbf{x}_0} \frac{q_{t|0}(\mathbf{x}|\mathbf{x}_0)}{q_{t|0}(\mathbf{y}|\mathbf{x}_0)} p_{0|t}^\theta(\mathbf{x}_0|\mathbf{y}) \approx \frac{q_t(\mathbf{x})}{q_t(\mathbf{y})}, \quad (24)$$

这是边际概率比率的估计量。

命题3.1. $g_t^\theta(\mathbf{x}|\mathbf{y})$ 的向量形式参数化可以通过解析方法简化为：

$$g_t^\theta(\cdot|\mathbf{y}) = \left[\left(1 - \frac{\bar{\alpha}_{t|0} \langle f_t^\theta(\mathbf{y}), \mathbf{y} \rangle}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle \mathbf{y}, \mathbf{m} \rangle} \right) \mathbf{m} + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} f_t^\theta(\mathbf{y}) \right] \odot \frac{\mathbf{1} - \mathbf{y}}{\langle \mathbf{y}, \mathbf{m} \rangle} + \mathbf{y} \quad (25)$$

证明见附录 §A.12，其扩展到多元素 $g_t^{\theta,d}(\cdot|\mathbf{y}^{1:D})$ 在公式(75)中。

3.3 贡献4：连续时间负VLB的简化

由于多元素情况下的推导难度更大，我们直接处理它，单元素可以作为特例推导出来。给定 $\mathbf{x}^{1:D}$ ，令 $\mathbf{x}^{\setminus d}$ 表示 $\mathbf{x}^{1:D \setminus d}$ ，即没有 d 元素的物体。由于我们假设不同元素的前向过程是独立的， r_t^d 表示前向CTMC过程在 d 元素处的转移率。

命题3.2. 在多元素情况下，等式(23)中的负VLB可以简化为

$$T \mathbb{E}_{\substack{t \sim \text{Uni}(0,T) \\ \mathbf{x}^{1:D} \sim q_{t|0}(\cdot|\mathbf{x}_0^{1:D}) \\ \mathbf{z}^{1:D} \sim S_t(\cdot|\mathbf{x}^{1:D})}} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D}) - \frac{1}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D}|\mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot|\mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d|\mathbf{x}_0^d)} \right] \quad (26)$$

其中 $S_t(\mathbf{z}^{1:D}|\mathbf{x}^{1:D})$ 是任何未归一化的分布（参见附录中的等式(82)）用于从 $\mathbf{x}^{1:D}$ 中采样辅助变量 $\mathbf{z}^{1:D}$ 。引入辅助变量是为了避免多次运行模型以计算等式(23)的第二项。 $\mathcal{M}_{S_t}$ 是一个归一化标量（参见附录中的等式(86)），它仅取决于 S_t 、 $\mathbf{z}^{1:D}$ 和 $\mathbf{x}_0^{1:D}$ 。

详细的证明在附录 §A.14中。这种公式简化并推广了Campbell等人中的结果 [2022]，使得任何 S_t 都可用于引入辅助变量 $\mathbf{z}^{1:D}$ 。重要的是，等式(26)表明更改 S_t 只影响一个标量权重。计算损失需要分别对 $\mathbf{z}^{1:D}$ 和 $\mathbf{x}^{1:D}$ 运行模型两次。现在我们证明它可以进一步简化为只需要一次模型运行。

命题3.3. 等式 (26) 中的损失可以进一步简化为

$$T \mathbb{E}_{\substack{t \sim \text{Uni}(0,T) \\ \mathbf{x}^{1:D} \sim q_{t|0}(\cdot|\mathbf{x}_0^{1:D})}} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d|\cdot)^\top \left(g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D}) - \frac{q_{t|0}(\cdot|\mathbf{x}_0^d) \odot \log g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D})}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \right) \right] \quad (27)$$

³As $q_t(\mathbf{x}) = \sum_{\mathbf{x}_0} q_{t|0}(\mathbf{x}|\mathbf{x}_0) q_{\text{data}}(\mathbf{x}_0)$ and $q_t(\mathbf{x}) = \frac{q_{t|0}(\mathbf{x}|\tilde{\mathbf{x}}_0) q_{\text{data}}(\tilde{\mathbf{x}}_0)}{q_{0|t}(\tilde{\mathbf{x}}_0|\mathbf{x})}$.

The detailed proof can be found in Appendix §A.15. Note that this loss requires only a single forward pass of the model, making it extremely efficient. Additionally, this loss aligns with the DWDSE loss (while presented in single-element case) in Lou et al. [2024], which is derived from the different concrete score matching direction. This indicates that the VLB is essentially the same as the ratio matching loss, as also demonstrated in the continuous-state setting, such that the score matching Song et al. [2021b] is shown to be equivalent to VLB computation in Ho et al. [2020a].

3.4 Contrib. 5: Fast Backward Sampling & Unification of Discrete Diffusion

Campbell et al. [2022] proposes to use the learned transition rate $\hat{r}_t^\theta$ of the backward CTMC to sample reversely for the target distribution $p_{\text{data}}(\mathbf{x}_0)$. However, different from the forward CTMC where all elements are sampled independently, the backward CTMC processes for all elements are coupled together and do not have the closed-form transition probability derived from the learned transition rate. Direct and exact sampling from a CTMC uses the algorithm by Gillespie [1977], which is intractable for multi-element objects. Campbell et al. [2022] proposes to use tau-leaping [Gillespie, 2001] that approximately samples all transitions from time t to $t - \tau$, assuming $R_t^{1:D}$ fixed during the time period. While being tractable, it introduces unknown errors that needs correction process.

We propose to avoid the costly sampling from using the transition rate of backward CTMC. We first observe that the estimated transition rate $\hat{r}_t^\theta$ is essentially derived from the estimated $p_{0|t}^\theta(\mathbf{x}_0|\mathbf{x}_t)$. Hence using $\hat{r}_t^\theta$ is equivalent to using $p_{0|t}^\theta(\mathbf{x}_0|\mathbf{x}_t)$ in an indirect way. We realize that the transition probability $q_{s|t}(\mathbf{x}_s|\mathbf{x}_t)$ can be computed easily from $p_{0|t}^\theta(\mathbf{x}_0|\mathbf{x}_t)$ directly, as shown in Eq. (10). With the help of Eq. (10), we can sample $\mathbf{x}_0$ or equivalently $\mathbf{x}_{0:T}$ through sampling $\mathbf{x}_{t_{n-1}|t_n}, \dots, \mathbf{x}_{t_1|t_2}, \mathbf{x}_{t_0|t_1}$ sequentially, with any $\{t_i\}_{i=0}^n$ that satisfies $0=t_0 < t_1 < \dots < t_n=T$. Although Eq. (10) is derived for discrete-time case, it applies directly in continuous-time case, thanks to the unification of the forward process for discrete- and continuous-time diffusion (§3.2). Hence, discrete-time and continuous-time diffusion also have the same **unified backward generation process**.

★ **Unified Discrete Diffusion.** All in all, through a series of mathematical simplifications, we have shown that both discrete&continuous-time discrete diffusion share (1) the same forward diffusion process; (2) the same parameterization for learning $p_{0|t}^\theta$; and (3) the same backward denoising process. In light of our reformulations, we propose USD3, a novel Unified and Simplified Discrete Denoising Diffusion model. Notably, USD3 utilizes the same source code for both discrete- and continuous-time, up to a single alteration in the loss function during training, and a shared sample generation process (resp. Algo. 1 and 2 in Appx. §A.16).

4 Experiments

Discrete diffusion has limited, no established guidelines on model training. Many prior work optimize a combined VLB and cross-entropy (CE) loss with a fixed weight, without deeper investigation. In this work we not only improve VLB loss mathematically, but also empirically explore an extensive testbed of training regimes for discrete diffusion—including various loss combinations, both discrete- and continuous-time training, as well as varying model sizes—toward a deeper understanding of which yield tractable optimization and higher generation quality. USD3 is a hybrid of our simplified exact VLB and CE, and USD3* refers to its approximation with further simplifications (§2.2.5 and §3.3). USD3-CE and USD3-VLB are variants, resp. with CE or VLB loss *only*.

Baselines. We compare USD3 and variants to three latest SOTA discrete diffusion models in the literature: D3PM in Austin et al. [2021], RDM in Zhang et al. [2023] (discrete-time), τ -LDR-0 in Campbell et al. [2022], SDDM in Sun et al. [2023] and SEDD in Lou et al. [2024] (continuous time).

Training Details. Thanks to unification, we can evaluate both discrete- and continuous-time models with both cosine and constant noise schedulers at ease. In USD3, we combine VLB and CE with weight 0.001 for the latter, following prior literature. For USD3*, we combine VLB with CE weight 1.0. As architecture, we parameterize $f_t^\theta(\mathbf{x}_t)$ with a sequence transformer model. By varying its depths and widths, we study the impact of model size. The details are described in Appx. §A.18.4.

4.1 Music Generation

Lakh Pianoroll Datasets. We evaluate monophonic music generation on PIANO, the cleaned Lakh pianoroll dataset [Raffel 2016, Dong et al. 2017], containing 6,000 training and 973 evaluation (or test) sequences of 256 notes each. Here we perform conditional generation; given the first 32

详细的证明可以在附录 § A.15 中找到。请注意，这个损失只需要模型进行一次前向传递，因此效率极高。此外，这个损失与 Lou 等人 [2024], 中提出的 DWDSE 损失（虽然是在单元素情况下）一致，该损失源自不同的具体分数匹配方向。这表明 VLB 本质上与比例匹配损失相同，正如在连续状态设置中所展示的那样，分数匹配 Song 等人 [2021b] 被证明等同于 Ho 等人 [2020a] 中的 VLB 计算。

3.4 贡献 5：快速反向采样与离散扩散统一

Campbell 等人 [2022] 提出使用反向 CTMC 学习到的转移率 $\hat{r}_t^\theta$ 对目标分布 $p_{\text{data}}(\mathbf{x}_0)$ 进行反向采样。然而，与所有元素独立采样的正向 CTMC 不同，反向 CTMC 的所有元素处理过程是耦合的，并且无法从学习到的转移率推导出封闭形式的转移概率。从 CTMC 进行直接和精确采样使用 Gillespie [1977], 算法，但对于多元素对象是不可行的。Campbell 等人 [2022] 提出使用 tau-leaping [Gillespie,2001] 大约从时间 t 到 $t - \tau$ 采样所有转移，假设 $R_t^{1:D}$ 在时间段内保持固定。虽然这是可行的，但它引入了未知误差，需要校正过程。

我们提出避免使用反向 CTMC 转移率进行昂贵的采样。我们首先观察到估计的转移率 $\hat{r}_t^\theta$ 实际上是从估计的 $p_{0|t}^\theta(\mathbf{x}_0|\mathbf{x}_t)$ 推导的。因此使用 $\hat{r}_t^\theta$ 等同于间接使用 $p_{0|t}^\theta(\mathbf{x}_0|\mathbf{x}_t)$ 。我们意识到转移概率 $q_{s|t}(\mathbf{x}_s|\mathbf{x}_t)$ 可以直接从 $p_{0|t}^\theta(\mathbf{x}_0|\mathbf{x}_t)$ 计算出来，如等式 (10) 所示。借助等式 (10)，我们可以通过依次采样 $\mathbf{x}_{t_{n-1}|t_n}$ 或等价地采样 $\mathbf{x}_{0:T}$ 来采样 $\mathbf{x}_{t_{n-1}|t_n} \dots \mathbf{x}_{t_1|t_2} \mathbf{x}_{t_0|t_1}, \dots$ ，只要 $\{t_i\}_{i=0}^n$ 满足 $0=t_0 < t_1 < \dots < t_n=T$ 。虽然等式 (10) 是针对离散时间情况推导的，但由于离散时间和连续时间扩散的正向过程的统一 (§ 3.2)，它也直接适用于连续时间情况。因此，离散时间和连续时间扩散具有相同的统一反向生成过程。

★ 统一离散扩散。总而言之，通过一系列数学简化，我们证明了离散时间与连续时间离散扩散共享 (1) 相同的正向扩散过程；(2) 相同的学习 $p_{0|t}^\theta$ 的参数化方式；(3) 相同的逆向去噪过程。基于我们的重新表述，我们提出了USD3，一种新型统一简化离散去噪扩散模型。值得注意的是，USD3对离散时间与连续时间使用相同的源代码，仅在训练过程中的损失函数进行单一修改，并共享样本生成过程（分别对应附录 § A.16中的算法1和算法2）。

4 实验部分

离散扩散在模型训练方面存在局限，尚无明确的指导方针。许多先前工作使用固定的权重优化组合的VLB和交叉熵（CE）损失，而未进行更深入的研究。在本工作中，我们不仅从数学上改进了VLB损失，还通过实证探索了离散扩散的广泛训练方案——包括各种损失组合、离散时间与连续时间训练，以及不同的模型尺寸——以更深入地理解哪些方案能实现可处理的优化和更高的生成质量。USD3是我们简化的精确VLB和CE的混合体，USD3* 指其进一步简化的近似版本 (§ 2.2.5和 § 3.3)。USD3-CE和USD3-VLB分别是仅使用CE损失或VLB损失的变体。

基线。我们将 USD3 及其变体与文献中三个最新的离散扩散模型进行比较：Austin 等人的 D3PM、张等人（离散时间）的 RDM、Campbell 等人的 [2021], τ -LDR-0、Sun 等人的 SDDM 以及 Lou 等人（连续时间）的 SEDD。

训练细节。得益于统一性，我们可以轻松地使用余弦和常数噪声调度器评估离散时间和连续时间模型。在 USD3 中，我们将 VLB 和 CE 结合，后者权重为 0.001，遵循先前文献。对于 USD3*，我们将 VLB 与权重为 1.0 的 CE 结合。在架构方面，我们使用序列转换器模型参数化 $f_t^\theta(\mathbf{x}_t)$ 。通过改变其深度和宽度，我们研究了模型大小的影响。详细信息见附录 § A.18.4。

4.1 音乐生成

拉赫钢琴卷数据集。我们在PIANO上评估单音音乐生成，该数据集是拉赫清洗后的钢琴卷数据集 [Raffel 2016, Dong等人 [2017]]，包含6,000个训练序列和973个评估（或测试）序列，每个序列包含256个音符。这里我们进行条件生成；给定前32个音符，模型需要生成其余224个音符。有趣的是，我们发现一些（124个）评估序列与一个或多个训练序列共享相同的前32个音符。这种重复为我们提供了一个独特的机会来研究我们称之为“鹦鹉学舌”——即模型在生成过程中只是简单地重放训练数据的现象。为此，我们创建了PIANO-P，这是一个仅包含这124个评估序列及其与前32个音符匹配的训练序列的子集。详情见附录 § A.18.1。

notes, the models are required to generate the rest 224 notes. Interestingly, we find that some (124) evaluation sequences share the *same* first 32 notes as one or more training sequences. Such repetition gives us a unique opportunity to investigate what-we-call “parroting”—the phenomenon that models are simply re-playing the exact training data during generation. To that end we create PIANO-P, a subset consisting only of these 124 evaluation sequences paired with their first-32-note-matching training sequences. Details are in Appx. §A.18.1.

Metrics. We use a number of new metrics for PIANO: 2- and 3-gram (↓) Hellinger Distance (↓) and (2) Proportion of Outliers (↓), in addition to previously used 1-gram counterparts based only on *single* note appearances in Campbell et al. [2022]; (3) Diverse Edit Distance (↑), which accounts for the creativity/novelty across multiple generated samples. On PIANO-P, we also report (4) Train-to-Test Ratios (↑) for {1,2,3}-gram Hellinger as well as Proportion of Outliers, which compare the weighted distance of a generated sample to its matching training vs. test sequence that share the same first 32 notes. Such ratios quantify the extent of “parroting”; a smaller ratio indicates a higher (lower) degree of memorization (generalization). Calculation details of all metrics are given in Appx. §A.18.3.

Results. Table 1 shows that USD3 and its variants perform better than the baseline methods across metrics. An exception is D3PM’s Diverse Edit Distance, which trades-off high novelty with low generation quality. USD3-CE, while performing well w.r.t. 1-gram metrics, does not compete with USD3-VLB and USD3 w.r.t. {2,3}-gram metrics, which indicates that *CE only* loss captures the least sequential information. Models with combined losses, USD3 and USD3*, achieve higher Train-to-Test ratio than pure CE or VLB based ones, suggesting that the combination of two losses can alleviate overfitting, i.e. “parroting” the training data. We find continuous-time USD3 to perform better than discrete-time counterparts models because its ability of denoising through any timestep.

Table 1: Conditional music gen. quality (3 samples avg.) on PIANO w.r.t. n-gram Hellinger, n-gram Prop. of Out., and Diverse Edit Dist. Also, Train-to-Test Ratio for on PIANO-P quantifies “parroting”. **First & Second** shown in color (Detailed version is in Table 4).

	Method	n-gram Hellinger(↓)			n-gram Prop. of Out.(↓)			Div. Edit Dist. (↑)	3g-Prop. Ratio(↑)
		1gram	2gram	3gram	1gram	2gram	3gram		
Discrete-time	D3PM	0.398	0.530	0.591	0.120	0.253	0.379	0.295	2.221
	RDM	0.412	0.596	0.620	0.140	0.246	0.386	0.289	2.521
	USD3-CE	0.375	0.483	0.574	0.107	0.209	0.303	0.047	2.888
	USD3-VLB	0.379	0.464	0.542	0.117	0.184	0.273	0.082	2.863
	USD3	0.377	0.469	0.552	0.107	0.186	0.286	0.064	3.083
	USD3*	0.375	0.470	0.555	0.110	0.191	0.283	0.066	2.959
Continuous-time	SDDM	0.375	0.485	0.577	0.110	0.205	0.340	0.060	2.901
	τ -LDR-0	0.379	0.481	0.571	0.114	0.207	0.320	0.050	2.965
	SEDD	0.381	0.471	0.547	0.105	0.182	0.271	0.061	3.042
	USD3-CE	0.373	0.483	0.577	0.116	0.222	0.346	0.043	2.637
	USD3-VLB	0.376	0.462	0.549	0.120	0.186	0.295	0.061	2.904
	USD3	0.374	0.462	0.543	0.112	0.184	0.270	0.066	3.083
	USD3*	0.374	0.462	0.549	0.114	0.184	0.271	0.052	2.923

Table 2: Image generation quality w.r.t. IS and FID over 50,000 unconditional generated CIFAR10 samples. **First & Second** shown in color.

	Method	IS (↑)	FID (↓)
Discrete-time	D3PM	8.13	18.08
	RDM	7.16	24.23
	USD3-CE	9.02	12.64
	USD3-VLB	9.27	12.07
	USD3	8.85	13.25
	USD3*		
Cont.-time	SDDM	8.72	14.17
	τ -LDR	8.37	17.61
	SEDD	8.51	17.01
	USD3-CE	9.23	11.97
	USD3*	8.83	14.03
	USD3	8.97	13.09
	VQGAN Recons.	10.42	7.68

4.2 Image Generation

VQCIFAR10 Dataset. CIFAR10 contains 50,000 training images in continuous values, which we convert to vectors from 64×512 -dimensional quantization hash-code space with a pre-trained VQGAN Esser et al. [2020]. This conversion allows us to (1) evaluate our methods on *nominal* data by breaking the neighboring orders in the image space, and (2) select a closed-form stationary distribution m . Details on VQCIFAR10 dataset are given in Appx. §A.18.2.

Metrics. After feeding the generated samples through the VQGAN decoder to obtain representations from the discretized space, we measure the Inception Score (IS) (↑) and Frechet Inception Distance (FID) (↓) against the original CIFAR10 dataset. Note that our training set, which are discretized images from VQGAN, achieves IS=10.42 and FID=7.68, which are optimistic limits for generation.

Results. Table 2 shows that our proposed approaches outperform existing baselines. Discrete-time D3PM and RDM fall short for the harder image generation task, whereas our simplified discrete-time loss boosts quality significantly. In continuous-time, τ -LDR with a similar loss to USD3 falls short due to its complicated generation process, whereas SDDM is most competitive among the baselines, although requires substantial compute resources while being limited to a specialized model architecture. USD3-CE achieves competitive IS and FID scores, which validates our finding (recall Eq. (15)) that CE loss is essential for diffusion loss minimization.

有趣的是，我们发现一些（124个）评估序列与一个或多个训练序列共享相同的前32个音符。这种重复为我们提供了一个独特的机会来研究我们称之为“鹦鹉学舌”——即模型在生成过程中只是简单地重放训练数据的现象。为此，我们创建了PIANO-P，这是一个仅包含这124个评估序列及其与前32个音符匹配的训练序列的子集。详情见附录 § A.18.1。

指标。我们为PIANO使用了一系列新的指标：2-和3-gram (1) Hellinger距离 (↓) 和 (2) 异常值比例 (↓)，此外还有基于Campbell等人 [2022]中仅基于单个音符出现的1-gram对应指标；(3) 多样化编辑距离 (↑)，它考虑了跨多个生成样本的创造力/新颖性。在PIANO-P上，我们还报告了 (4) {1,2,3}-gram Hellinger以及异常值比例的训练集-测试集比率 (↑)，这些比率比较了生成样本与其匹配的训练集和测试序列的加权距离，这些序列共享前32个音符。这些比率量化了“鹦鹉学舌”的程度；比率越小，表示记忆化（泛化）程度越高（越低）。所有指标的计算细节见附录 § A.18.3。

结果。表1显示，USD3及其变体在各项指标上均优于基线方法。一个例外是D3PM的多样性编辑距离，它在新颖性与生成质量之间进行了权衡。USD3-CE虽然在一元指标上表现良好，但在 {2,3}-元指标上则不与USD3-VLB和USD3竞争，这表明CE仅损失捕获了最少的序列信息。结合损失的模型，USD3和USD3*，其训练集到测试集的比率高于纯CE或VLB基础的模型，表明两种损失的组合可以缓解过拟合，即“模仿”训练数据。我们发现连续时间的USD3比离散时间对应模型表现更好，因为其通过任何时间步的降噪能力更强。

表1: 条件音乐生成质量（3个样本平均）在PIANO上关于n元Hellinger距离、n元输出比例和多样性编辑距离。此外，PIANO-P的训练集到测试集比率量化了“模仿”现象。第一项和第二项以颜色显示（详细版本见表4）。

	方法	n-gram Hellinger(↓)			n-gram 输出比例(↓)			编辑距离分布 (↑)	3g-Prop. Ratio(↑)
		1gram	2gram	3gram	1gram	2gram	3gram		
Discrete-time	D3PM	0.398	0.530	0.591	0.120	0.253	0.379	0.295	2.221
	RDM	0.412	0.596	0.620	0.140	0.246	0.386	0.289	2.521
	USD3-CE	0.375	0.483	0.574	0.107	0.209	0.303	0.047	2.888
	USD3-VLB	0.379	0.464	0.542	0.117	0.184	0.273	0.082	2.863
	USD3	0.377	0.469	0.552	0.107	0.186	0.286	0.064	3.083
	USD3*	0.375	0.470	0.555	0.110	0.191	0.283	0.066	2.959
Continuous-time	SDDM	0.375	0.485	0.577	0.110	0.205	0.340	0.060	2.901
	τ -LDR-0	0.379	0.481	0.571	0.114	0.207	0.320	0.050	2.965
	SEDD	0.381	0.471	0.547	0.105	0.182	0.271	0.061	3.042
	USD3-CE	0.373	0.483	0.577	0.116	0.222	0.346	0.043	2.637
	USD3-VLB	0.376	0.462	0.549	0.120	0.186	0.295	0.061	2.904
	USD3	0.374	0.462	0.543	0.112	0.184	0.270	0.066	3.083
	USD3*	0.374	0.462	0.549	0.114	0.184	0.271	0.052	2.923

表2: 在50,000个无条件生成的CIFAR10样本上，IS和FID的图像生成质量。第一项和第二项以彩色显示。

	方法	IS (↑)	FID (↓)
Discrete-time	D3PM	8.13	18.08
	RDM	7.16	24.23
	USD3-CE	9.02	12.64
	USD3*	9.27	12.07
	USD3	8.85	13.25
	USD3*		
Cont.-time	SDDM	8.72	14.17
	τ -LDR	8.37	17.61
	SEDD	8.51	17.01
	USD3-CE	9.23	11.97
	USD3*	8.83	14.03
	USD3	8.97	13.09
	VQGAN 重构。	10.42	7.68

4.2 图像生成

VQCIFAR10数据集。CIFAR10包含50,000张连续值的训练图像，我们使用预训练的VQGAN Esser等人 [2020]将其从 64×512 维量化哈希码空间转换为向量。这种转换使我们能够 (1) 通过打破图像空间中的相邻顺序，在名义数据上评估我们的方法，以及 (2) 选择闭式平稳分布 m 。VQCIFAR10数据集的详细信息见附录 § A.18.2。

指标。在将生成的样本通过VQGAN解码器以获得离散化空间的表示后，我们针对原始CIFAR10数据集测量了Inception Score (IS) (↑)和Frechet Inception Distance (FID) (↓)。请注意，我们的训练集（来自VQGAN的离散化图像）实现了IS=10.42 和FID=7.68，这些是生成结果的乐观上限。

结果。表2显示我们提出的方法优于现有基线。离散时间D3PM和RDM在更难图像生成任务上表现不足，而我们的简化离散时间损失显著提升了质量。在连续时间中， τ -LDR由于生成过程复杂，尽管损失与USD3相似，但仍表现不足，而SDDM在基线模型中最为有竞争力，尽管它需要大量计算资源且限于专门的模型架构。USD3-CE实现了有竞争力的IS和FID分数，这验证了我们的发现（参考公式(15)）即CE损失对于扩散损失最小化至关重要。

Additional results. We provide memory and runtimes of models in Table 7 in Appx. §A.19.4, generated images in Fig.2 in Appx. §A.19.5, and ablations of MCMC sampling in Appx. §A.19.6.

5 Conclusion

We introduce two fundamental contributions for both discrete-time and continuous-time diffusion for categorical data. First, we presented extensive *mathematical simplifications for the loss functions*, including exact closed-form derivations as well as novel easy-to-optimize approximations. Second, we established a *mathematical unification* of discrete-time and continuous-time discrete diffusion, enabling mutual benefits. Our proposed approach USD3 for discrete diffusion achieved significant improvements on established datasets across a suite of generation quality metrics.

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附加结果。我们在附录 § A.19.4的表7中提供了模型内存和运行时间，在附录 § A.19.5的图2中提供了生成图像，以及在附录 § A.19.6中提供了MCMC采样的消融实验结果。

5 结论

我们为离散时间和连续时间的分类数据扩散提出了两项基本贡献。首先，我们为损失函数提供了广泛的数学简化，包括精确的闭式推导以及易于优化的近似方法。其次，我们建立了离散时间和连续时间离散扩散的数学统一，实现了相互促进。我们提出的离散扩散方法 USD3在多个生成质量指标上，于已建立的数据集上取得了显著改进。

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A Appendix

A.1 Variational Lower Bound Derivation

$$\begin{aligned}
\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} &\geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_{0:T}) - \log q(\mathbf{x}_{1:T}|\mathbf{x}_0)] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_T) + \sum_{t=1}^T \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_T) + \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \sum_{t=2}^T \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)}] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_T) + \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \sum_{t=2}^T \log \left(\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \right)] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_T) + \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{q(\mathbf{x}_T|\mathbf{x}_0)} + \sum_{t=2}^T \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1) + \log \frac{p_\theta(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} - \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}] \\
&= \underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]}_{-\mathcal{L}_1(\theta)} - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\mathcal{L}_t(\theta)} - \text{const.} \quad (28)
\end{aligned}$$

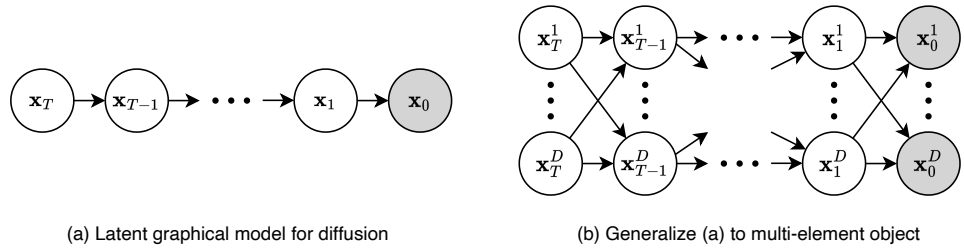


Figure 1: The graphical model diagram for both element-wise diffusion and multi-element diffusion.

A.2 Definition of nominal data

Nominal data, or categorical data, refers to data that can be categorized but not ranked or ordered. In mathematics, nominal data can be represented as a set of discrete categories or distinct labels without inherent ordering. As there is no ordering inside, operations of comparing categories is meaningless except checking equality. In the real world, most categories of data are nominal in nature, such as nationality, blood type, colors, among others.

A.3 Derivation of $q(x_{t-1}|x_t, x_0)$

First, let us define $\bar{Q}_{t|s} = Q_{s+1} \dots Q_t$. Note that $\bar{Q}_{t|0} = \bar{Q}_t$ and $\bar{Q}_{t|t-1} = Q_t$. Accordingly, we can derive the following two equalities.

$$q(\mathbf{x}_t|\mathbf{x}_s) = \text{Cat}(\mathbf{x}_t; \bar{Q}_{t|s}^\top \mathbf{x}_s) \quad (29)$$

$$\begin{aligned}
q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) &= \frac{q(\mathbf{x}_t|\mathbf{x}_s)q(\mathbf{x}_s|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} = \frac{\text{Cat}(\mathbf{x}_t; \bar{Q}_{t|s}^\top \mathbf{x}_s) \text{Cat}(\mathbf{x}_s; \bar{Q}_s^\top \mathbf{x}_0)}{\text{Cat}(\mathbf{x}_t; \bar{Q}_t^\top \mathbf{x}_0)} \\
&= \frac{\mathbf{x}_s^\top \bar{Q}_{t|s} \mathbf{x}_t \cdot \mathbf{x}_s^\top \bar{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \bar{Q}_t^\top \mathbf{x}_0} = \mathbf{x}_s^\top \frac{\bar{Q}_{t|s} \mathbf{x}_t \odot \bar{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \bar{Q}_t^\top \mathbf{x}_0} = \text{Cat}(\mathbf{x}_s; \frac{\bar{Q}_{t|s} \mathbf{x}_t \odot \bar{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \bar{Q}_t^\top \mathbf{x}_0}) \quad (30)
\end{aligned}$$

A 附录

A.1 变分下界推导

$$\begin{aligned}
\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} &\geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_{0:T}) - \log q(\mathbf{x}_{1:T}|\mathbf{x}_0)] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_T) + \sum_{t=1}^T \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_T) + \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \sum_{t=2}^T \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)}] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_T) + \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \sum_{t=2}^T \log \left(\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \right)] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_T) + \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{q(\mathbf{x}_T|\mathbf{x}_0)} + \sum_{t=2}^T \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}] \\
&= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1) + \log \frac{p_\theta(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} - \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}] \\
&= \underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]}_{-\mathcal{L}_1(\theta)} - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\mathcal{L}_t(\theta)} - \text{const.} \quad (28)
\end{aligned}$$

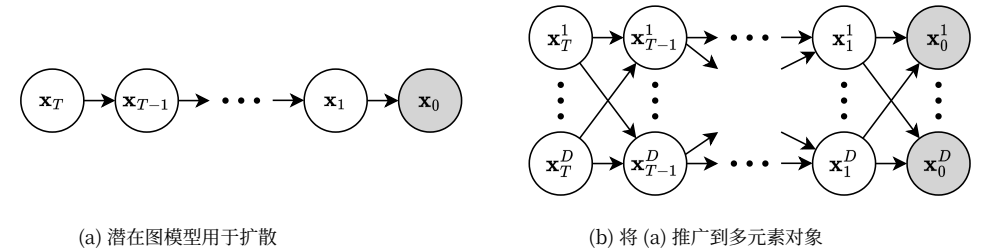


图1: 用于元素级扩散和多元素扩散的图模型图

A.2 名义数据的定义

名义数据，或称分类数据，是指可以分类但不能排序或排序的数据。在数学中，名义数据可以表示为一组离散类别或不同标签，而没有固有的顺序。由于内部没有顺序，比较类别的操作除了检查相等性之外是无意义的。在现实生活中，大多数数据类别都是名义性质的，例如国籍、血型、颜色等。

A.3 $q(x_{t-1}|x_t, x_0)$ 的推导

首先，让我们定义 $\bar{Q}_{t|s} = Q_{s+1} \dots Q_t$ 。请注意 $\bar{Q}_{t|0} = \bar{Q}_t$ 和 $\bar{Q}_{t|t-1} = Q_t$ 。因此，我们可以推导出以下两个等式。

$$q(\mathbf{x}_t|\mathbf{x}_s) = \text{Cat}(\mathbf{x}_t; \bar{Q}_{t|s}^\top \mathbf{x}_s) \quad (29)$$

$$\begin{aligned}
q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) &= \frac{q(\mathbf{x}_t|\mathbf{x}_s)q(\mathbf{x}_s|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} = \frac{\text{Cat}(\mathbf{x}_t; \bar{Q}_{t|s}^\top \mathbf{x}_s) \text{Cat}(\mathbf{x}_s; \bar{Q}_s^\top \mathbf{x}_0)}{\text{Cat}(\mathbf{x}_t; \bar{Q}_t^\top \mathbf{x}_0)} \\
&= \frac{\mathbf{x}_s^\top \bar{Q}_{t|s} \mathbf{x}_t \cdot \mathbf{x}_s^\top \bar{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \bar{Q}_t^\top \mathbf{x}_0} = \mathbf{x}_s^\top \frac{\bar{Q}_{t|s} \mathbf{x}_t \odot \bar{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \bar{Q}_t^\top \mathbf{x}_0} = \text{Cat}(\mathbf{x}_s; \frac{\bar{Q}_{t|s} \mathbf{x}_t \odot \bar{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \bar{Q}_t^\top \mathbf{x}_0}) \quad (30)
\end{aligned}$$

Using the formulation of Q_t in Eq. (7), $\bar{Q}_{t|s}$ can be written as

$$\bar{Q}_{t|s} = \bar{\alpha}_{t|s}I + (1 - \bar{\alpha}_{t|s})\mathbf{1}\mathbf{m}^\top \quad (31)$$

where $\bar{\alpha}_{t|s} = \prod_{i=s+1}^t \alpha_i$. Note that $\bar{\alpha}_t = \bar{\alpha}_{t|s}\bar{\alpha}_s$.

Next, we can simplify Eq. (5) using the above formulations as

$$\begin{aligned} \frac{\bar{Q}_{t|s}\mathbf{x}_t \odot \bar{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \bar{Q}_t^\top \mathbf{x}_0} &= \frac{(\bar{\alpha}_{t|s}\mathbf{x}_t + (1 - \bar{\alpha}_{t|s})\langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{1}) \odot (\bar{\alpha}_s \mathbf{x}_0 + (1 - \bar{\alpha}_s)\mathbf{m})}{\bar{\alpha}_t \langle \mathbf{x}_t, \mathbf{x}_0 \rangle + (1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle} \\ &= \frac{\bar{\alpha}_t \mathbf{x}_t \odot \mathbf{x}_0 + (\bar{\alpha}_s - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{x}_0 + (\bar{\alpha}_{t|s} - \bar{\alpha}_t) \mathbf{x}_t \odot \mathbf{m} + (1 - \bar{\alpha}_s)(1 - \bar{\alpha}_{t|s}) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{m}}{\bar{\alpha}_t \langle \mathbf{x}_t, \mathbf{x}_0 \rangle + (1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle} \end{aligned} \quad (32)$$

We can simplify the above equation further by considering two cases: (1) $\mathbf{x}_t = \mathbf{x}_0$ and (2) $\mathbf{x}_t \neq \mathbf{x}_0$.

(Case 1) When $\mathbf{x}_t = \mathbf{x}_0$, using the fact that both $\mathbf{x}_t$ and $\mathbf{x}_0$ are one-hot encoded, we observe

$$\begin{aligned} \text{Eq. (32)} &= \frac{\bar{\alpha}_t \mathbf{x}_t + (\bar{\alpha}_s - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{x}_t + (\bar{\alpha}_{t|s} - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{x}_t + (1 - \bar{\alpha}_s)(1 - \bar{\alpha}_{t|s}) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{m}}{\bar{\alpha}_t + (1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle} \\ &= \frac{\bar{\alpha}_t + (\bar{\alpha}_{t|s} + \bar{\alpha}_s - 2\bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle}{\bar{\alpha}_t + (1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle} \mathbf{x}_t + \frac{(1 - \bar{\alpha}_s)(1 - \bar{\alpha}_{t|s}) \langle \mathbf{m}, \mathbf{x}_t \rangle}{\bar{\alpha}_t + (1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle} \mathbf{m} \\ &= (1 - \lambda_{t|s}) \cdot \mathbf{x}_t + \lambda_{t|s} \cdot \mathbf{m}, \end{aligned} \quad (33)$$

where $\lambda_{t|s}$ is defined as

$$\lambda_{t|s} = \frac{(1 - \bar{\alpha}_s)(1 - \bar{\alpha}_{t|s}) \langle \mathbf{m}, \mathbf{x}_t \rangle}{\bar{\alpha}_t + (1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle}. \quad (34)$$

(Case 2) When $\mathbf{x}_t \neq \mathbf{x}_0$, we can similarly derive

$$\begin{aligned} \text{Eq. (32)} &= \frac{(\bar{\alpha}_s - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{x}_0 + (\bar{\alpha}_{t|s} - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{x}_t + (1 - \bar{\alpha}_s)(1 - \bar{\alpha}_{t|s}) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{m}}{(1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle} \\ &= \frac{\bar{\alpha}_s - \bar{\alpha}_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 + \frac{1 - \bar{\alpha}_s}{1 - \bar{\alpha}_t} \bar{\alpha}_{t|s} \mathbf{x}_t + \frac{1 - \bar{\alpha}_s}{1 - \bar{\alpha}_t} (1 - \bar{\alpha}_{t|s}) \mathbf{m} \\ &= (1 - \mu_{t|s}) \cdot \mathbf{x}_0 + \mu_{t|s} \bar{\alpha}_{t|s} \cdot \mathbf{x}_t + \mu_{t|s} (1 - \bar{\alpha}_{t|s}) \cdot \mathbf{m}, \end{aligned} \quad (35)$$

where $\mu_{t|s}$ is defined as

$$\mu_{t|s} = \frac{1 - \bar{\alpha}_s}{1 - \bar{\alpha}_t} \quad (36)$$

Combining the results from both cases together, we can write $q(\mathbf{x}_s | \mathbf{x}_t, \mathbf{x}_0)$ in the following form.

$$q(\mathbf{x}_s | \mathbf{x}_t, \mathbf{x}_0) = \begin{cases} \text{Cat}(\mathbf{x}_s; (1 - \lambda_{t|s}) \cdot \mathbf{x}_t + \lambda_{t|s} \cdot \mathbf{m}) & \text{when } \mathbf{x}_t = \mathbf{x}_0 \\ \text{Cat}(\mathbf{x}_s; (1 - \mu_{t|s}) \cdot \mathbf{x}_0 + \mu_{t|s} \bar{\alpha}_{t|s} \cdot \mathbf{x}_t + \mu_{t|s} (1 - \bar{\alpha}_{t|s}) \cdot \mathbf{m}) & \text{when } \mathbf{x}_t \neq \mathbf{x}_0 \end{cases} \quad (37)$$

A.4 Parameterization of $p_\theta(x_s | x_t)$

We first provide the formulation as follows.

$$p_\theta(\mathbf{x}_s | \mathbf{x}_t) = \sum_{\mathbf{x}_0} q(\mathbf{x}_s | \mathbf{x}_t, \mathbf{x}_0) p_\theta(\mathbf{x}_0 | \mathbf{x}_t) \quad (38)$$

Using $q(\mathbf{x}_s | \mathbf{x}_t, \mathbf{x}_0)$ in Eq. (9) and $p_\theta(\mathbf{x}_0 | \mathbf{x}_t) = \text{Cat}(\mathbf{x}_0; f_t^\theta(\mathbf{x}_t))$, we can expand it as

使用公式 (7) 中的 Q_t , $Q_{t|s}$ 可以表示为

$$\bar{Q}_{t|s} = \bar{\alpha}_{t|s}I + (1 - \bar{\alpha}_{t|s})\mathbf{1}\mathbf{m}^\top \quad (31)$$

在 $\alpha_{t|s} = \prod_{i=s+1}^t \alpha_i$ 处。请注意 $\alpha_t = \alpha_{t|s}\alpha_s$

接下来, 我们可以使用上述公式简化公式(5)为

$$\begin{aligned} \frac{\bar{Q}_{t|s}\mathbf{x}_t \odot \bar{Q}_s^\top \mathbf{x}_0}{\mathbf{x}_t^\top \bar{Q}_t^\top \mathbf{x}_0} &= \frac{(\bar{\alpha}_{t|s}\mathbf{x}_t + (1 - \bar{\alpha}_{t|s})\langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{1}) \odot (\bar{\alpha}_s \mathbf{x}_0 + (1 - \bar{\alpha}_s)\mathbf{m})}{\bar{\alpha}_t \langle \mathbf{x}_t, \mathbf{x}_0 \rangle + (1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle} \\ &= \frac{\bar{\alpha}_t \mathbf{x}_t \odot \mathbf{x}_0 + (\bar{\alpha}_s - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{x}_0 + (\bar{\alpha}_{t|s} - \bar{\alpha}_t) \mathbf{x}_t \odot \mathbf{m} + (1 - \bar{\alpha}_s)(1 - \bar{\alpha}_{t|s}) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{m}}{\bar{\alpha}_t \langle \mathbf{x}_t, \mathbf{x}_0 \rangle + (1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle} \end{aligned} \quad (32)$$

我们可以通过考虑两种情况进一步简化上述方程: (1) $\mathbf{x}_t = \mathbf{x}_0$ 和 (2) $\mathbf{x}_t \neq \mathbf{x}_0$. (情况1) 当

$\mathbf{x}_t = \mathbf{x}_0$ 时, 利用 $\mathbf{x}_t$ 和 $\mathbf{x}_0$ 都是独热编码这一事实, 我们观察到

$$\begin{aligned} \text{Eq. (32)} &= \frac{\bar{\alpha}_t \mathbf{x}_t + (\bar{\alpha}_s - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{x}_t + (\bar{\alpha}_{t|s} - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{x}_t + (1 - \bar{\alpha}_s)(1 - \bar{\alpha}_{t|s}) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{m}}{\bar{\alpha}_t + (1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle} \\ &= \frac{\bar{\alpha}_t + (\bar{\alpha}_{t|s} + \bar{\alpha}_s - 2\bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle}{\bar{\alpha}_t + (1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle} \mathbf{x}_t + \frac{(1 - \bar{\alpha}_s)(1 - \bar{\alpha}_{t|s}) \langle \mathbf{m}, \mathbf{x}_t \rangle}{\bar{\alpha}_t + (1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle} \mathbf{m} \\ &= (1 - \lambda_{t|s}) \cdot \mathbf{x}_t + \lambda_{t|s} \cdot \mathbf{m}, \end{aligned} \quad (33)$$

其中 $\lambda_{t|s}$ 定义为

$$\lambda_{t|s} = \frac{(1 - \bar{\alpha}_s)(1 - \bar{\alpha}_{t|s}) \langle \mathbf{m}, \mathbf{x}_t \rangle}{\bar{\alpha}_t + (1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle}. \quad (34)$$

(情况2) 当 $\mathbf{x}_t \neq \mathbf{x}_0$ 时, 我们可以类似地推导出

$$\begin{aligned} \text{Eq. (32)} &= \frac{(\bar{\alpha}_s - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{x}_0 + (\bar{\alpha}_{t|s} - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{x}_t + (1 - \bar{\alpha}_s)(1 - \bar{\alpha}_{t|s}) \langle \mathbf{m}, \mathbf{x}_t \rangle \mathbf{m}}{(1 - \bar{\alpha}_t) \langle \mathbf{m}, \mathbf{x}_t \rangle} \\ &= \frac{\bar{\alpha}_s - \bar{\alpha}_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 + \frac{1 - \bar{\alpha}_s}{1 - \bar{\alpha}_t} \bar{\alpha}_{t|s} \mathbf{x}_t + \frac{1 - \bar{\alpha}_s}{1 - \bar{\alpha}_t} (1 - \bar{\alpha}_{t|s}) \mathbf{m} \\ &= (1 - \mu_{t|s}) \cdot \mathbf{x}_0 + \mu_{t|s} \bar{\alpha}_{t|s} \cdot \mathbf{x}_t + \mu_{t|s} (1 - \bar{\alpha}_{t|s}) \cdot \mathbf{m}, \end{aligned} \quad (35)$$

其中 $\mu_{t|s}$ 定义为

$$\mu_{t|s} = \frac{1 - \bar{\alpha}_s}{1 - \bar{\alpha}_t} \quad (36)$$

将两种情况的结果结合起来, 我们可以写成以下形式。

$$q(\mathbf{x}_s | \mathbf{x}_t, \mathbf{x}_0) = \begin{cases} \text{Cat}(\mathbf{x}_s; (1 - \lambda_{t|s}) \cdot \mathbf{x}_t + \lambda_{t|s} \cdot \mathbf{m}) & \text{when } \mathbf{x}_t = \mathbf{x}_0 \\ \text{Cat}(\mathbf{x}_s; (1 - \mu_{t|s}) \cdot \mathbf{x}_0 + \mu_{t|s} \bar{\alpha}_{t|s} \cdot \mathbf{x}_t + \mu_{t|s} (1 - \bar{\alpha}_{t|s}) \cdot \mathbf{m}) & \text{when } \mathbf{x}_t \neq \mathbf{x}_0 \end{cases} \quad (37)$$

A.4 $p_\theta(x_s | x_t)$ 参数化

我们首先提供如下公式。

$$p_\theta(\mathbf{x}_s | \mathbf{x}_t) = \sum_{\mathbf{x}_0} q(\mathbf{x}_s | \mathbf{x}_t, \mathbf{x}_0) p_\theta(\mathbf{x}_0 | \mathbf{x}_t) \quad (38)$$

在公式(9)中使用 $q(\mathbf{x}_s | \mathbf{x}_t, \mathbf{x}_0)$ 和 $p_\theta(\mathbf{x}_0 | \mathbf{x}_t) = \text{Cat}(\mathbf{x}_0; f_t^\theta(\mathbf{x}_t))$, 我们可以将其展开为

$$\begin{aligned}
p_\theta(\mathbf{x}_s|\mathbf{x}_t) &= q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_t)p_\theta(\mathbf{x}_t|\mathbf{x}_t) + \sum_{\mathbf{x} \neq \mathbf{x}_t} q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x})p_\theta(\mathbf{x}|\mathbf{x}_t) \\
&= \mathbf{x}_s^\top \left((1 - \lambda_{t|s})\mathbf{x}_t + \lambda_{t|s}\mathbf{m} \right) \mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) \\
&\quad + \sum_{\mathbf{x} \neq \mathbf{x}_t} \mathbf{x}_s^\top \left((1 - \mu_{t|s})\mathbf{x} + \mu_{t|s}\bar{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1 - \bar{\alpha}_{t|s})\mathbf{m} \right) \mathbf{x}^\top f_t^\theta(\mathbf{x}_t) \\
&= \mathbf{x}_s^\top \left[((1 - \lambda_{t|s})\mathbf{x}_t + \lambda_{t|s}\mathbf{m})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) + (1 - \mu_{t|s}) \left(\sum_{\mathbf{x} \neq \mathbf{x}_t} \mathbf{x}\mathbf{x}^\top \right) f_t^\theta(\mathbf{x}_t) \right. \\
&\quad \left. + (\mu_{t|s}\bar{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1 - \bar{\alpha}_{t|s})\mathbf{m}) \left(\sum_{\mathbf{x} \neq \mathbf{x}_t} \mathbf{x} \right)^\top f_t^\theta(\mathbf{x}_t) \right] \\
&= \mathbf{x}_s^\top \left[((1 - \lambda_{t|s})\mathbf{x}_t + \lambda_{t|s}\mathbf{m})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) + (1 - \mu_{t|s})(I - \mathbf{x}_t\mathbf{x}_t^\top) f_t^\theta(\mathbf{x}_t) \right. \\
&\quad \left. + (\mu_{t|s}\bar{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1 - \bar{\alpha}_{t|s})\mathbf{m})(\mathbf{1} - \mathbf{x}_t)^\top f_t^\theta(\mathbf{x}_t) \right] \\
&= \mathbf{x}_s^\top \left[(1 - \lambda_{t|s})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)\mathbf{x}_t + \lambda_{t|s}\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)\mathbf{m} + (1 - \mu_{t|s})(f_t^\theta(\mathbf{x}_t) - \mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)\mathbf{x}_t) \right. \\
&\quad \left. + (\mu_{t|s}\bar{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1 - \bar{\alpha}_{t|s})\mathbf{m})(1 - \mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)) \right] \\
&= \mathbf{x}_s^\top \left[(1 - \mu_{t|s}) \cdot f_t^\theta(\mathbf{x}_t) + \left((\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) + \mu_{t|s}\bar{\alpha}_{t|s} \right) \cdot \mathbf{x}_t \right. \\
&\quad \left. + \left(-(\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) + \mu_{t|s}(1 - \bar{\alpha}_{t|s}) \right) \cdot \mathbf{m} \right] \\
&= \text{Cat}(\mathbf{x}_s; (1 - \mu_{t|s}) \cdot f_t^\theta(\mathbf{x}_t) + (\mu_{t|s}\bar{\alpha}_{t|s} + \gamma_{t|s}^\theta) \cdot \mathbf{x}_t + (\mu_{t|s}(1 - \bar{\alpha}_{t|s}) - \gamma_{t|s}^\theta) \cdot \mathbf{m}) \\
&\quad (39)
\end{aligned}$$

A.5 Proof of Proposition 3

As $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ has two different categorical distributions based on whether $\mathbf{x}_t$ is equal to $\mathbf{x}_0$, we prove the Proposition 3 based on two cases.

Case 1: $\mathbf{x}_t = \mathbf{x}_0$. In this case, let $h_t^\theta := f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0$, now replace f_t^θ with g_t^θ in Eq. (2.2) and Eq. (10), we get

$$\gamma_{t|s}^\theta = (\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s})\langle h_t^\theta + \mathbf{x}_0, \mathbf{x}_t \rangle = (\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s})\langle h_t^\theta, \mathbf{x}_t \rangle + (\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s}) \quad (40)$$

Taking the formulation into Eq. (10) and cancel out terms, we can simplify $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ as

$$\begin{aligned}
p_\theta(\mathbf{x}_s|\mathbf{x}_t) &= (1 - \mu_{t|s})h_t^\theta + (1 - \lambda_{t|s})\mathbf{x}_t + \lambda_{t|s}\mathbf{m} + (\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s})\langle h_t^\theta, \mathbf{x}_t \rangle(\mathbf{x}_t - \mathbf{m}) \\
&= q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) + (1 - \mu_{t|s}) \left[h_t^\theta + \frac{\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s}}{1 - \mu_{t|s}} \langle h_t^\theta, \mathbf{x}_t \rangle (\mathbf{x}_t - \mathbf{m}) \right] \\
&= q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) + (1 - \mu_{t|s}) \left[h_t^\theta + \phi_{t|s} \langle h_t^\theta, \mathbf{x}_t \rangle (\mathbf{x}_t - \mathbf{m}) \right] \quad (41)
\end{aligned}$$

Case 2: $\mathbf{x}_t \neq \mathbf{x}_0$. Based on Eq. (9) and Eq. (10), it's easy to find that

$$\begin{aligned}
p_\theta(\mathbf{x}_s|\mathbf{x}_t) &= (1 - \mu_{t|s})(f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0) + \gamma_{t|s}^\theta(\mathbf{x}_t - \mathbf{m}) + (1 - \mu_{t|s})\mathbf{x}_0 + \mu_{t|s}\bar{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1 - \bar{\alpha}_{t|s})\mathbf{m} \\
&= (1 - \mu_{t|s})(f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0) + \gamma_{t|s}^\theta(\mathbf{x}_t - \mathbf{m}) + q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) \\
&= q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) + (1 - \mu_{t|s}) \left[f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0 + \frac{\gamma_{t|s}^\theta}{(1 - \mu_{t|s})}(\mathbf{x}_t - \mathbf{m}) \right] \quad (42)
\end{aligned}$$

Taking the definition of $\gamma_{t|s}^\theta$ in Eq. (2.2) and using the fact $\langle \mathbf{x}_0, \mathbf{x}_t \rangle = 0$ we get the formulation in proposition 3.

For both cases, we proved that the formulation in proposition 3 is correct.

$$\begin{aligned}
p_\theta(\mathbf{x}_s|\mathbf{x}_t) &= q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_t)p_\theta(\mathbf{x}_t|\mathbf{x}_t) + \sum_{\mathbf{x} \neq \mathbf{x}_t} q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x})p_\theta(\mathbf{x}|\mathbf{x}_t) \\
&= \mathbf{x}_s^\top \left((1 - \lambda_{t|s})\mathbf{x}_t + \lambda_{t|s}\mathbf{m} \right) \mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) \\
&\quad + \sum_{\mathbf{x} \neq \mathbf{x}_t} \mathbf{x}_s^\top \left((1 - \mu_{t|s})\mathbf{x} + \mu_{t|s}\bar{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1 - \bar{\alpha}_{t|s})\mathbf{m} \right) \mathbf{x}^\top f_t^\theta(\mathbf{x}_t) \\
&= \mathbf{x}_s^\top \left[((1 - \lambda_{t|s})\mathbf{x}_t + \lambda_{t|s}\mathbf{m})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) + (1 - \mu_{t|s}) \left(\sum_{\mathbf{x} \neq \mathbf{x}_t} \mathbf{x}\mathbf{x}^\top \right) f_t^\theta(\mathbf{x}_t) \right. \\
&\quad \left. + (\mu_{t|s}\bar{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1 - \bar{\alpha}_{t|s})\mathbf{m}) \left(\sum_{\mathbf{x} \neq \mathbf{x}_t} \mathbf{x} \right)^\top f_t^\theta(\mathbf{x}_t) \right] \\
&= \mathbf{x}_s^\top \left[((1 - \lambda_{t|s})\mathbf{x}_t + \lambda_{t|s}\mathbf{m})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) + (1 - \mu_{t|s})(I - \mathbf{x}_t\mathbf{x}_t^\top) f_t^\theta(\mathbf{x}_t) \right. \\
&\quad \left. + (\mu_{t|s}\bar{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1 - \bar{\alpha}_{t|s})\mathbf{m})(\mathbf{1} - \mathbf{x}_t)^\top f_t^\theta(\mathbf{x}_t) \right] \\
&= \mathbf{x}_s^\top \left[(1 - \lambda_{t|s})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)\mathbf{x}_t + \lambda_{t|s}\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)\mathbf{m} + (1 - \mu_{t|s})(f_t^\theta(\mathbf{x}_t) - \mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)\mathbf{x}_t) \right. \\
&\quad \left. + (\mu_{t|s}\bar{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1 - \bar{\alpha}_{t|s})\mathbf{m})(1 - \mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t)) \right] \\
&= \mathbf{x}_s^\top \left[(1 - \mu_{t|s}) \cdot f_t^\theta(\mathbf{x}_t) + \left((\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) + \mu_{t|s}\bar{\alpha}_{t|s} \right) \cdot \mathbf{x}_t \right. \\
&\quad \left. + \left(-(\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s})\mathbf{x}_t^\top f_t^\theta(\mathbf{x}_t) + \mu_{t|s}(1 - \bar{\alpha}_{t|s}) \right) \cdot \mathbf{m} \right] \\
&= \text{Cat}(\mathbf{x}_s; (1 - \mu_{t|s}) \cdot f_t^\theta(\mathbf{x}_t) + (\mu_{t|s}\bar{\alpha}_{t|s} + \gamma_{t|s}^\theta) \cdot \mathbf{x}_t + (\mu_{t|s}(1 - \bar{\alpha}_{t|s}) - \gamma_{t|s}^\theta) \cdot \mathbf{m}) \\
&\quad (39)
\end{aligned}$$

A.5 命题 3 的证明

由于 $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ 根据 $\mathbf{x}_t$ 是否等于 $\mathbf{x}_0$, 有两种不同的分类分布, 我们基于两种情况证明命题3。

情况 1: $\mathbf{x}_t = \mathbf{x}_0$. 在这种情况下, 令 $h_t^\theta := f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0$, 现在将 f_t^θ 替换 Eq. (2.2) 和 Eq. (10) 中的 g_t^θ , 我们得到

$$\gamma_{t|s}^\theta = (\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s})\langle h_t^\theta + \mathbf{x}_0, \mathbf{x}_t \rangle = (\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s})\langle h_t^\theta, \mathbf{x}_t \rangle + (\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s}) \quad (40)$$

将公式代入 Eq. (10) 并消去项, 我们可以将 $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ 简化为

$$\begin{aligned}
p_\theta(\mathbf{x}_s|\mathbf{x}_t) &= (1 - \mu_{t|s})h_t^\theta + (1 - \lambda_{t|s})\mathbf{x}_t + \lambda_{t|s}\mathbf{m} + (\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s})\langle h_t^\theta, \mathbf{x}_t \rangle(\mathbf{x}_t - \mathbf{m}) \\
&= q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) + (1 - \mu_{t|s}) \left[h_t^\theta + \frac{\mu_{t|s} - \lambda_{t|s} - \mu_{t|s}\bar{\alpha}_{t|s}}{1 - \mu_{t|s}} \langle h_t^\theta, \mathbf{x}_t \rangle (\mathbf{x}_t - \mathbf{m}) \right] \\
&= q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) + (1 - \mu_{t|s}) \left[h_t^\theta + \phi_{t|s} \langle h_t^\theta, \mathbf{x}_t \rangle (\mathbf{x}_t - \mathbf{m}) \right] \quad (41)
\end{aligned}$$

情况 2: $\mathbf{x}_t \neq \mathbf{x}_0$. 根据 Eq. (9) 和 Eq. (10), 很容易发现

$$\begin{aligned}
p_\theta(\mathbf{x}_s|\mathbf{x}_t) &= (1 - \mu_{t|s})(f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0) + \gamma_{t|s}^\theta(\mathbf{x}_t - \mathbf{m}) + (1 - \mu_{t|s})\mathbf{x}_0 + \mu_{t|s}\bar{\alpha}_{t|s}\mathbf{x}_t + \mu_{t|s}(1 - \bar{\alpha}_{t|s})\mathbf{m} \\
&= (1 - \mu_{t|s})(f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0) + \gamma_{t|s}^\theta(\mathbf{x}_t - \mathbf{m}) + q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) \\
&= q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0) + (1 - \mu_{t|s}) \left[f_t^\theta(\mathbf{x}_t) - \mathbf{x}_0 + \frac{\gamma_{t|s}^\theta}{(1 - \mu_{t|s})}(\mathbf{x}_t - \mathbf{m}) \right] \quad (42)
\end{aligned}$$

将 Eq. (2.2) 中 $\gamma_{t|s}^\theta$ 的定义和事实 $\langle \mathbf{x}_0, \mathbf{x}_t \rangle = 0$ 结合, 我们得到命题 3 中的公式。

对于这两种情况, 我们都证明了命题3中的表述是正确的。

A.6 Proof of KL divergence approximation

Assume that $p(\mathbf{x}) = q(\mathbf{x}) + \Delta p(\mathbf{x})$, and both $p(\mathbf{x})$ and $q(\mathbf{x})$ are valid categorical distributions. Then

$$D_{\text{KL}}(q(\mathbf{x})\|p(\mathbf{x})) = -\sum_{\mathbf{x}} q(\mathbf{x}) \log \frac{p(\mathbf{x})}{q(\mathbf{x})} = -\sum_{\mathbf{x}} q(\mathbf{x}) \log(1 + \frac{\Delta p(\mathbf{x})}{q(\mathbf{x})}) \quad (43)$$

By Taylor expansion, $\log(1+x) \approx x - \frac{x^2}{2}$, we ignore larger than 2 order terms. Then apply this we get

$$\begin{aligned} D_{\text{KL}}(q(\mathbf{x})\|p(\mathbf{x})) &= -\sum_{\mathbf{x}} q(\mathbf{x}) [\frac{\Delta p(\mathbf{x})}{q(\mathbf{x})} - (\frac{\Delta p(\mathbf{x})}{q(\mathbf{x})})^2] \\ &= -\sum_{\mathbf{x}} \Delta p(\mathbf{x}) + \sum_{\mathbf{x}} \frac{\Delta p(\mathbf{x})^2}{q(\mathbf{x})} \\ &= \sum_{\mathbf{x}} \frac{\Delta p(\mathbf{x})^2}{q(\mathbf{x})} \end{aligned} \quad (44)$$

Change probabilities q and p to $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ and $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ we get the targeted formulation.

A.7 VLB with partial time steps

Assume that we only observe $\mathbf{x}_t$, $\mathbf{x}_s$, and $\mathbf{x}_0$, and assume that we have learned the prior $p_\theta(\mathbf{x}_t)$.

$$\begin{aligned} \log p_\theta(\mathbf{x}_0) &= \log \mathbb{E}_{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} \frac{p_\theta(\mathbf{x}_t, \mathbf{x}_s, \mathbf{x}_0)}{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} \geq \mathbb{E}_{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} \log \frac{p_\theta(\mathbf{x}_t, \mathbf{x}_s, \mathbf{x}_0)}{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} \\ &= \mathbb{E}_{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} [\log \frac{p_\theta(\mathbf{x}_t)p_\theta(\mathbf{x}_s|\mathbf{x}_t)p_\theta(\mathbf{x}_0|\mathbf{x}_s)}{q(\mathbf{x}_t|\mathbf{x}_0)q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)}] \\ &= \mathbb{E}_{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} \left[\log \frac{p_\theta(\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_0)} + \log p_\theta(\mathbf{x}_t) + \log \frac{p_\theta(\mathbf{x}_s|\mathbf{x}_t)}{q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)} \right] \\ &= -D_{\text{KL}}[q(\mathbf{x}_t|\mathbf{x}_0)\|p_\theta(\mathbf{x}_t)] + \mathbb{E}_{q(\mathbf{x}_s|\mathbf{x}_0)} \log p_\theta(\mathbf{x}_0|\mathbf{x}_s) - D_{\text{KL}}[q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)\|p_\theta(\mathbf{x}_s|\mathbf{x}_t)] \end{aligned} \quad (45)$$

A.8 Reparameterization Form for Sampling

In practice, we need to sample $\mathbf{x}_{t|0} \sim q(\mathbf{x}_t|\mathbf{x}_0)$ for training and $\mathbf{x}_{s|t} \sim p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ for generation (backward denoising). In what follows, we provide the reparameterization form for these to facilitate fast sampling. Given $q(\mathbf{x}_t|\mathbf{x}_0) = \text{Cat}(\mathbf{x}_t; \bar{\alpha}_t \mathbf{x}_0 + (1 - \bar{\alpha}_t) \mathbf{m})$ and $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ as in Eq. (10), we can rewrite the corresponding variables as

$$\mathbf{x}_{t|0} = \delta_{1, \mathbf{b}_t} \mathbf{x}_0 + (1 - \delta_{1, \mathbf{b}_t}) \mathbf{m}_0, \text{ where } \mathbf{b}_t \sim \text{Bernoulli}(\bar{\alpha}_t) \quad (46)$$

$$\mathbf{x}_{s|t} = \delta_{1, \mathbf{b}_{s|t}} \tilde{\mathbf{x}}_{0|t} + \delta_{2, \mathbf{b}_{s|t}} \mathbf{x}_t + \delta_{3, \mathbf{b}_{s|t}} \mathbf{m}_0, \quad (47)$$

where $\tilde{\mathbf{x}}_{0|t} \sim \text{Cat}(f_t^\theta(\mathbf{x}_t))$ and $\mathbf{b}_{s|t} \sim \text{Cat}(\cdot; [1 - \mu_{t|s}, \mu_{t|s} \bar{\alpha}_{t|s} + \gamma_{t|s}^\theta, \mu_{t|s}(1 - \bar{\alpha}_{t|s}) - \gamma_{t|s}^\theta])$.

Eq. (46) and Eq. (47) essentially show that the sampling process can be divided into two steps: first, sample the branch indicator $\mathbf{b}_t$ (or $\mathbf{b}_{s|t}$), and then sample from the categorical distribution of that branch, i.e. $\mathbf{x}_t$, $\mathbf{m}_0$, or $\tilde{\mathbf{x}}_{0|t}$. Moreover, the three terms in Eq. (47) highlight that the denoising step of generating $\mathbf{x}_s$ from $\mathbf{x}_t$ essentially draws samples via three levers: (1) use the predicted sample $\mathbf{x}_0$ from the trained network $f_t^\theta(\mathbf{x}_t)$ directly, (2) keep it unchanged as $\mathbf{x}_t$, or (3) roll it back to noise $\mathbf{m}_0$, offering an intuitive understanding.

A.9 Discrete-Time Multi-element Object Extension

We first extend $q(\mathbf{x}_t|\mathbf{x}_0)$, $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$, and $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ to multi-element object, and then present the negative VLB loss extension. First, for the forward process conditional on $\mathbf{x}_0$, each element $\mathbf{x}_t^d$ of the multi-element object $\mathbf{x}_t^{1:D}$ has its own diffusion process without interactions with others. Formally,

$$q(\mathbf{x}_t^{1:D}|\mathbf{x}_0^{1:D}) = \prod_{d=1}^D q(\mathbf{x}_t^d|\mathbf{x}_0^d) \quad , \text{ and } \quad q(\mathbf{x}_s^{1:D}|\mathbf{x}_t^{1:D}, \mathbf{x}_0^{1:D}) = \prod_{d=1}^D q(\mathbf{x}_s^d|\mathbf{x}_t^d, \mathbf{x}_0^d) . \quad (48)$$

A.6 KL散度近似证明

假设 $p(\mathbf{x}) = q(\mathbf{x}) + \Delta p(\mathbf{x})$, 并且 $p(\mathbf{x})$ 和 $q(\mathbf{x})$ 都是有效的分类分布。那么

$$D_{\text{KL}}(q(\mathbf{x})\|p(\mathbf{x})) = -\sum_{\mathbf{x}} q(\mathbf{x}) \log \frac{p(\mathbf{x})}{q(\mathbf{x})} = -\sum_{\mathbf{x}} q(\mathbf{x}) \log(1 + \frac{\Delta p(\mathbf{x})}{q(\mathbf{x})}) \quad (43)$$

通过泰勒展开, $\log(1+x) \approx x - \frac{x^2}{2}$, 我们忽略高于2阶的项。然后应用这个方法我们得到

$$\begin{aligned} D_{\text{KL}}(q(\mathbf{x})\|p(\mathbf{x})) &= -\sum_{\mathbf{x}} q(\mathbf{x}) [\frac{\Delta p(\mathbf{x})}{q(\mathbf{x})} - (\frac{\Delta p(\mathbf{x})}{q(\mathbf{x})})^2] \\ &= -\sum_{\mathbf{x}} \Delta p(\mathbf{x}) + \sum_{\mathbf{x}} \frac{\Delta p(\mathbf{x})^2}{q(\mathbf{x})} \\ &= \sum_{\mathbf{x}} \frac{\Delta p(\mathbf{x})^2}{q(\mathbf{x})} \end{aligned} \quad (44)$$

将概率 q 和 p 转换为 $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$ 和 $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$, 我们得到目标公式。

A.7 带部分时间步长的VLB

假设我们只观察到 $\mathbf{x}_t$, $\mathbf{x}_s$ 和 $\mathbf{x}_0$, 并假设我们已经学习了先验 $p_\theta(\mathbf{x}_t)$ 。

$$\begin{aligned} \log p_\theta(\mathbf{x}_0) &= \log \mathbb{E}_{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} \frac{p_\theta(\mathbf{x}_t, \mathbf{x}_s, \mathbf{x}_0)}{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} \geq \mathbb{E}_{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} \log \frac{p_\theta(\mathbf{x}_t, \mathbf{x}_s, \mathbf{x}_0)}{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} \\ &= \mathbb{E}_{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} [\log \frac{p_\theta(\mathbf{x}_t)p_\theta(\mathbf{x}_s|\mathbf{x}_t)p_\theta(\mathbf{x}_0|\mathbf{x}_s)}{q(\mathbf{x}_t|\mathbf{x}_0)q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)}] \\ &= \mathbb{E}_{q(\mathbf{x}_t, \mathbf{x}_s|\mathbf{x}_0)} \left[\log \frac{p_\theta(\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_0)} + \log p_\theta(\mathbf{x}_t) + \log \frac{p_\theta(\mathbf{x}_s|\mathbf{x}_t)}{q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)} \right] \\ &= -D_{\text{KL}}[q(\mathbf{x}_t|\mathbf{x}_0)\|p_\theta(\mathbf{x}_t)] + \mathbb{E}_{q(\mathbf{x}_s|\mathbf{x}_0)} \log p_\theta(\mathbf{x}_0|\mathbf{x}_s) - D_{\text{KL}}[q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)\|p_\theta(\mathbf{x}_s|\mathbf{x}_t)] \end{aligned} \quad (45)$$

A.8 采样重参数化公式

在实际应用中, 我们需要采样 $\mathbf{x}_{t|0} \sim q(\mathbf{x}_t|\mathbf{x}_0)$ 用于训练和 $\mathbf{x}_{s|t} \sim p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ 用于生成 (反向去噪)。在以下内容中, 我们提供了这些的重新参数化形式, 以促进快速采样。给定与公式 (10) 中相同的 $q(\mathbf{x}_t|\mathbf{x}_0) = \text{Cat}(\mathbf{x}_t; \alpha_t \mathbf{x}_0 + (1 - \alpha_t) \mathbf{m})$ 和 $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$, 我们可以将相应的变量重写为

$$\mathbf{x}_{t|0} = \delta_{1, \mathbf{b}_t} \mathbf{x}_0 + (1 - \delta_{1, \mathbf{b}_t}) \mathbf{m}_0, \text{ 其中 } \mathbf{b}_t \sim \text{Bernoulli}(\alpha_t) \quad (46)$$

$$\mathbf{x}_{s|t} = \delta_{1, \mathbf{b}_{s|t}} \tilde{\mathbf{x}}_{0|t} + \delta_{2, \mathbf{b}_{s|t}} \mathbf{x}_t + \delta_{3, \mathbf{b}_{s|t}} \mathbf{m}_0, \quad (47)$$

其中 $\tilde{\mathbf{x}}_{0|t} \sim \text{Cat}(f_t^\theta(\mathbf{x}_t))$ 和 $\mathbf{b}_{s|t} \sim \text{Cat}(\cdot; [1 - \mu_{t|s}, \mu_{t|s} \alpha_{t|s} + \gamma_{t|s}^\theta, \mu_{t|s}(1 - \alpha_{t|s}) - \gamma_{t|s}^\theta])$ 。

式(46)和式(47)本质上表明采样过程可以分为两个步骤: 首先, 采样分支指示符 $\mathbf{b}_t$ (或 $\mathbf{b}_{s|t}$), 然后从该分支的离散分布中采样, 即 $\mathbf{x}_t$, $\mathbf{m}_0$, 或 $\tilde{\mathbf{x}}_{0|t}$ 。此外, 式(47)中的三项突出了从 $\mathbf{x}_t$ 生成 $\mathbf{x}_s$ 的去噪步骤本质上通过三个杠杆进行采样: (1) 直接使用训练好的网络 $f_t^\theta(\mathbf{x}_t)$ 预测的样本 $\mathbf{x}_0$, (2) 将其保持为 $\mathbf{x}_t$ 不变, 或(3) 将其回滚为噪声 $\mathbf{m}_0$, 这提供了直观的理解。

A.9 离散时间多元素对象扩展

我们首先将 $q(\mathbf{x}_t|\mathbf{x}_0)$, $q(\mathbf{x}_s|\mathbf{x}_t, \mathbf{x}_0)$, $p_\theta(\mathbf{x}_s|\mathbf{x}_t)$ 扩展到多元素对象, 然后提出负 VLB 损失的扩展。首先, 对于正向过程条件于 $\mathbf{x}_0$, 多元素对象 $\mathbf{x}_t^{1:D}$ 的每个元素 $\mathbf{x}_t^d$ 都有其自身的扩散过程, 而与其他元素无交互。形式上,

$$q(\mathbf{x}_t^{1:D}|\mathbf{x}_0^{1:D}) = \prod_{d=1}^D q(\mathbf{x}_t^d|\mathbf{x}_0^d) \quad , \text{ and } \quad q(\mathbf{x}_s^{1:D}|\mathbf{x}_t^{1:D}, \mathbf{x}_0^{1:D}) = \prod_{d=1}^D q(\mathbf{x}_s^d|\mathbf{x}_t^d, \mathbf{x}_0^d) . \quad (48)$$

The corresponding forward reparameterization form for generating $\mathbf{x}_{t|0}^{1:D}$ can be updated as

$$\mathbf{x}_{t|0}^{1:D} = \delta_{1,\mathbf{b}_t^{1:D}}^{1:D} \odot \mathbf{x}_0^{1:D} + (1 - \delta_{1,\mathbf{b}_t^{1:D}}^{1:D}) \odot \mathbf{m}_0^{1:D}, \quad (49)$$

where $\delta_{1,\mathbf{b}_t^{1:D}}^{1:D}$ is a D -dimensional vector with d -th element being $\delta_{1,\mathbf{b}_t^d}$ and $\mathbf{b}_t^d \sim \text{Bernoulli}(\bar{\alpha}_t)$.

For the backward process, all elements' transition processes are coupled together, as shown in the graphical model in Appx. Fig. 1(b). We start with the parameterization $p_\theta(\mathbf{x}_0^{1:D}|\mathbf{x}_t^{1:D})$, with the form

$$p_\theta(\mathbf{x}_0^{1:D}|\mathbf{x}_t^{1:D}) = \prod_{d=1}^D p_\theta(\mathbf{x}_0^d|\mathbf{x}_t^{1:D}) \quad \text{with} \quad p_\theta(\mathbf{x}_0^d|\mathbf{x}_t^{1:D}) = \text{Cat}(\mathbf{x}_0^d|f_t^{\theta,d}(\mathbf{x}_t^{1:D})) \quad (50)$$

Then, $p_\theta(\mathbf{x}_s^{1:D}|\mathbf{x}_t^{1:D}) = \prod_{d=1}^D p_\theta(\mathbf{x}_s^d|\mathbf{x}_t^{1:D})$, with the component $p_\theta(\mathbf{x}_s^d|\mathbf{x}_t^{1:D})$ has the form

$$\text{Cat}\left(\mathbf{x}_s^d; (1 - \mu_{t|s})f_t^{\theta,d}(\mathbf{x}_t^{1:D}) + (\mu_{t|s}\bar{\alpha}_{t|s} + \gamma_{t|s}^{\theta,d})\mathbf{x}_t^d + (\mu_{t|s}(1 - \bar{\alpha}_{t|s}) - \gamma_{t|s}^{\theta,d})\mathbf{m}^d\right), \quad (51)$$

$$\text{where } \gamma_{t|s}^{\theta,d} = (\mu_{t|s} - \lambda_{t|s}^d - \mu_{t|s}\bar{\alpha}_{t|s})(f_t^{\theta,d}(\mathbf{x}_t^{1:D}), \mathbf{x}_t^d), \quad \lambda_{t|s}^d = \frac{(1 - \bar{\alpha}_s)(1 - \bar{\alpha}_{t|s})\langle \mathbf{m}^d, \mathbf{x}_t^d \rangle}{\bar{\alpha}_t + (1 - \bar{\alpha}_t)\langle \mathbf{m}^d, \mathbf{x}_t^d \rangle}.$$

Similarly, the reparameterization form can be expressed as

$$\mathbf{x}_{s|t}^{1:D} = \delta_{1,\mathbf{b}_{s|t}^{1:D}}^{1:D} \odot \tilde{\mathbf{x}}_{0|t}^{1:D} + \delta_{2,\mathbf{b}_{s|t}^{1:D}}^{1:D} \odot \mathbf{x}_t^{1:D} + \delta_{3,\mathbf{b}_{s|t}^{1:D}}^{1:D} \odot \mathbf{m}_0^{1:D}, \quad (52)$$

where $\tilde{\mathbf{x}}_{0|t}^d \sim \text{Cat}(f_t^{\theta,d}(\mathbf{x}_t^{1:D}))$ and $\mathbf{b}_{s|t}^d \sim \text{Cat}(1 - \mu_{t|s}, \mu_{t|s}\bar{\alpha}_{t|s} + \gamma_{t|s}^{\theta,d}, \mu_{t|s}(1 - \bar{\alpha}_{t|s}) - \gamma_{t|s}^{\theta,d})$.

Finally, the following reformulate the loss functions $\mathcal{L}_t(\theta)$ for multi-element objects, where we drop the constant term for simplicity.

$$\begin{aligned} \mathcal{L}_t^{1:D}(\theta) = & \mathbb{E}_{q(\mathbf{x}_t^{1:D}|\mathbf{x}_0^{1:D})} \sum_{d=1}^D \left[\delta_{\mathbf{x}_t^d, \mathbf{x}_0^d} \mathbb{E}_{q(\mathbf{x}_{t-1}^d|\mathbf{x}_t^d=\mathbf{x}_0^d)} [-\log p_\theta(\mathbf{x}_{t-1}^d|\mathbf{x}_t^{1:D})] + \right. \\ & \left. (1 - \delta_{\mathbf{x}_t^d, \mathbf{x}_0^d}) \mathbb{E}_{q(\mathbf{x}_{t-1}^d|\mathbf{x}_t^d \neq \mathbf{x}_0^d)} [-\log p_\theta(\mathbf{x}_{t-1}^d|\mathbf{x}_t^{1:D})] \right] \end{aligned} \quad (53)$$

A.10 CTMC introduction

The CTMC process can go with either increasing t or decreasing t , and we use increasing t as the default direction to introduce CTMC. For any CTMC, its *transition rate matrix* R_t with $[R_t]_{ij} = r_t(\mathbf{e}_j|\mathbf{e}_i)$ fully determines the underlying stochastic process. CTMC is categorized into time-homogeneous and time-inhomogeneous based on whether R_t is static with respect to t . In this paper we work with time-inhomogeneous CTMC.

Based on the definition in Eq. (17), we can derive some properties as follows

$$\forall \mathbf{x}, r_t(\mathbf{x}|\mathbf{x}) \leq 0; \quad \forall \mathbf{y} \neq \mathbf{x}, r_t(\mathbf{y}|\mathbf{x}) \geq 0; \quad \forall \mathbf{x}, \sum_{\mathbf{y}} r_t(\mathbf{y}|\mathbf{x}) = 0 \quad (54)$$

$$q_{t|t-\Delta t}(\mathbf{y}|\mathbf{x}) = \delta_{\mathbf{x},\mathbf{y}} + r_t(\mathbf{y}|\mathbf{x})\Delta t + o(\Delta t) \quad (55)$$

Eq. (54) shows the properties of transition rate, which imply $r_t(\mathbf{x}|\mathbf{x}) = -\sum_{\mathbf{y} \neq \mathbf{x}} r_t(\mathbf{y}|\mathbf{x})$. Eq. (55) characterizes the infinitesimal change of transition probability, and can be used to derive the relationship between transition matrix $\bar{Q}_{t|s}$ and transition rate matrix R_t . Formally, a CTMC's transition probabilities satisfies the Kolmogorov forward and backward equations [Kolmogoroff, 1931]:

$$\frac{d}{dt} q_{t|s}(\mathbf{y}|\mathbf{x}) = \sum_{\mathbf{z}} q_{t|s}(\mathbf{z}|\mathbf{x}) r_t(\mathbf{y}|\mathbf{z}) \quad \text{or} \quad \frac{d}{dt} \bar{Q}_{t|s} = \bar{Q}_{t|s} R_t \quad \text{Kolmogorov Forward} \quad (56)$$

$$\frac{d}{ds} q_{t|s}(\mathbf{y}|\mathbf{x}) = -\sum_{\mathbf{z}} r_s(\mathbf{z}|\mathbf{x}) q_{t|s}(\mathbf{y}|\mathbf{z}) \quad \text{or} \quad \frac{d}{ds} \bar{Q}_{t|s} = -R_s \bar{Q}_{t|s} \quad \text{Kolmogorov Backward} \quad (57)$$

The above equations are Ordinary Different Equations (ODEs) and have unique solutions. Rindos et al. [1995] mentioned that when R_{t_1} and R_{t_2} commute (i.e. $R_{t_1}R_{t_2} = R_{t_2}R_{t_1}$) for any t_1, t_2 , the transition probability matrix can be written as

$$\bar{Q}_{t|s} = \exp\left(\int_s^t R_a da\right) \quad (58)$$

生成 $\mathbf{x}_{t|0}^{1:D}$ 的相应正向重参数化形式可以更新为

$$\mathbf{x}_{t|0}^{1:D} = \delta_{1,\mathbf{b}_t^{1:D}}^{1:D} \odot \mathbf{x}_0^{1:D} + (1 - \delta_{1,\mathbf{b}_t^{1:D}}^{1:D}) \odot \mathbf{m}_0^{1:D}, \quad (49)$$

$\delta_{1,\mathbf{b}_t^{1:D}}^{1:D}$ 在哪里, 是一个 D -维向量, 其第 d 个元素为 $\delta_{1,\mathbf{b}_t^d}$ 和 $\mathbf{b}_t^d \sim \text{服从伯努利}(\alpha_t)$ 分布。

对于反向过程, 所有元素的转换过程是耦合在一起的, 如图Appx.图1(b)所示。我们从参数化 $p_\theta(\mathbf{x}_0^{1:D}|\mathbf{x}_t^{1:D})$ 开始, 其形式为

$$p_\theta(\mathbf{x}_0^{1:D}|\mathbf{x}_t^{1:D}) = \prod_{d=1}^D p_\theta(\mathbf{x}_0^d|\mathbf{x}_t^{1:D}) \quad \text{with} \quad p_\theta(\mathbf{x}_0^d|\mathbf{x}_t^{1:D}) = \text{Cat}(\mathbf{x}_0^d|f_t^{\theta,d}(\mathbf{x}_t^{1:D})) \quad (50)$$

然后, 具有分量 $p_\theta(\mathbf{x}_s^d|\mathbf{x}_t^{1:D})$ 的 $p_\theta(\mathbf{x}_s^{1:D}|\mathbf{x}_t^{1:D}) = \prod_{d=1}^D p_\theta(\mathbf{x}_s^d|\mathbf{x}_t^{1:D})$ 的形式为

$$\text{Cat}\left(\mathbf{x}_s^d; (1 - \mu_{t|s})f_t^{\theta,d}(\mathbf{x}_t^{1:D}) + (\mu_{t|s}\bar{\alpha}_{t|s} + \gamma_{t|s}^{\theta,d})\mathbf{x}_t^d + (\mu_{t|s}(1 - \bar{\alpha}_{t|s}) - \gamma_{t|s}^{\theta,d})\mathbf{m}^d\right), \quad (51)$$

$$\text{where } \gamma_{t|s}^{\theta,d} = (\mu_{t|s} - \lambda_{t|s}^d - \mu_{t|s}\bar{\alpha}_{t|s})(f_t^{\theta,d}(\mathbf{x}_t^{1:D}), \mathbf{x}_t^d), \quad \lambda_{t|s}^d = \frac{(1 - \bar{\alpha}_s)(1 - \bar{\alpha}_{t|s})\langle \mathbf{m}^d, \mathbf{x}_t^d \rangle}{\bar{\alpha}_t + (1 - \bar{\alpha}_t)\langle \mathbf{m}^d, \mathbf{x}_t^d \rangle}.$$

类似地, 重新参数化的形式可以表示为

$$\mathbf{x}_{s|t}^{1:D} = \delta_{1,\mathbf{b}_{s|t}^{1:D}}^{1:D} \odot \tilde{\mathbf{x}}_{0|t}^{1:D} + \delta_{2,\mathbf{b}_{s|t}^{1:D}}^{1:D} \odot \mathbf{x}_t^{1:D} + \delta_{3,\mathbf{b}_{s|t}^{1:D}}^{1:D} \odot \mathbf{m}_0^{1:D}, \quad (52)$$

其中 $\tilde{\mathbf{x}}_{0|t}^d \sim \text{Cat}(f_t^{\theta,d}(\mathbf{x}_t^{1:D}))$ and $\mathbf{b}_{s|t}^d \sim \text{Cat}(1 - \mu_{t|s}, \mu_{t|s}\bar{\alpha}_{t|s} + \gamma_{t|s}^{\theta,d}, \mu_{t|s}(1 - \bar{\alpha}_{t|s}) - \gamma_{t|s}^{\theta,d})$.

最后, 以下重新表述了用于多元素对象的损失函数 $\mathcal{L}_t(\theta)$, 为简化起见, 我们省略了常数项。

$$\begin{aligned} \mathcal{L}_t^{1:D}(\theta) = & \mathbb{E}_{q(\mathbf{x}_t^{1:D}|\mathbf{x}_0^{1:D})} \sum_{d=1}^D \left[\delta_{\mathbf{x}_t^d, \mathbf{x}_0^d} \mathbb{E}_{q(\mathbf{x}_{t-1}^d|\mathbf{x}_t^d=\mathbf{x}_0^d)} [-\log p_\theta(\mathbf{x}_{t-1}^d|\mathbf{x}_t^{1:D})] + \right. \\ & \left. (1 - \delta_{\mathbf{x}_t^d, \mathbf{x}_0^d}) \mathbb{E}_{q(\mathbf{x}_{t-1}^d|\mathbf{x}_t^d \neq \mathbf{x}_0^d)} [-\log p_\theta(\mathbf{x}_{t-1}^d|\mathbf{x}_t^{1:D})] \right] \end{aligned} \quad (53)$$

A.10 CTMC 简介

CTMC 过程可以随着 t 的增加或 t 的减少进行, 我们使用增加 t 作为引入 CTMC 的默认方向。对于任何 CTMC, 其转移率矩阵 R_t 与 $[R_t]_{ij} = r_t(\mathbf{e}_j|\mathbf{e}_i)$ 完全决定了底层随机过程。根据 R_t 是否相对于 t 保持静态, CTMC 被分为时齐和非时齐两类。在本文中我们处理的是非时齐 CTMC。

根据公式 (17) 的定义, 我们可以推导出以下性质

$$\forall \mathbf{x}, r_t(\mathbf{x}|\mathbf{x}) \leq 0; \quad \forall \mathbf{y} \neq \mathbf{x}, r_t(\mathbf{y}|\mathbf{x}) \geq 0; \quad \forall \mathbf{x}, \sum_{\mathbf{y}} r_t(\mathbf{y}|\mathbf{x}) = 0 \quad (54)$$

$$q_{t|t-\Delta t}(\mathbf{y}|\mathbf{x}) = \delta_{\mathbf{x},\mathbf{y}} + r_t(\mathbf{y}|\mathbf{x})\Delta t + o(\Delta t) \quad (55)$$

公式 (54) 展示了转移率的性质, 这表明了 $r_t(\mathbf{x}|\mathbf{x}) = -\sum_{\mathbf{y} \neq \mathbf{x}} r_t(\mathbf{y}|\mathbf{x})$ 。公式 (55) 描述了转移概率的微分变化, 可用于推导转移矩阵 $\bar{Q}_{t|s}$ 与转移率矩阵 R_t 之间的关系。形式上, CTMC 的转移概率满足 Kolmogorov 前向和后向方程 [Kolmogoroff, 1931]:

$$\frac{d}{dt} q_{t|s}(\mathbf{y}|\mathbf{x}) = \sum_{\mathbf{z}} q_{t|s}(\mathbf{z}|\mathbf{x}) r_t(\mathbf{y}|\mathbf{z}) \quad \text{or} \quad \frac{d}{dt} \bar{Q}_{t|s} = \bar{Q}_{t|s} R_t \quad \text{科尔莫哥洛夫前向} \quad (56)$$

$$\frac{d}{ds} q_{t|s}(\mathbf{y}|\mathbf{x}) = -\sum_{\mathbf{z}} r_s(\mathbf{z}|\mathbf{x}) q_{t|s}(\mathbf{y}|\mathbf{z}) \quad \text{or} \quad \frac{d}{ds} \bar{Q}_{t|s} = -R_s \bar{Q}_{t|s} \quad \text{科尔莫哥洛夫后向} \quad (57)$$

上述方程是常微分方程 (ODE), 具有唯一解。Rindos 等人 [1995] 提到, 当 R_{t_1} 和 R_{t_2} 可交换 (即 $R_{t_1}R_{t_2} = R_{t_2}R_{t_1}$) 对任何 t_1, t_2 时, 转移概率矩阵可以表示为

$$\bar{Q}_{t|s} = \exp\left(\int_s^t R_a da\right) \quad (58)$$

where $\exp(M) := \sum_{k=0}^{\infty} \frac{M^k}{k!}$ is the matrix exponential operation. The commutative property of R_t can be achieved by choosing $R_t = \beta(t)R_b$ where $R_b \in \mathbb{R}^{K \times K}$ is a time-independent base transition rate matrix satisfying properties in Eq. (54) and $\beta(t)$ is a positive scalar dependent on time. Now assume R_b is diagonalizable and $R_b = U\Sigma U^{-1}$ by eigen-decomposition, then we can simplify Eq. (58) as

$$\bar{Q}_{t|s} = \exp\left(\int_s^t \beta(t)R_b da\right) = \exp(\bar{\beta}_{t|s}R_b) = \exp(\bar{\beta}_{t|s}U\Sigma U^{-1}) = U \exp(\bar{\beta}_{t|s}\Sigma)U^{-1} \quad (59)$$

where $\bar{\beta}_{t|s} := \int_s^t \beta(a)da$.

A.11 Derivation of Eq. (18)

$$\begin{aligned} \bar{Q}_{t|s} &= \exp(\bar{\beta}_{t|s}R_b) = I + \sum_{k=1}^{\infty} \frac{(-\bar{\beta}_{t|s})^k (-R_b)^k}{k!} = I - \left(\sum_{k=1}^{\infty} \frac{(-\bar{\beta}_{t|s})^k}{k!}\right) R_b \\ &= I - (e^{-\bar{\beta}_{t|s}} - 1)R_b = e^{-\bar{\beta}_{t|s}}I + (1 - e^{-\bar{\beta}_{t|s}})\mathbf{1}\mathbf{m}^\top \end{aligned} \quad (60)$$

A.12 Derivation of Eq. (25)

With the help of Eq. (19), we first show that g_t^θ can be simplified as follows.

$$\begin{aligned} g_t^\theta(z|x) &= \frac{1}{\langle x, \mathbf{m} \rangle} \left[\langle z, \mathbf{m} \rangle + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} p_{0|t}^\theta(z|x) - \frac{\bar{\alpha}_{t|0} \langle z, \mathbf{m} \rangle}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle x, \mathbf{m} \rangle} p_{0|t}^\theta(x|x) \right] \\ &= \frac{1}{\langle x, \mathbf{m} \rangle} \left[\left(1 - \frac{\bar{\alpha}_{t|0} p_{0|t}^\theta(x|x)}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle x, \mathbf{m} \rangle}\right) \langle z, \mathbf{m} \rangle + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} p_{0|t}^\theta(z|x) \right] \quad \forall z \neq x \end{aligned} \quad (61)$$

Proof.

$$\frac{q_{t|0}(z|x_0)}{q_{t|0}(x|x_0)} = \frac{\mathbf{z}^\top \bar{Q}_{t|0}^\top \mathbf{x}_0}{\mathbf{x}^\top \bar{Q}_{t|0}^\top \mathbf{x}_0} = \frac{\bar{\alpha}_{t|0} \mathbf{z}^\top \mathbf{x}_0 + (1 - \bar{\alpha}_{t|0}) \langle z, \mathbf{m} \rangle}{\bar{\alpha}_{t|0} \mathbf{x}^\top \mathbf{x}_0 + (1 - \bar{\alpha}_{t|0}) \langle x, \mathbf{m} \rangle} \quad (62)$$

Then when $z \neq x$, we can write it as

$$\frac{q_{t|0}(z|x_0)}{q_{t|0}(x|x_0)} = \begin{cases} \frac{\langle z, \mathbf{m} \rangle}{\langle x, \mathbf{m} \rangle} & \text{if } x_0 \neq x \text{ and } x_0 \neq z \\ \frac{\frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} + \langle z, \mathbf{m} \rangle}{\langle x, \mathbf{m} \rangle} & \text{if } x_0 = z \\ \frac{\langle z, \mathbf{m} \rangle}{\frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} + \langle x, \mathbf{m} \rangle} & \text{if } x_0 = x \end{cases} \quad (63)$$

Hence,

$$\begin{aligned} g_t^\theta(z|x) &= \sum_{x_0} \frac{q_{t|0}(z|x_0)}{q_{t|0}(x|x_0)} p_{0|t}^\theta(x_0|x) \\ &= \frac{q_{t|0}(z|x)}{q_{t|0}(x|x)} p_{0|t}^\theta(x|x) + \frac{q_{t|0}(z|z)}{q_{t|0}(x|z)} p_{0|t}^\theta(z|x) + \sum_{x_0 \neq z, x} \frac{q_{t|0}(z|x_0)}{q_{t|0}(x|x_0)} p_{0|t}^\theta(x_0|x) \\ &= \frac{\langle z, \mathbf{m} \rangle}{\frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} + \langle x, \mathbf{m} \rangle} p_{0|t}^\theta(x|x) + \frac{\frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} + \langle z, \mathbf{m} \rangle}{\langle x, \mathbf{m} \rangle} p_{0|t}^\theta(z|x) + \frac{\langle z, \mathbf{m} \rangle}{\langle x, \mathbf{m} \rangle} (1 - p_{0|t}^\theta(z|x) - p_{0|t}^\theta(x|x)) \\ &= \frac{1}{\langle x, \mathbf{m} \rangle} \left[\left(1 - \frac{\bar{\alpha}_{t|0} p_{0|t}^\theta(x|x)}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle x, \mathbf{m} \rangle}\right) \langle z, \mathbf{m} \rangle + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} p_{0|t}^\theta(z|x) \right] \quad \forall z \neq x \end{aligned} \quad (64)$$

Additionally, when $z = x$, $g_t^\theta(z|x) = \sum_{x_0} p_{0|t}^\theta(x_0|x) = 1$. $\square$

We can also compute the vectorization $g_t^\theta(\cdot|x)$ directly as

$$g_t^\theta(\cdot|x) = \frac{1}{\langle x, \mathbf{m} \rangle} \left[\left(1 - \frac{\bar{\alpha}_{t|0} \langle f_t^\theta(x), x \rangle}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle x, \mathbf{m} \rangle}\right) \mathbf{m} + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} f_t^\theta(x) \right] \odot (\mathbf{1} - x) + x \quad (65)$$

其中 $\exp(M) := \sum_{k=0}^{\infty} \frac{M^k}{k!}$ 是矩阵指数运算。 R_t 的可交换性可以通过选择 $R_t = \beta(t)R_b$ 来实现，其中 $R_b \in \mathbb{R}^{K \times K}$ 是一个满足 Eq. (54) 中性质的时间无关基本转移率矩阵，而 $\beta(t)$ 是一个随时间变化的正标量。现在假设 R_b 是可对角化的，通过特征分解得到

$R_b = U\Sigma U^{-1}$ ，我们可以将 Eq. (58) 简化为

$$\bar{Q}_{t|s} = \exp\left(\int_s^t \beta(t)R_b da\right) = \exp(\bar{\beta}_{t|s}R_b) = \exp(\bar{\beta}_{t|s}U\Sigma U^{-1}) = U \exp(\bar{\beta}_{t|s}\Sigma)U^{-1} \quad (59)$$

在 $\beta_{t|s} := \int_s^t \beta(a)da$ 处

A.11 公式(18)的推导

$$\begin{aligned} \bar{Q}_{t|s} &= \exp(\bar{\beta}_{t|s}R_b) = I + \sum_{k=1}^{\infty} \frac{(-\bar{\beta}_{t|s})^k (-R_b)^k}{k!} = I - \left(\sum_{k=1}^{\infty} \frac{(-\bar{\beta}_{t|s})^k}{k!}\right) R_b \\ &= I - (e^{-\bar{\beta}_{t|s}} - 1)R_b = e^{-\bar{\beta}_{t|s}}I + (1 - e^{-\bar{\beta}_{t|s}})\mathbf{1}\mathbf{m}^\top \end{aligned} \quad (60)$$

A.12 公式(25)的推导

借助公式(19)，我们首先证明 g_t^θ 可以简化如下。

$$\begin{aligned} g_t^\theta(z|x) &= \frac{1}{\langle x, \mathbf{m} \rangle} \left[\langle z, \mathbf{m} \rangle + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} p_{0|t}^\theta(z|x) - \frac{\bar{\alpha}_{t|0} \langle z, \mathbf{m} \rangle}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle x, \mathbf{m} \rangle} p_{0|t}^\theta(x|x) \right] \\ &= \frac{1}{\langle x, \mathbf{m} \rangle} \left[\left(1 - \frac{\bar{\alpha}_{t|0} p_{0|t}^\theta(x|x)}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle x, \mathbf{m} \rangle}\right) \langle z, \mathbf{m} \rangle + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} p_{0|t}^\theta(z|x) \right] \quad \forall z \neq x \end{aligned} \quad (61)$$

证明。

$$\frac{q_{t|0}(z|x_0)}{q_{t|0}(x|x_0)} = \frac{\mathbf{z}^\top \bar{Q}_{t|0}^\top \mathbf{x}_0}{\mathbf{x}^\top \bar{Q}_{t|0}^\top \mathbf{x}_0} = \frac{\bar{\alpha}_{t|0} \mathbf{z}^\top \mathbf{x}_0 + (1 - \bar{\alpha}_{t|0}) \langle z, \mathbf{m} \rangle}{\bar{\alpha}_{t|0} \mathbf{x}^\top \mathbf{x}_0 + (1 - \bar{\alpha}_{t|0}) \langle x, \mathbf{m} \rangle} \quad (62)$$

然后当 $z \neq x$ 时，我们可以将其写为

$$\frac{q_{t|0}(z|x_0)}{q_{t|0}(x|x_0)} = \begin{cases} \frac{\langle z, \mathbf{m} \rangle}{\langle x, \mathbf{m} \rangle} & \text{if } x_0 \neq x \text{ and } x_0 \neq z \\ \frac{\frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} + \langle z, \mathbf{m} \rangle}{\langle x, \mathbf{m} \rangle} & \text{if } x_0 = z \\ \frac{\langle z, \mathbf{m} \rangle}{\frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} + \langle x, \mathbf{m} \rangle} & \text{if } x_0 = x \end{cases} \quad (63)$$

因此，

$$\begin{aligned} g_t^\theta(z|x) &= \sum_{x_0} \frac{q_{t|0}(z|x_0)}{q_{t|0}(x|x_0)} p_{0|t}^\theta(x_0|x) \\ &= \frac{q_{t|0}(z|x)}{q_{t|0}(x|x)} p_{0|t}^\theta(x|x) + \frac{q_{t|0}(z|z)}{q_{t|0}(x|z)} p_{0|t}^\theta(z|x) + \sum_{x_0 \neq z, x} \frac{q_{t|0}(z|x_0)}{q_{t|0}(x|x_0)} p_{0|t}^\theta(x_0|x) \\ &= \frac{\langle z, \mathbf{m} \rangle}{\frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} + \langle x, \mathbf{m} \rangle} p_{0|t}^\theta(x|x) + \frac{\frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} + \langle z, \mathbf{m} \rangle}{\langle x, \mathbf{m} \rangle} p_{0|t}^\theta(z|x) + \frac{\langle z, \mathbf{m} \rangle}{\langle x, \mathbf{m} \rangle} (1 - p_{0|t}^\theta(z|x) - p_{0|t}^\theta(x|x)) \\ &= \frac{1}{\langle x, \mathbf{m} \rangle} \left[\left(1 - \frac{\bar{\alpha}_{t|0} p_{0|t}^\theta(x|x)}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle x, \mathbf{m} \rangle}\right) \langle z, \mathbf{m} \rangle + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} p_{0|t}^\theta(z|x) \right] \quad \forall z \neq x \end{aligned} \quad (64)$$

此外，当 $z = x$ $g_t^\theta(z|x) = \sum_{x_0} p_{0|t}^\theta(x_0|x) = 1$ 时， $\circ$

$\square$

我们也可以直接计算向量化 $g_t^\theta(\cdot|x)$ ，

$$g_t^\theta(\cdot|x) = \frac{1}{\langle x, \mathbf{m} \rangle} \left[\left(1 - \frac{\bar{\alpha}_{t|0} \langle f_t^\theta(x), x \rangle}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle x, \mathbf{m} \rangle}\right) \mathbf{m} + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} f_t^\theta(x) \right] \odot (\mathbf{1} - x) + x \quad (65)$$

where $f_t^\theta(\mathbf{x})$ is the parameterized neural network for $p_{0|t}^\theta(\cdot|\mathbf{x})$ that outputs the distribution of $\mathbf{x}_{0|t}$. Eq. (25) combines Eq. (61) and $g_t^\theta(\mathbf{x}|\mathbf{x}) = 1, \forall \mathbf{x}$.

A.13 Derivation of Forward and Backward Transition Rate in Multi-element Case

In this section, we show how to extend transition rates, and the ratio g_t^θ , into multi-element case. We let $\mathbf{x}^{\setminus d}$ represent $\mathbf{x}^{1:D \setminus d}$, i.e. the object without d -th element, for simplicity.

Forward Transition Rates: First, the transition rates for forward sampling has a specific decomposition formulation in multi-element case as proven by Campbell et al. [2022], thus, we summarize the result as follows. The key assumption for CTMC is that at a single time, only one dimension can change.

$$r_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) = \sum_{d=1}^D r_t^d(\mathbf{y}^d|\mathbf{x}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \quad (66)$$

where $\delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}}$ is the Kronecker delta and it is 1 if and only if $\mathbf{x}^{\setminus d} = \mathbf{y}^{\setminus d}$. As we also assume that all dimension processes are independent, r_t^d denotes the transition rate of the CTMC process at d -th element/dimension.

Backward Transition Rates: Now let us work on $\hat{r}_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D})$. Notice that as the backward process is also a CTMC, it also satisfies that only one dimension can change at a time. We summarize two equivalent formulations as follows.

$$\hat{r}_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) = \sum_{d=1}^D r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} q_{0|t}(\mathbf{x}_0^d|\mathbf{x}^{1:D}) \quad (67)$$

$$\hat{r}_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) = \sum_{d=1}^D \frac{r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}}}{\sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)} q_{0|t}(\mathbf{x}_0^d|\mathbf{y}^{1:D})} \quad (68)$$

Notice that these two formulations should be equivalent. In practice, we use the first formulation to parameterize the reverse transition rate in learning.

Proof.

$$\frac{q_t(\mathbf{y}^{1:D})}{q_t(\mathbf{x}^{1:D})} = \sum_{\mathbf{x}_0^{1:D}} \frac{q_{t|0}(\mathbf{y}^{1:D}|\mathbf{x}_0^{1:D})}{q_{t|0}(\mathbf{x}^{1:D}|\mathbf{x}_0^{1:D})} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{x}^{1:D}) = \sum_{\mathbf{x}_0^{1:D}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{x}^{1:D}) \prod_{d=1}^D \frac{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \quad (69)$$

$$\sum_{\mathbf{x}_0^{\setminus d}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{x}^{1:D}) = \sum_{\mathbf{x}_0^{\setminus d}} q_{0|t}(\mathbf{x}_0^d|\mathbf{x}^{1:D}) q_{0|t}(\mathbf{x}_0^{\setminus d}|\mathbf{x}^{1:D}, \mathbf{x}_0^d) = q_{0|t}(\mathbf{x}_0^d|\mathbf{x}^{1:D}) \quad (70)$$

其中 $f_t^\theta(\mathbf{x})$ 是用于 $p_{0|t}^\theta(\cdot|\mathbf{x})$ 并输出 $\mathbf{x}_{0|t}$ 分布的参数化神经网络。公式 (25) 结合了公式 (61) 和 $g_t^\theta(\mathbf{x}|\mathbf{x}) = 1, \forall \mathbf{x}$ 。

A.13 多元素情况下的正向和反向转换率推导

在本节中，我们展示了如何将转换率以及比率 g_t^θ 扩展到多元素情况。我们让 $\mathbf{x}^{\setminus d}$ 表示 $\mathbf{x}^{1:D \setminus d}$ ，即不包含 d -th 元素的物体，以便简化。

正向转换率：首先，正向采样的转换率在多元素情况下具有特定的分解公式，这一点由 Campbell 等人证明。因此，我们总结结果如下。CTMC 的关键假设是，在单个时间点，只有一个维度可以改变。

$$r_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) = \sum_{d=1}^D r_t^d(\mathbf{y}^d|\mathbf{x}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \quad (66)$$

其中 $\delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}}$ 是克罗内克δ函数，当且仅当 $\mathbf{x}^{\setminus d} = \mathbf{y}^{\setminus d}$ 时为 1。由于我们还假设所有维度过程都是独立的， r_t^d 表示 CTMC 过程在 d -th 元素/维度上的转换率。

反向转换率：现在让我们处理 $\hat{r}_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D})$ 。注意，由于反向过程也是 CTMC，它同样满足每次只有一个维度可以改变。我们总结两个等价公式如下。

$$\hat{r}_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) = \sum_{d=1}^D r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} q_{0|t}(\mathbf{x}_0^d|\mathbf{x}^{1:D}) \quad (67)$$

$$\hat{r}_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) = \sum_{d=1}^D \frac{r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}}}{\sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)} q_{0|t}(\mathbf{x}_0^d|\mathbf{y}^{1:D})} \quad (68)$$

请注意，这两个公式应该等效。在实践中，我们使用第一个公式来参数化学习中的反向转换率。

证明。

$$\frac{q_t(\mathbf{y}^{1:D})}{q_t(\mathbf{x}^{1:D})} = \sum_{\mathbf{x}_0^{1:D}} \frac{q_{t|0}(\mathbf{y}^{1:D}|\mathbf{x}_0^{1:D})}{q_{t|0}(\mathbf{x}^{1:D}|\mathbf{x}_0^{1:D})} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{x}^{1:D}) = \sum_{\mathbf{x}_0^{1:D}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{x}^{1:D}) \prod_{d=1}^D \frac{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \quad (69)$$

$$\sum_{\mathbf{x}_0^{\setminus d}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{x}^{1:D}) = \sum_{\mathbf{x}_0^{\setminus d}} q_{0|t}(\mathbf{x}_0^d|\mathbf{x}^{1:D}) q_{0|t}(\mathbf{x}_0^{\setminus d}|\mathbf{x}^{1:D}, \mathbf{x}_0^d) = q_{0|t}(\mathbf{x}_0^d|\mathbf{x}^{1:D}) \quad (70)$$

(Case 1)

$$\begin{aligned}
\hat{r}_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) &= r_t^{1:D}(\mathbf{x}^{1:D}|\mathbf{y}^{1:D}) \frac{q_t(\mathbf{y}^{1:D})}{q_t(\mathbf{x}^{1:D})} \\
&= \left(\sum_{d=1}^D r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \right) \left(\sum_{\mathbf{x}_0^{1:D}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{x}^{1:D}) \prod_{d=1}^D \frac{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \right) \\
&= \sum_{d=1}^D \sum_{\mathbf{x}_0^{1:D}} r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{x}^{1:D}) \frac{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \\
&= \sum_{d=1}^D r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \sum_{\mathbf{x}_0^{\setminus d}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{x}^{1:D}) \\
&= \sum_{d=1}^D r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} q_{0|t}(\mathbf{x}_0^d|\mathbf{x}^{1:D}) \quad (71)
\end{aligned}$$

(Case 2)

$$\begin{aligned}
\hat{r}_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) &= r_t^{1:D}(\mathbf{x}^{1:D}|\mathbf{y}^{1:D}) / \frac{q_t(\mathbf{x}^{1:D})}{q_t(\mathbf{y}^{1:D})} \\
&= \sum_{d=1}^D \frac{r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}}}{\sum_{\mathbf{x}_0^{1:D}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{y}^{1:D}) \prod_{i=1}^D \frac{q_{t|0}(\mathbf{x}^i|\mathbf{x}_0^i)}{q_{t|0}(\mathbf{y}^i|\mathbf{x}_0^i)}} \\
&= \sum_{d=1}^D \frac{r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}}}{\sum_{\mathbf{x}_0^{1:D}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{y}^{1:D}) \frac{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}} \\
&= \sum_{d=1}^D \frac{r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}}}{\sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)} q_{0|t}(\mathbf{x}_0^d|\mathbf{y}^{1:D})} \quad (72)
\end{aligned}$$

□

Ratio: We now define an estimator $g_t^{\theta, d}(\mathbf{x}^d|\mathbf{y}^{1:D})$ as follows.

$$g_t^{\theta, d}(\mathbf{x}^d|\mathbf{y}^{1:D}) := \sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d) p_{0|t}^\theta(\mathbf{x}_0^d|\mathbf{y}^{1:D})}{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)} \approx \sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)} q_{0|t}(\mathbf{x}_0^d|\mathbf{y}^{1:D}) = \frac{q_t(\mathbf{x}^d|\mathbf{y}^{\setminus d})}{q_t(\mathbf{y}^d|\mathbf{y}^{\setminus d})} \quad (73)$$

$$g_t^\theta(\mathbf{x}^{1:D}|\mathbf{y}^{1:D}) := \prod_{d=1}^D g_t^{\theta, d}(\mathbf{x}^d|\mathbf{y}^{1:D}) = \sum_{\mathbf{x}_0^{1:D}} \frac{q_{t|0}(\mathbf{x}^{1:D}|\mathbf{x}_0^{1:D})}{q_{t|0}(\mathbf{y}^{1:D}|\mathbf{x}_0^{1:D})} p_{0|t}^\theta(\mathbf{x}_0^{1:D}|\mathbf{y}^{1:D}) \approx \frac{q_t(\mathbf{x}^{1:D})}{q_t(\mathbf{y}^{1:D})} \quad (74)$$

We can extend the vector formulation Eq. (25) in Proposition 4 to $g_t^{\theta, d}(\mathbf{x}^d|\mathbf{y}^{1:D})$:

$$g_t^{\theta, d}(\cdot|\mathbf{x}^{1:D}) = \frac{1}{\langle \mathbf{x}^d, \mathbf{m}^d \rangle} \left[\left(1 - \frac{\bar{\alpha}_{t|0} \langle f_t^{\theta, d}(\mathbf{x}^{1:D}), \mathbf{x}^d \rangle}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle \mathbf{x}^d, \mathbf{m}^d \rangle} \right) \mathbf{m}^d + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} f_t^{\theta, d}(\mathbf{x}^{1:D}) \right] \odot (\mathbf{1} - \mathbf{x}^d) + \mathbf{x}^d \quad (75)$$

Then, we can derive two approximators for transition rate $\hat{r}_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D})$ as follows.

$$[\hat{r}_t^{\theta, 1:D}]^1(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) := \sum_{d=1}^D r_t^d(\mathbf{x}^d|\mathbf{y}^d) g_t^{\theta, d}(\mathbf{y}^d|\mathbf{x}^{1:D}) \cdot \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \approx \text{Eq. (67)} \quad (76)$$

$$[\hat{r}_t^{\theta, 1:D}]^2(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) := \sum_{d=1}^D \frac{r_t^d(\mathbf{x}^d|\mathbf{y}^d)}{g_t^{\theta, d}(\mathbf{x}^d|\mathbf{y}^{1:D})} \cdot \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \approx \text{Eq. (68)} \quad (77)$$

(情况1)

$$\begin{aligned}
\hat{r}_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) &= r_t^{1:D}(\mathbf{x}^{1:D}|\mathbf{y}^{1:D}) \frac{q_t(\mathbf{y}^{1:D})}{q_t(\mathbf{x}^{1:D})} \\
&= \left(\sum_{d=1}^D r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \right) \left(\sum_{\mathbf{x}_0^{1:D}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{x}^{1:D}) \prod_{d=1}^D \frac{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \right) \\
&= \sum_{d=1}^D \sum_{\mathbf{x}_0^{1:D}} r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{x}^{1:D}) \frac{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \\
&= \sum_{d=1}^D r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \sum_{\mathbf{x}_0^{\setminus d}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{x}^{1:D}) \\
&= \sum_{d=1}^D r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} q_{0|t}(\mathbf{x}_0^d|\mathbf{x}^{1:D}) \quad (71)
\end{aligned}$$

(案例2)

$$\begin{aligned}
\hat{r}_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) &= r_t^{1:D}(\mathbf{x}^{1:D}|\mathbf{y}^{1:D}) / \frac{q_t(\mathbf{x}^{1:D})}{q_t(\mathbf{y}^{1:D})} \\
&= \sum_{d=1}^D \frac{r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}}}{\sum_{\mathbf{x}_0^{1:D}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{y}^{1:D}) \prod_{i=1}^D \frac{q_{t|0}(\mathbf{x}^i|\mathbf{x}_0^i)}{q_{t|0}(\mathbf{y}^i|\mathbf{x}_0^i)}} \\
&= \sum_{d=1}^D \frac{r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}}}{\sum_{\mathbf{x}_0^{1:D}} q_{0|t}(\mathbf{x}_0^{1:D}|\mathbf{y}^{1:D}) \frac{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)}} \\
&= \sum_{d=1}^D \frac{r_t^d(\mathbf{x}^d|\mathbf{y}^d) \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}}}{\sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)} q_{0|t}(\mathbf{x}_0^d|\mathbf{y}^{1:D})} \quad (72)
\end{aligned}$$

□

比率：我们现在定义一个估计器 $g_t^{\theta, d}(\mathbf{x}^d|\mathbf{y}^{1:D})$ ，如下所示。

$$g_t^{\theta, d}(\mathbf{x}^d|\mathbf{y}^{1:D}) := \sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d) p_{0|t}^\theta(\mathbf{x}_0^d|\mathbf{y}^{1:D})}{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)} \approx \sum_{\mathbf{x}_0^d} \frac{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)}{q_{t|0}(\mathbf{y}^d|\mathbf{x}_0^d)} q_{0|t}(\mathbf{x}_0^d|\mathbf{y}^{1:D}) = \frac{q_t(\mathbf{x}^d|\mathbf{y}^{\setminus d})}{q_t(\mathbf{y}^d|\mathbf{y}^{\setminus d})} \quad (73)$$

$$g_t^\theta(\mathbf{x}^{1:D}|\mathbf{y}^{1:D}) := \prod_{d=1}^D g_t^{\theta, d}(\mathbf{x}^d|\mathbf{y}^{1:D}) = \sum_{\mathbf{x}_0^{1:D}} \frac{q_{t|0}(\mathbf{x}^{1:D}|\mathbf{x}_0^{1:D})}{q_{t|0}(\mathbf{y}^{1:D}|\mathbf{x}_0^{1:D})} p_{0|t}^\theta(\mathbf{x}_0^{1:D}|\mathbf{y}^{1:D}) \approx \frac{q_t(\mathbf{x}^{1:D})}{q_t(\mathbf{y}^{1:D})} \quad (74)$$

我们可以将命题4中的向量公式(25)扩展到 $g_t^{\theta, d}(\mathbf{x}^d|\mathbf{y}^{1:D})$ ，：

$$g_t^{\theta, d}(\cdot|\mathbf{x}^{1:D}) = \frac{1}{\langle \mathbf{x}^d, \mathbf{m}^d \rangle} \left[\left(1 - \frac{\bar{\alpha}_{t|0} \langle f_t^{\theta, d}(\mathbf{x}^{1:D}), \mathbf{x}^d \rangle}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle \mathbf{x}^d, \mathbf{m}^d \rangle} \right) \mathbf{m}^d + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} f_t^{\theta, d}(\mathbf{x}^{1:D}) \right] \odot (\mathbf{1} - \mathbf{x}^d) + \mathbf{x}^d \quad (75)$$

然后，我们可以按如下方式推导出两个用于转换率 $\hat{r}_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D})$ 的近似器。

$$[\hat{r}_t^{\theta, 1:D}]^1(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) := \sum_{d=1}^D r_t^d(\mathbf{x}^d|\mathbf{y}^d) g_t^{\theta, d}(\mathbf{y}^d|\mathbf{x}^{1:D}) \cdot \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \approx \text{Eq. (67)} \quad (76)$$

$$[\hat{r}_t^{\theta, 1:D}]^2(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) := \sum_{d=1}^D \frac{r_t^d(\mathbf{x}^d|\mathbf{y}^d)}{g_t^{\theta, d}(\mathbf{x}^d|\mathbf{y}^{1:D})} \cdot \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \approx \text{Eq. (68)} \quad (77)$$

A.14 Derivation of Negative VLB Loss in Multi-element Case

As forward process is defined in Eq. (19), in multi-element case we can easily get

$$r_t^d(\mathbf{x}^d|\cdot) = R_t^d \mathbf{x}^d = \beta(t)(\langle \mathbf{x}^d, \mathbf{m}^d \rangle \mathbf{1} - \mathbf{x}^d) \quad (78)$$

$$r_t^d(\cdot|\mathbf{x}^d) = (R_t^d)^\top \mathbf{x}^d = \beta(t)(\mathbf{m}^d - \mathbf{x}^d) \quad (79)$$

The $r_t^d(\mathbf{x}|\cdot)$ and $r_t^d(\cdot|\mathbf{x})$ are essentially the $\mathbf{x}$ -th column and row of the transition rate matrix R_t^d .

In Multi-element case, the negative VLB loss in Eq. (23) can be written as

$$T \mathbb{E}_{t \sim \text{Uni}(0,T)} \left[\sum_{\mathbf{x}^{1:D} \sim q_{t|0}} \hat{r}_t^{\theta,1:D}(\mathbf{z}^{1:D}|\mathbf{x}^{1:D}) - \sum_{\mathbf{z}^{1:D} \neq \mathbf{x}^{1:D}} r_t^{1:D}(\mathbf{z}^{1:D}|\mathbf{x}^{1:D}) \log \hat{r}_t^{\theta,1:D}(\mathbf{x}^{1:D}|\mathbf{z}^{1:D}) \right] \quad (80)$$

As there are two terms, let's work on each term separately.

A.14.1 Term 1

Based on the formulation of $r_t^{1:D}$ and $\hat{r}_t^{\theta,1:D}$, we can rewrite the first term as

$$\begin{aligned} \text{Term1} &= T \mathbb{E}_{t, \mathbf{x}^{1:D}} \sum_{\mathbf{z}^{1:D} \neq \mathbf{x}^{1:D}} \hat{r}_t^{\theta,1:D}(\mathbf{z}^{1:D}|\mathbf{x}^{1:D}) \\ &= T \mathbb{E}_{t, \mathbf{x}^{1:D}} \sum_{\mathbf{z}^{1:D} \neq \mathbf{x}^{1:D}} \sum_{d=1}^D r_t^d(\mathbf{x}^d|\mathbf{z}^d) g_t^{\theta,d}(\mathbf{z}^d|\mathbf{x}^{1:D}) \cdot \delta_{\mathbf{x} \setminus d, \mathbf{z} \setminus d} \\ &= T \mathbb{E}_{t, \mathbf{x}^{1:D}} \left[\sum_{d=1}^D \sum_{\mathbf{z}^d \neq \mathbf{x}^d} r_t^d(\mathbf{x}^d|\mathbf{z}^d) g_t^{\theta,d}(\mathbf{z}^d|\mathbf{x}^{1:D}) \right] \\ &= T \mathbb{E}_{t, \mathbf{x}^{1:D}} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D}) \right] + \text{const.} \\ &= T \mathbb{E}_{t \sim \text{Uni}(0,T)} \beta(t) \sum_{\mathbf{x}^{1:D} \sim q_{t|0}} \langle \mathbf{x}^d, \mathbf{m}^d \rangle \left[\mathbf{1}^\top g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D}) \right] + \text{const.} \quad (81) \end{aligned}$$

A.14.2 Term 2

As the evaluation of $\hat{r}_t^{\theta}(\cdot|\mathbf{x})$ for any $\mathbf{x}$ requires a single forward pass of the parameterized network $p_{0|t}^{\theta}(\cdot|\mathbf{x})$, the second term within the expectation of Eq. (23) requires multiple passes of the model. This complexity is even greatly amplified in cases with multi-element objects. Campbell et al. [2022] avoids the multiple passes by changing the expectation variable through importance sampling. We take a similar approach to simplify the second term. Differently, Campbell et al. [2022] uses a specific sampling distribution (same as the forward transition rate) to introduce the auxiliary variable for changing the expectation variable, we generalize it to use a general sampling process S_t defined below.

Let $\mathbf{y}^{1:D}$ be the new variable upon which the exchanged expectation is based, and assume that $\mathbf{y}^{1:D}$ is sampled from an unnormalized joint distribution $S_t(\mathbf{y}^{1:D}|\mathbf{x}^{1:D})$. We restrict $S_t(\mathbf{y}^{1:D}|\mathbf{x}^{1:D})$ to be a unnormalized probability that is nonzero if and only if $\mathbf{y}^{1:D}$ and $\mathbf{x}^{1:D}$ are different at a single element. Formally, we can write the unnormalized distribution as

$$\begin{aligned} S_t(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) &= (1 - \delta_{\mathbf{y}^{1:D}, \mathbf{x}^{1:D}}) \sum_{d=1}^D S_t^d(\mathbf{y}^d|\mathbf{x}^d) \delta_{\mathbf{y} \setminus d, \mathbf{x} \setminus d} \\ \text{with normalizer } \mathcal{S}_t(\mathbf{x}^{1:D}) &= \sum_{\mathbf{y}^{1:D}} S_t(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) = \sum_{d=1}^D \sum_{\mathbf{y}^d \neq \mathbf{x}^d} S_t^d(\mathbf{y}^d|\mathbf{x}^d) \quad (82) \end{aligned}$$

where $S_t^d(\mathbf{y}^d|\mathbf{x}^d)$ is any unnormalized probability at dimension d .

A.14 多元素情况下的负VLB损失推导

由于正向过程在公式 (19) 中定义，在多元素情况下我们可以轻松得到

$$r_t^d(\mathbf{x}^d|\cdot) = R_t^d \mathbf{x}^d = \beta(t)(\langle \mathbf{x}^d, \mathbf{m}^d \rangle \mathbf{1} - \mathbf{x}^d) \quad (78)$$

$$r_t^d(\cdot|\mathbf{x}^d) = (R_t^d)^\top \mathbf{x}^d = \beta(t)(\mathbf{m}^d - \mathbf{x}^d) \quad (79)$$

$r_t^d(\mathbf{x}|\cdot)$ 和 $r_t^d(\cdot|\mathbf{x})$ 本质上是转移率矩阵 R_t^d 的第 $\mathbf{x}$ 列和行。

在多元素情况下，公式(23)中的负VLB损失可以表示为

$$T \mathbb{E}_{t \sim \text{Uni}(0,T)} \left[\sum_{\mathbf{x}^{1:D} \sim q_{t|0}} \hat{r}_t^{\theta,1:D}(\mathbf{z}^{1:D}|\mathbf{x}^{1:D}) - \sum_{\mathbf{z}^{1:D} \neq \mathbf{x}^{1:D}} r_t^{1:D}(\mathbf{z}^{1:D}|\mathbf{x}^{1:D}) \log \hat{r}_t^{\theta,1:D}(\mathbf{x}^{1:D}|\mathbf{z}^{1:D}) \right] \quad (80)$$

由于有两个项，我们分别处理每个项。

A.14.1 第1项

基于 $r_t^{1:D}$ 和 $\hat{r}_t^{\theta,1:D}$ 的公式，我们可以将第一项重写为

$$\begin{aligned} \text{Term1} &= T \mathbb{E}_{t, \mathbf{x}^{1:D}} \sum_{\mathbf{z}^{1:D} \neq \mathbf{x}^{1:D}} \hat{r}_t^{\theta,1:D}(\mathbf{z}^{1:D}|\mathbf{x}^{1:D}) \\ &= T \mathbb{E}_{t, \mathbf{x}^{1:D}} \sum_{\mathbf{z}^{1:D} \neq \mathbf{x}^{1:D}} \sum_{d=1}^D r_t^d(\mathbf{x}^d|\mathbf{z}^d) g_t^{\theta,d}(\mathbf{z}^d|\mathbf{x}^{1:D}) \cdot \delta_{\mathbf{x} \setminus d, \mathbf{z} \setminus d} \\ &= T \mathbb{E}_{t, \mathbf{x}^{1:D}} \left[\sum_{d=1}^D \sum_{\mathbf{z}^d \neq \mathbf{x}^d} r_t^d(\mathbf{x}^d|\mathbf{z}^d) g_t^{\theta,d}(\mathbf{z}^d|\mathbf{x}^{1:D}) \right] \\ &= T \mathbb{E}_{t, \mathbf{x}^{1:D}} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D}) \right] + \text{const.} \\ &= T \mathbb{E}_{t \sim \text{Uni}(0,T)} \beta(t) \sum_{\mathbf{x}^{1:D} \sim q_{t|0}} \langle \mathbf{x}^d, \mathbf{m}^d \rangle \left[\mathbf{1}^\top g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D}) \right] + \text{const.} \quad (81) \end{aligned}$$

A.14.2 第二项

由于对任何 $\mathbf{x}$ 的 $\hat{r}_t^{\theta}(\cdot|\mathbf{x})$ 评估只需要参数化网络的单次前向传递 $p_{0|t}^{\theta}(\cdot|\mathbf{x})$ ，因此等式 (23) 的期望中的第二项需要多次传递模型。这种复杂性在涉及多元素对象的情况下会被进一步放大。Campbell 等人 [2022] 通过改变期望变量进行重要性采样来避免多次传递。我们采用类似的方法来简化第二项。不同之处在于，Campbell 等人 [2022] 使用特定的采样分布（与正向转换率相同）来引入辅助变量以改变期望变量，而我们将其推广为使用下面定义的通用采样过程 S_t 。

设 $\mathbf{y}^{1:D}$ 是基于其进行交换的期望的新变量，并假设 $\mathbf{y}^{1:D}$ 从一个未归一化的联合分布 $S_t(\mathbf{y}^{1:D}|\mathbf{x}^{1:D})$ 中采样。我们将 $S_t(\mathbf{y}^{1:D}|\mathbf{x}^{1:D})$ 限制为一个未归一化的概率，它在且仅在 $\mathbf{y}^{1:D}$ 和 $\mathbf{x}^{1:D}$ 在单个元素处不同时才为零。形式上，我们可以将未归一化分布写为

$$\begin{aligned} S_t(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) &= (1 - \delta_{\mathbf{y}^{1:D}, \mathbf{x}^{1:D}}) \sum_{d=1}^D S_t^d(\mathbf{y}^d|\mathbf{x}^d) \delta_{\mathbf{y} \setminus d, \mathbf{x} \setminus d} \\ \text{with normalizer } \mathcal{S}_t(\mathbf{x}^{1:D}) &= \sum_{\mathbf{y}^{1:D}} S_t(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) = \sum_{d=1}^D \sum_{\mathbf{y}^d \neq \mathbf{x}^d} S_t^d(\mathbf{y}^d|\mathbf{x}^d) \quad (82) \end{aligned}$$

$S_t^d(\mathbf{y}^d|\mathbf{x}^d)$ 在哪里是任何未归一化的概率，维度为 d 。

With this definition, the Term 2 can be further rewritten as

$$\begin{aligned}
\text{Term2} &= \mathbb{E}_{t, \mathbf{z}^{1:D} \sim S_t} \left[\frac{1}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D} | \mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{1}{q_{t|0}(\mathbf{z}^d | \mathbf{x}_0^d)} \sum_{\mathbf{x}^d \neq \mathbf{z}^d} q_{t|0}(\mathbf{x}^d | \mathbf{x}_0^d) \cdot r_t^d(\mathbf{z}^d | \mathbf{x}^d) \cdot \log g_t^{\theta, d}(\mathbf{x}^d | \mathbf{z}^{1:D}) \right] \\
&= \mathbb{E}_{t, \mathbf{z}^{1:D} \sim S_t} \left[\frac{1}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D} | \mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot | \mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d | \cdot) \odot \log g_t^{\theta, d}(\cdot | \mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d | \mathbf{x}_0^d)} \right] + \text{const.} \\
&= \mathbb{E}_{t, \mathbf{z}^{1:D} \sim S_t} \left[\frac{\beta(t)}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D} | \mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\langle \mathbf{z}^d, \mathbf{m}^d \rangle}{q_{t|0}(\mathbf{z}^d | \mathbf{x}_0^d)} \mathbf{1}^\top [q_{t|0}(\cdot | \mathbf{x}_0^d) \odot \log g_t^{\theta, d}(\cdot | \mathbf{z}^{1:D})] + \text{const.} \right] \quad (87)
\end{aligned}$$

A.14.3 All Terms

Combine term 1 and term 2 together, we can write the negative VLB loss as

$$\begin{aligned}
T \mathbb{E}_{t \sim \text{Uni}(0, T)} \left[\sum_{\mathbf{x}^{1:D} \sim q_{t|0}} \hat{r}_t^{\theta, 1:D}(\mathbf{z}^{1:D} | \mathbf{x}^{1:D}) - \sum_{\mathbf{z}^{1:D} \neq \mathbf{x}^{1:D}} r_t^{1:D}(\mathbf{z}^{1:D} | \mathbf{x}^{1:D}) \log \hat{r}_t^{\theta, 1:D}(\mathbf{x}^{1:D} | \mathbf{z}^{1:D}) \right] \\
= T \mathbb{E}_{\substack{t \sim \text{Uni}(0, T) \\ \mathbf{x}^{1:D} \sim q_{t|0}(\mathbf{x}_0^{1:D}) \\ \mathbf{z}^{1:D} \sim S_t(\mathbf{x}^{1:D})}} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d | \cdot)^\top g_t^{\theta, d}(\cdot | \mathbf{x}^{1:D}) \right. \\
\left. - \frac{1}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D} | \mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot | \mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d | \cdot) \odot \log g_t^{\theta, d}(\cdot | \mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d | \mathbf{x}_0^d)} \right] + \text{const.} \quad (88)
\end{aligned}$$

A.15 Further Simplification of Continuous VLB

Proposition A.1. The loss in Eq. (26) can be further simplified as

$$T \mathbb{E}_{\substack{t \sim \text{Uni}(0, T) \\ \mathbf{x}^{1:D} \sim q_{t|0}(\cdot | \mathbf{x}_0^{1:D})}} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d | \cdot)^\top \left(g_t^{\theta, d}(\cdot | \mathbf{x}^{1:D}) - \frac{q_{t|0}(\cdot | \mathbf{x}_0^d) \odot \log g_t^{\theta, d}(\cdot | \mathbf{x}^{1:D})}{q_{t|0}(\mathbf{x}^d | \mathbf{x}_0^d)} \right) \right] \quad (89)$$

where the loss only requires a single forward pass of the model and no auxiliary variable is needed.

Proof. We use $q_{t|0}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D})$ to denote $P(\mathbf{x}_t^{1:D} = \mathbf{x}^{1:D} | \mathbf{x}_0^{1:D})$. Similarly, let $\tilde{q}_{t|0}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D})$ denotes $P(\mathbf{z}_t^{1:D} = \mathbf{x}^{1:D} | \mathbf{x}_0^{1:D})$. Based on the definition of S_t in Eq. (82) and the procedure of sampling $\mathbf{z}_t^{1:D}$ conditional on $\mathbf{x}_t^{1:D}$ described in Proposition 3.2, we can compute the conditional probability $\tilde{q}_{t|0}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D})$ as follows:

$$\begin{aligned}
\tilde{q}_{t|0}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D}) &= \sum_{\mathbf{x}_t^{1:D}} P(\mathbf{z}_t^{1:D} = \mathbf{x}^{1:D}, \mathbf{x}_t^{1:D} | \mathbf{x}_0^{1:D}) = \sum_{\mathbf{x}_t^{1:D}} \frac{S_t(\mathbf{x}^{1:D} | \mathbf{x}_t^{1:D})}{S_t(\mathbf{x}_t^{1:D})} q_{t|0}(\mathbf{x}_t^{1:D} | \mathbf{x}_0^{1:D}) \\
&= \sum_{\mathbf{x}_t^{1:D}} (1 - \delta_{\mathbf{x}^{1:D}, \mathbf{x}_t^{1:D}}) \left(\sum_{d=1}^D \delta_{\mathbf{x} \setminus^d, \mathbf{x}_t \setminus^d} \cdot \frac{S_t^d(\mathbf{x}^d | \mathbf{x}_t^d)}{S_t(\mathbf{x} \setminus^d \circ \mathbf{x}_t^d)} q_{t|0}(\mathbf{x} \setminus^d \circ \mathbf{x}_t^d | \mathbf{x}_0^{1:D}) \right) \\
&= q_{t|0}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D}) \sum_{\mathbf{x}_t^{1:D}} (1 - \delta_{\mathbf{x}^{1:D}, \mathbf{x}_t^{1:D}}) \left(\sum_{d=1}^D \delta_{\mathbf{x} \setminus^d, \mathbf{x}_t \setminus^d} \cdot \frac{S_t^d(\mathbf{x}^d | \mathbf{x}_t^d)}{S_t(\mathbf{x} \setminus^d \circ \mathbf{x}_t^d)} \frac{q_{t|0}(\mathbf{x}_t^d | \mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d | \mathbf{x}_0^d)} \right) \\
&= q_{t|0}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D}) \mathcal{M}_{S_t}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D}) \quad (\text{Based on } \mathcal{M}_{S_t} \text{ in Eq. (86)}). \quad (90)
\end{aligned}$$

The original loss in Eq. (26) can be written as

$$\begin{aligned}
Eq. (26) &= T \mathbb{E}_{\substack{t \sim \text{Uni}(0, T) \\ \mathbf{x}^{1:D} \sim q_{t|0}(\cdot | \mathbf{x}_0^{1:D})}} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d | \cdot)^\top g_t^{\theta, d}(\cdot | \mathbf{x}^{1:D}) \right] - \\
&\quad T \mathbb{E}_{\substack{t \sim \text{Uni}(0, T) \\ \mathbf{x}^{1:D} \sim q_{t|0}(\cdot | \mathbf{x}_0^{1:D}) \\ \mathbf{z}^{1:D} \sim S_t(\cdot | \mathbf{x}^{1:D})}} \left[\frac{1}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D} | \mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot | \mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d | \cdot) \odot \log g_t^{\theta, d}(\cdot | \mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d | \mathbf{x}_0^d)} \right]
\end{aligned}$$

根据此定义，项2可以进一步重写为

$$\begin{aligned}
\text{Term2} &= \mathbb{E}_{t, \mathbf{z}^{1:D} \sim S_t} \left[\frac{1}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D} | \mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{1}{q_{t|0}(\mathbf{z}^d | \mathbf{x}_0^d)} \sum_{\mathbf{x}^d \neq \mathbf{z}^d} q_{t|0}(\mathbf{x}^d | \mathbf{x}_0^d) \cdot r_t^d(\mathbf{z}^d | \mathbf{x}^d) \cdot \log g_t^{\theta, d}(\mathbf{x}^d | \mathbf{z}^{1:D}) \right] \\
&= \mathbb{E}_{t, \mathbf{z}^{1:D} \sim S_t} \left[\frac{1}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D} | \mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot | \mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d | \cdot) \odot \log g_t^{\theta, d}(\cdot | \mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d | \mathbf{x}_0^d)} \right] + \text{const.} \\
&= \mathbb{E}_{t, \mathbf{z}^{1:D} \sim S_t} \left[\frac{\beta(t)}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D} | \mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\langle \mathbf{z}^d, \mathbf{m}^d \rangle}{q_{t|0}(\mathbf{z}^d | \mathbf{x}_0^d)} \mathbf{1}^\top [q_{t|0}(\cdot | \mathbf{x}_0^d) \odot \log g_t^{\theta, d}(\cdot | \mathbf{z}^{1:D})] + \text{const.} \right] \quad (87)
\end{aligned}$$

A.14.3 所有术语

将术语1和术语2结合起来，我们可以将负VLB损失表示为

$$\begin{aligned}
T \mathbb{E}_{t \sim \text{Uni}(0, T)} \left[\sum_{\mathbf{x}^{1:D} \sim q_{t|0}} \hat{r}_t^{\theta, 1:D}(\mathbf{z}^{1:D} | \mathbf{x}^{1:D}) - \sum_{\mathbf{z}^{1:D} \neq \mathbf{x}^{1:D}} r_t^{1:D}(\mathbf{z}^{1:D} | \mathbf{x}^{1:D}) \log \hat{r}_t^{\theta, 1:D}(\mathbf{x}^{1:D} | \mathbf{z}^{1:D}) \right] \\
= T \mathbb{E}_{\substack{t \sim \text{Uni}(0, T) \\ \mathbf{x}^{1:D} \sim q_{t|0}(\mathbf{x}_0^{1:D}) \\ \mathbf{z}^{1:D} \sim S_t(\mathbf{x}^{1:D})}} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d | \cdot)^\top g_t^{\theta, d}(\cdot | \mathbf{x}^{1:D}) \right. \\
\left. - \frac{1}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D} | \mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot | \mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d | \cdot) \odot \log g_t^{\theta, d}(\cdot | \mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d | \mathbf{x}_0^d)} \right] + \text{const.} \quad (88)
\end{aligned}$$

A.15 连续VLB的进一步简化

命题A.1. 等式(26)中的损失可以进一步简化为

$$T \mathbb{E}_{\substack{t \sim \text{Uni}(0, T) \\ \mathbf{x}^{1:D} \sim q_{t|0}(\cdot | \mathbf{x}_0^{1:D})}} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d | \cdot)^\top \left(g_t^{\theta, d}(\cdot | \mathbf{x}^{1:D}) - \frac{q_{t|0}(\cdot | \mathbf{x}_0^d) \odot \log g_t^{\theta, d}(\cdot | \mathbf{x}^{1:D})}{q_{t|0}(\mathbf{x}^d | \mathbf{x}_0^d)} \right) \right] \quad (89)$$

其中损失只需要模型的一次前向传递，并且不需要辅助变量。

证明。我们使用 $q_{t|0}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D})$ 表示 $P(\mathbf{x}_t^{1:D} = \mathbf{x}^{1:D} | \mathbf{x}_0^{1:D})$ 。类似地，令 $\tilde{q}_{t|0}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D})$ 表示 $P(\mathbf{z}_t^{1:D} = \mathbf{x}^{1:D} | \mathbf{x}_0^{1:D})$ 。根据等式(82)中 S_t 的定义以及命题3.2中描述的关于 $\mathbf{x}_t^{1:D}$ 条件下采样 $\mathbf{z}_t^{1:D} | \tilde{q}_{t|0}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D})$ 的步骤，我们可以按如下方式计算条件概率：

$$\begin{aligned}
\tilde{q}_{t|0}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D}) &= \sum_{\mathbf{x}_t^{1:D}} P(\mathbf{z}_t^{1:D} = \mathbf{x}^{1:D}, \mathbf{x}_t^{1:D} | \mathbf{x}_0^{1:D}) = \sum_{\mathbf{x}_t^{1:D}} \frac{S_t(\mathbf{x}^{1:D} | \mathbf{x}_t^{1:D})}{S_t(\mathbf{x}_t^{1:D})} q_{t|0}(\mathbf{x}_t^{1:D} | \mathbf{x}_0^{1:D}) \\
&= \sum_{\mathbf{x}_t^{1:D}} (1 - \delta_{\mathbf{x}^{1:D}, \mathbf{x}_t^{1:D}}) \left(\sum_{d=1}^D \delta_{\mathbf{x} \setminus^d, \mathbf{x}_t \setminus^d} \cdot \frac{S_t^d(\mathbf{x}^d | \mathbf{x}_t^d)}{S_t(\mathbf{x} \setminus^d \circ \mathbf{x}_t^d)} q_{t|0}(\mathbf{x} \setminus^d \circ \mathbf{x}_t^d | \mathbf{x}_0^{1:D}) \right) \\
&= q_{t|0}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D}) \sum_{\mathbf{x}_t^{1:D}} (1 - \delta_{\mathbf{x}^{1:D}, \mathbf{x}_t^{1:D}}) \left(\sum_{d=1}^D \delta_{\mathbf{x} \setminus^d, \mathbf{x}_t \setminus^d} \cdot \frac{S_t^d(\mathbf{x}^d | \mathbf{x}_t^d)}{S_t(\mathbf{x} \setminus^d \circ \mathbf{x}_t^d)} \frac{q_{t|0}(\mathbf{x}_t^d | \mathbf{x}_0^d)}{q_{t|0}(\mathbf{x}^d | \mathbf{x}_0^d)} \right) \\
&= q_{t|0}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D}) \mathcal{M}_{S_t}(\mathbf{x}^{1:D} | \mathbf{x}_0^{1:D}) \quad (\text{Based on } \mathcal{M}_{S_t} \text{ in Eq. (86)}). \quad (90)
\end{aligned}$$

公式(26)中的原始损失可以表示为

$$\begin{aligned}
Eq. (26) &= T \mathbb{E}_{\substack{t \sim \text{Uni}(0, T) \\ \mathbf{x}^{1:D} \sim q_{t|0}(\cdot | \mathbf{x}_0^{1:D})}} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d | \cdot)^\top g_t^{\theta, d}(\cdot | \mathbf{x}^{1:D}) \right] - \\
&\quad T \mathbb{E}_{\substack{t \sim \text{Uni}(0, T) \\ \mathbf{x}^{1:D} \sim q_{t|0}(\cdot | \mathbf{x}_0^{1:D}) \\ \mathbf{z}^{1:D} \sim S_t(\cdot | \mathbf{x}^{1:D})}} \left[\frac{1}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D} | \mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot | \mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d | \cdot) \odot \log g_t^{\theta, d}(\cdot | \mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d | \mathbf{x}_0^d)} \right]
\end{aligned}$$

Notice that the second part of the loss can rewritten to

$$\begin{aligned}
& \mathbb{E}_{\mathbf{z}^{1:D} \sim q_{t|0}(\cdot|\mathbf{x}_0^{1:D})} \left[\frac{1}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D}|\mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot|\mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d|\mathbf{x}_0^d)} \right] \\
&= \mathbb{E}_t \sum_{\mathbf{z}^{1:D}} \left[\frac{\tilde{q}_{t|0}(\mathbf{z}^{1:D}|\mathbf{x}_0^{1:D})}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D}|\mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot|\mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d|\mathbf{x}_0^d)} \right] \text{ (now take Eq. (90) into)} \\
&= \mathbb{E}_t \sum_{\mathbf{z}^{1:D}} \left[q_{t|0}(\mathbf{z}^{1:D}|\mathbf{x}_0^{1:D}) \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot|\mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d|\mathbf{x}_0^d)} \right] \\
&= \mathbb{E}_{\mathbf{z}^{1:D} \sim q_{t|0}(\cdot|\mathbf{x}_0^{1:D})} \left[\sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot|\mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d|\mathbf{x}_0^d)} \right] \tag{91}
\end{aligned}$$

Taking Eq. (91) back into Eq. (26) we can get the following further simplified loss that only have a single forward pass of the model:

$$\begin{aligned}
& T \mathbb{E}_{\mathbf{x}^{1:D} \sim q_{t|0}(\cdot|\mathbf{x}_0^{1:D})} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D}) - \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot|\mathbf{x}_0^d) \odot r_t^d(\mathbf{x}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D})]}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \right] \\
&= T \mathbb{E}_{\mathbf{x}^{1:D} \sim q_{t|0}(\cdot|\mathbf{x}_0^{1:D})} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d|\cdot)^\top \left(g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D}) - \frac{q_{t|0}(\cdot|\mathbf{x}_0^d) \odot \log g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D})}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \right) \right] \tag{92}
\end{aligned}$$

□

A.16 Algorithm for Unified Training and Generation

Alg. 1 and Alg. 2 demonstrate our unified training and generation framework.

Algorithm 1 USD3 Unified Training: **Red: discrete-time step**, and **blue: continuous-time step**.

- 1: **Input:** A stationary distribution $\mathbf{m}$, samples $\sim p_{\text{data}}$, weight factor λ , **max time T**
 - 2: **repeat**
 - 3: Draw $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x}_0)$
 - 4: Draw $t \sim \text{Uniform}(0, \dots, T)$ or $t \sim \text{Uniform}(0, 1)$
 - 5: Compute $\bar{\alpha}_t$ from Table 3
 - 6: Draw $\mathbf{m}_0^{1:D} \sim \text{Cat}(\mathbf{m}_0^{1:D}; \mathbf{m})$
 - 7: $\mathbf{x}_{t|0}^{1:D} = \delta_{1, \mathbf{b}_t^{1:D}}^{1:D} \odot \mathbf{x}_0^{1:D} + (1 - \delta_{1, \mathbf{b}_t^{1:D}}^{1:D}) \odot \mathbf{m}_0^{1:D}$ where $\mathbf{b}_t \sim \text{Bernoulli}(\bar{\alpha}_t)$, from Eq. (49)
 - 8: Compute $f_t^\theta(\mathbf{x}_t^{1:D})$ as parameterization of $p_\theta(\mathbf{x}_0^{1:D}|\mathbf{x}_t^{1:D})$
 - 9: Take gradient descent step on $\nabla_\theta \{\mathcal{L}_t(\theta) + \lambda \mathcal{L}_t^{CE}(\theta)\}$ (from Eq. (11) + Eq. (12))
 - 10: or $\nabla_\theta \{\mathcal{L}_t^{CTMC}(\theta) + \lambda \mathcal{L}_t^{CE}(\theta)\}$ (from Eq. (23) + Eq. (12))
 - 11: **until convergence**
-

Algorithm 2 USD3 Unified Sampling

- 1: **Input:** A stationary distribution $\mathbf{m}$, $\{t_i\}_{i=0}^n$ s.t. $0 = t_0 < t_1 < \dots < t_n = T$, learned $f_{t_i}^\theta$.
 - 2: **MCMC Input:** use MCMC, step size Δn , total steps N .
 - 3: Set $\mathbf{x}_n^{1:D} \sim \text{Cat}(\mathbf{x}_n^{1:D}; \mathbf{m})$
 - 4: **for** $i \in \{n, \dots, 0\}$ **do**
 - 5: Compute $f_{t_i}^\theta(\mathbf{x}_i^{1:D})$ and $p_\theta(\mathbf{x}_{i-1}^{1:D}|\mathbf{x}_i^{1:D})$ from Eq. (10)
 - 6: Draw $\mathbf{x}_{i-1}^{1:D} \sim \text{Cat}(\mathbf{x}_{i-1}^{1:D}; p_\theta(\mathbf{x}_{i-1}^{1:D}|\mathbf{x}_i^{1:D}))$
 - 7: **If** use MCMC:
 - 8: $\mathbf{x}_{i-1}^{1:D} \leftarrow \text{MCMC_Corrector}(t_i, \mathbf{x}_{i-1}^{1:D}, \Delta n, f_{t_i}^\theta, N)$ (Algo. 3)
 - 9: **return** $\mathbf{x}_0$
-

请注意，损失的第二部分可以重写为

$$\begin{aligned}
& \mathbb{E}_{\mathbf{z}^{1:D} \sim q_{t|0}(\cdot|\mathbf{x}_0^{1:D})} \left[\frac{1}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D}|\mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot|\mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d|\mathbf{x}_0^d)} \right] \\
&= \mathbb{E}_t \sum_{\mathbf{z}^{1:D}} \left[\frac{\tilde{q}_{t|0}(\mathbf{z}^{1:D}|\mathbf{x}_0^{1:D})}{\mathcal{M}_{S_t}(\mathbf{z}^{1:D}|\mathbf{x}_0^{1:D})} \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot|\mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d|\mathbf{x}_0^d)} \right] \text{ (now take Eq. (90) into)} \\
&= \mathbb{E}_t \sum_{\mathbf{z}^{1:D}} \left[q_{t|0}(\mathbf{z}^{1:D}|\mathbf{x}_0^{1:D}) \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot|\mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d|\mathbf{x}_0^d)} \right] \\
&= \mathbb{E}_{\mathbf{z}^{1:D} \sim q_{t|0}(\cdot|\mathbf{x}_0^{1:D})} \left[\sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot|\mathbf{x}_0^d) \odot r_t^d(\mathbf{z}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\mathbf{z}^{1:D})]}{q_{t|0}(\mathbf{z}^d|\mathbf{x}_0^d)} \right] \tag{91}
\end{aligned}$$

将式(91)代入式(26)后，我们可以得到一个进一步简化的损失函数，该函数仅需模型单次前向传递即可计算：

$$\begin{aligned}
& T \mathbb{E}_{\mathbf{x}^{1:D} \sim q_{t|0}(\cdot|\mathbf{x}_0^{1:D})} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d|\cdot)^\top g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D}) - \sum_{d=1}^D \frac{\mathbf{1}^\top [q_{t|0}(\cdot|\mathbf{x}_0^d) \odot r_t^d(\mathbf{x}^d|\cdot) \odot \log g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D})]}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \right] \\
&= T \mathbb{E}_{\mathbf{x}^{1:D} \sim q_{t|0}(\cdot|\mathbf{x}_0^{1:D})} \left[\sum_{d=1}^D r_t^d(\mathbf{x}^d|\cdot)^\top \left(g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D}) - \frac{q_{t|0}(\cdot|\mathbf{x}_0^d) \odot \log g_t^{\theta,d}(\cdot|\mathbf{x}^{1:D})}{q_{t|0}(\mathbf{x}^d|\mathbf{x}_0^d)} \right) \right] \tag{92}
\end{aligned}$$

□

A.16 统一训练与生成算法

算法1和算法2展示了我们统一的训练和生成框架。

Algorithm 1 USD3 Unified Training: **Red: discrete-time step**, and **blue: continuous-time step**.

- 1: **Input:** A stationary distribution $\mathbf{m}$, samples $\sim p_{\text{data}}$, weight factor λ , **max time T**
 - 2: **repeat**
 - 3: Draw $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x}_0)$
 - 4: Draw $t \sim \text{Uniform}(0, \dots, T)$ or $t \sim \text{Uniform}(0, 1)$
 - 5: Compute $\bar{\alpha}_t$ from Table 3
 - 6: Draw $\mathbf{m}_0^{1:D} \sim \text{Cat}(\mathbf{m}_0^{1:D}; \mathbf{m})$
 - 7: $\mathbf{x}_{t|0}^{1:D} = \delta_{1, \mathbf{b}_t^{1:D}}^{1:D} \odot \mathbf{x}_0^{1:D} + (1 - \delta_{1, \mathbf{b}_t^{1:D}}^{1:D}) \odot \mathbf{m}_0^{1:D}$ where $\mathbf{b}_t \sim \text{Bernoulli}(\bar{\alpha}_t)$, from Eq. (49)
 - 8: Compute $f_t^\theta(\mathbf{x}_t^{1:D})$ as parameterization of $p_\theta(\mathbf{x}_0^{1:D}|\mathbf{x}_t^{1:D})$
 - 9: Take gradient descent step on $\nabla_\theta \{\mathcal{L}_t(\theta) + \lambda \mathcal{L}_t^{CE}(\theta)\}$ (from Eq. (11) + Eq. (12))
 - 10: or $\nabla_\theta \{\mathcal{L}_t^{CTMC}(\theta) + \lambda \mathcal{L}_t^{CE}(\theta)\}$ (from Eq. (23) + Eq. (12))
 - 11: **until convergence**
-

Algorithm 2 USD3 Unified Sampling

- 1: **Input:** A stationary distribution $\mathbf{m}$, $\{t_i\}_{i=0}^n$ s.t. $0 = t_0 < t_1 < \dots < t_n = T$, learned $f_{t_i}^\theta$.
 - 2: **MCMC Input:** use MCMC, step size Δn , total steps N .
 - 3: Set $\mathbf{x}_n^{1:D} \sim \text{Cat}(\mathbf{x}_n^{1:D}; \mathbf{m})$
 - 4: **for** $i \in \{n, \dots, 0\}$ **do**
 - 5: Compute $f_{t_i}^\theta(\mathbf{x}_i^{1:D})$ and $p_\theta(\mathbf{x}_{i-1}^{1:D}|\mathbf{x}_i^{1:D})$ from Eq. (10)
 - 6: Draw $\mathbf{x}_{i-1}^{1:D} \sim \text{Cat}(\mathbf{x}_{i-1}^{1:D}; p_\theta(\mathbf{x}_{i-1}^{1:D}|\mathbf{x}_i^{1:D}))$
 - 7: **If** use MCMC:
 - 8: $\mathbf{x}_{i-1}^{1:D} \leftarrow \text{MCMC_Corrector}(t_i, \mathbf{x}_{i-1}^{1:D}, \Delta n, f_{t_i}^\theta, N)$ (Algo. 3)
 - 9: **return** $\mathbf{x}_0$
-

A.17 Continuous-Time MCMC Sampling Corrector & Noise Scheduling

Campbell et al. [2022] show that the MCMC for discrete data can be done by a predictor step to simulate $\hat{r}_t^\theta$ and a corrector step using $r_t + \hat{r}_t^\theta$. We extend this result with improved derivation and show that, thanks to the shared parameterized form, *both* discrete- and continuous-time discrete diffusion can leverage the same transition probability calculation (see Eq. (98)), leading to a shared MCMC scheme, with detailed derivation provided in Appx. §A.17.

A.17.1 The MCMC Sampling Corrector

Song et al. [2021b] introduced a predictor-corrector step to further improve the quality of generated data based on score-based Markov Chain Monte Carlo (MCMC) for continuous-time diffusion over continuous distribution. Campbell et al. [2022] showed that there is a similar MCMC based corrector that can be used for CTMC to improve reverse sampling at any time t . Although we use different reverse sampling than Campbell et al. [2022], the similar corrector step can also be developed to improve the quality of reverse sampling introduced in §3.4. In this section, we derive the corrector formally and simplify it based the multi-element formulations summarized in Eq. (98).

Formally, at any time t , Campbell et al. [2022] proved that a *time-homogeneous* CTMC with transition rate being $c_t := r_t + \hat{r}_t$ has its stationary distribution being $q_t(\mathbf{x}_t^{1:D})$. To avoid ambiguity, we use $n \in [0, +\infty)$ as the time variable for that CTMC with stationary distribution $q_t(\mathbf{x}_t^{1:D})$. Then for any sample $\mathbf{z}_t^{1:D}$ generated from reverse sampling process at time t , we can push it closer to the target marginal distribution $q_t(\mathbf{x}_t^{1:D})$ by sampling from the corrector CTMC with initial value being $\mathbf{z}_t^{1:D}$, named as $\mathbf{z}_{t,n=0}^{1:D}$. Let N be the maximum time allocated in the corrector CTMC, then after the corrector step $\mathbf{z}_{t,n=N}^{1:D}$ is used to replace the original $\mathbf{z}_t^{1:D}$.

We now introduce how to sampling from the CTMC. Let Δn be the time incremental for each sampling step of the corrector CTMC. Solving the Kolmogorov forward equation of this time-homogeneous CTMC can derive the transition probability at any time n as

$$\forall n \text{ and } \Delta n, p_{n+\Delta n|n}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) = \exp(\Delta n \cdot C_t^{1:D})[\mathbf{x}^{1:D}, \mathbf{y}^{1:D}] \quad (93)$$

Where C_t is the transition rate matrix of the corrector CTMC, and $\exp(\Delta n \cdot C_t^{1:D})$ is the transition probability matrix at time n . Notice that this matrix exponential does not have analytical formulation. Instead, we propose to control Δn to be small enough such that

$$\exp(\Delta n \cdot C_t^{1:D}) \approx I + \Delta n \cdot C_t^{1:D} \quad (94)$$

Then taking it back we can obtain

$$\begin{aligned} p_{n+\Delta n|n}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) &\approx \delta_{\mathbf{x}^{1:D}, \mathbf{y}^{1:D}} + \Delta n \cdot C_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) \\ &= \delta_{\mathbf{x}^{1:D}, \mathbf{y}^{1:D}} + \Delta n \sum_{d=1}^D \left[r_t^d(\mathbf{y}^d|\mathbf{x}^d) + r_t^d(\mathbf{x}^d|\mathbf{y}^d) \cdot g_t^d(\mathbf{y}^d|\mathbf{x}^{1:D}) \right] \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \end{aligned} \quad (95)$$

Instead of sampling all elements jointly, we propose to sample each element of the object independently from their individual marginal distribution, which can be analytically formulated as

$$\begin{aligned} p_{n+\Delta n|n}(\mathbf{y}^d|\mathbf{x}^{1:D}) &= \sum_{\mathbf{y}^{\setminus d}} p_{n+\Delta n|n}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) \\ &= \Delta n \left[r_t^d(\mathbf{y}^d|\mathbf{x}^d) + r_t^d(\mathbf{x}^d|\mathbf{y}^d) \cdot g_t^d(\mathbf{y}^d|\mathbf{x}^{1:D}) \right] \text{ if } \mathbf{y}^d \neq \mathbf{x}^d \\ &= \Delta n (\mathbf{y}^d)^\top \left[r_t^d(\cdot|\mathbf{x}^d) + r_t^d(\mathbf{x}^d|\cdot) \odot g_t^d(\cdot|\mathbf{x}^{1:D}) \right] \text{ if } \mathbf{y}^d \neq \mathbf{x}^d \\ &= \Delta n \beta(t)(\mathbf{y}^d)^\top \left[\mathbf{m}^d - \mathbf{x}^d + (\langle \mathbf{x}^d, \mathbf{m}^d \rangle \mathbf{1} - \mathbf{x}^d) \odot g_t^d(\cdot|\mathbf{x}^{1:D}) \right] \text{ if } \mathbf{y}^d \neq \mathbf{x}^d \\ &= \Delta n \beta(t)(\mathbf{y}^d)^\top \left[\mathbf{m}^d + \langle \mathbf{x}^d, \mathbf{m}^d \rangle g_t^d(\cdot|\mathbf{x}^{1:D}) \right] \text{ if } \mathbf{y}^d \neq \mathbf{x}^d \\ &\approx \Delta n \beta(t)(\mathbf{y}^d)^\top \left[\mathbf{m}^d + \langle \mathbf{x}^d, \mathbf{m}^d \rangle g_t^{\theta, d}(\cdot|\mathbf{x}^{1:D}) \right] \text{ if } \mathbf{y}^d \neq \mathbf{x}^d \end{aligned} \quad (96)$$

A.17 连续时间 MCMC 采样校正器&噪声调度

Campbell等人 [2022] 表明，用于离散数据的MCMC可以通过预测步骤来模拟 $\hat{r}_t^\theta$ ，并使用 $r_t + \hat{r}_t^\theta$ 进行校正步骤。我们通过改进的推导扩展了这一结果，并表明，由于共享的参数化形式，离散时间和连续时间的离散扩散可以利用相同的转移概率计算（参见公式(98)），从而得到一个共享的MCMC方案，详细的推导过程在附录 § A.17中提供。

A.17.1 MCMC 采样校正器

Song 等人 [2021b]为基于评分的连续时间扩散连续分布的连续时间 MCMC 引入了一个预测-校正步骤，以进一步提高生成数据的质量。Campbell 等人. [2022] 表明存在一个类似的 MCMC 校正器，可用于 CTMC 以在任何时间 t 改进逆向采样。尽管我们使用的逆向采样与 Campbell 等人. [2022], 不同，但类似的校正步骤也可以开发出来，以改进 § 3.4 中引入的逆向采样质量。在本节中，我们正式推导校正器，并根据 Eq. (98) 总结的多元素公式进行简化。

正式地，在任何时间 t ，Campbell 等人. [2022] 证明了具有转换率 $c_t := r_t + \hat{r}_t$ 的时间齐次 CTMC 的平稳分布为 $q_t(\mathbf{x}_t^{1:D})$ 。为了避免歧义，我们使用 $n \in [0, +\infty)$ 作为具有平稳分布 $q_t(\mathbf{x}_t^{1:D})$ 的该 CTMC 的时间变量。然后，对于从时间 t 生成的任何样本 $\mathbf{z}_t^{1:D}$ ，我们可以通过从初始值为 $\mathbf{z}_t^{1:D}$ 、命名为 $\mathbf{z}_{t,n=0}^{1:D}$ 的校正器 CTMC 中采样，将其推向目标边缘分布 $q_t(\mathbf{x}_t^{1:D})$ 。令 N 为校正器 CTMC 中分配的最大时间，然后在校正步骤 $\mathbf{z}_{t,n=N}^{1:D}$ 之后，用其替换原始 $\mathbf{z}_t^{1:D}$ 。

我们现在介绍如何从CTMC中进行采样。设 Δn 为校正CTMC每次采样步骤的时间增量。求解该时齐CTMC的Kolmogorov前向方程可以推导出在任何时间 n 的转移概率为

$$\forall n \text{ and } \Delta n, p_{n+\Delta n|n}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) = \exp(\Delta n \cdot C_t^{1:D})[\mathbf{x}^{1:D}, \mathbf{y}^{1:D}] \quad (93)$$

其中 C_t 是校正CTMC的转移率矩阵， $\exp(\Delta n \cdot C_t^{1:D})$ 是时间 n 的转移概率矩阵。请注意，该矩阵指数没有解析公式。相反，我们建议控制 Δn 足够小，使得

$$\exp(\Delta n \cdot C_t^{1:D}) \approx I + \Delta n \cdot C_t^{1:D} \quad (94)$$

然后将其代回，我们可以得到

$$\begin{aligned} p_{n+\Delta n|n}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) &\approx \delta_{\mathbf{x}^{1:D}, \mathbf{y}^{1:D}} + \Delta n \cdot C_t^{1:D}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) \\ &= \delta_{\mathbf{x}^{1:D}, \mathbf{y}^{1:D}} + \Delta n \sum_{d=1}^D \left[r_t^d(\mathbf{y}^d|\mathbf{x}^d) + r_t^d(\mathbf{x}^d|\mathbf{y}^d) \cdot g_t^d(\mathbf{y}^d|\mathbf{x}^{1:D}) \right] \delta_{\mathbf{x}^{\setminus d}, \mathbf{y}^{\setminus d}} \end{aligned} \quad (95)$$

与其联合采样所有元素，我们提出从每个元素的独立边缘分布中独立采样对象中的每个元素，这可以解析地表示为

$$\begin{aligned} p_{n+\Delta n|n}(\mathbf{y}^d|\mathbf{x}^{1:D}) &= \sum_{\mathbf{y}^{\setminus d}} p_{n+\Delta n|n}(\mathbf{y}^{1:D}|\mathbf{x}^{1:D}) \\ &= \Delta n \left[r_t^d(\mathbf{y}^d|\mathbf{x}^d) + r_t^d(\mathbf{x}^d|\mathbf{y}^d) \cdot g_t^d(\mathbf{y}^d|\mathbf{x}^{1:D}) \right] \text{ if } \mathbf{y}^d \neq \mathbf{x}^d \\ &= \Delta n (\mathbf{y}^d)^\top \left[r_t^d(\cdot|\mathbf{x}^d) + r_t^d(\mathbf{x}^d|\cdot) \odot g_t^d(\cdot|\mathbf{x}^{1:D}) \right] \text{ if } \mathbf{y}^d \neq \mathbf{x}^d \\ &= \Delta n \beta(t)(\mathbf{y}^d)^\top \left[\mathbf{m}^d - \mathbf{x}^d + (\langle \mathbf{x}^d, \mathbf{m}^d \rangle \mathbf{1} - \mathbf{x}^d) \odot g_t^d(\cdot|\mathbf{x}^{1:D}) \right] \text{ if } \mathbf{y}^d \neq \mathbf{x}^d \\ &= \Delta n \beta(t)(\mathbf{y}^d)^\top \left[\mathbf{m}^d + \langle \mathbf{x}^d, \mathbf{m}^d \rangle g_t^d(\cdot|\mathbf{x}^{1:D}) \right] \text{ if } \mathbf{y}^d \neq \mathbf{x}^d \\ &\approx \Delta n \beta(t)(\mathbf{y}^d)^\top \left[\mathbf{m}^d + \langle \mathbf{x}^d, \mathbf{m}^d \rangle g_t^{\theta, d}(\cdot|\mathbf{x}^{1:D}) \right] \text{ if } \mathbf{y}^d \neq \mathbf{x}^d \end{aligned} \quad (96)$$

Now we define the notation $\overline{p_{\Delta n}(\cdot|\mathbf{x}^{1:D})}$ to derive the distributional form of $p_{n+\Delta n|n}(\mathbf{y}^d|\mathbf{x}^{1:D})$.

$$\overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})} := \Delta n \beta(t) \left[\left(2 - \frac{\bar{\alpha}_{t|0} \langle f_t^{\theta,d}(\mathbf{x}^{1:D}), \mathbf{x}^d \rangle}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle \mathbf{x}^d, \mathbf{m}^d \rangle} \right) \mathbf{m}^d + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} f_t^{\theta,d}(\mathbf{x}^{1:D}) \right] \odot (\mathbf{1} - \mathbf{x}^d) \quad (97)$$

With the above notation, the sampling probability can be further simplified as

$$p_{n+\Delta n|n}(\mathbf{y}^d|\mathbf{x}^{1:D}) = \text{Cat}\left(\mathbf{y}^d; \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})} + (1 - \mathbf{1}^\top \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})}) \mathbf{x}^d\right) \quad (98)$$

Notice that Δn should be set small enough such that $\mathbf{1}^\top \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})} \leq 1$. This condition can be used to derive Δn dynamically. In practice, we can also easily clip the scale of $\overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})}$ to 1 when $\mathbf{1}^\top \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})} > 1$ to prevent illness condition. Intuitively, $1 - \mathbf{1}^\top \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})}$ defines the keeping rate of the d -th element during correction step, and it should be larger with increasing t and n during the reverse sampling and correction period.

Algorithm 3 The MCMC correcting algorithm at time t

Input: The sample at time t from reverse sampling, $\mathbf{z}_t^{1:D}$; step size Δn ; learned f_t^θ ; total steps N .

Initialize $\mathbf{x}_{t,0}^{1:D} \leftarrow \mathbf{z}_t^{1:D}$

for i from 1 to N **do**

 Compute $\overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}_{t,i-1}^{1:D})}$ from Eq. (97);

$\forall d$, Draw $\mathbf{x}_{t,i}^d \sim \text{Cat}\left(\cdot; \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}_{t,i-1}^{1:D})} + (1 - \mathbf{1}^\top \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}_{t,i-1}^{1:D})}) \mathbf{x}_{t,i-1}^d\right)$

end for

Output: the improved (corrected) sample $\mathbf{x}_{t,N}^{1:D}$

A.17.2 Noise Scheduling in Training

Notice that $\beta(t) = -\frac{1}{\bar{\alpha}_t} \cdot \frac{d\bar{\alpha}_t}{dt}$ as $\bar{\alpha}_{t|s} = \exp(-\int_s^t \beta(a) da)$. We now first present a general way to design the scheduling of $\bar{\alpha}_t$ based on Chen [2023]. For any continuous function $h(t)$, we define $\bar{\alpha}_t$ as the follows that satisfies $\bar{\alpha}_0 = 1$ and $\bar{\alpha}_T = 0$:

$$\bar{\alpha}_t = \frac{h(T) - h(t)}{h(T) - h(0)} \quad (99)$$

We can easily derive that

$$\beta(t) = -\frac{1}{\bar{\alpha}_t} \cdot \frac{h'(t)}{h(0) - h(T)} = \frac{h'(t)}{h(T) - h(t)} \quad (100)$$

$$\bar{\alpha}_{t|s} = \exp\left(-\int_s^t \frac{dh(t)}{h(T) - h(t)}\right) = \exp\left(\int_s^t \frac{d(h(T) - h(t))}{h(T) - h(t)}\right) = \frac{h(T) - h(t)}{h(T) - h(s)} \quad (101)$$

Based on the above general formulation, we now present some widely used noise schedules

Table 3: Widely used noise scheduling with maximum time T

Type	Param.	Reference	$h(t)$	$\bar{\alpha}_t$	$\beta(t)$	$\bar{\alpha}_{t s}$
Cosine	$a = 0.008$	[Hoogeboom et al., 2021]	$\cos(\frac{t/T+a}{1+a} \frac{\pi}{2})$	$\frac{h(t)}{h(0)}$	$\frac{\pi \tan(\frac{t/T+a}{1+a} \frac{\pi}{2})}{2T(1+a)}$	$\frac{h(t)}{h(s)}$
Linear	-	[Ho et al., 2020a]	t	$1 - \frac{t}{T}$	$\frac{1}{T-t}$	$\frac{T-t}{T-s}$
Exp.	a, b	[Campbell et al., 2022]	-	$\exp(Ta(1 - b^{\frac{t}{T}}))$	$ab^{\frac{t}{T}} \log b$	$\exp(Ta(b^{\frac{s}{T}} - b^{\frac{t}{T}}))$

A.18 Experiment Details

A.18.1 Lakh Piano Dataset Details

The Lakh pianoroll dataset contains 6,000 training and 973 evaluating piano sequences, with each music sequence spanning a length of 256 in total. Each music note in the sequence can take on a value

现在我们定义符号 $p_{\Delta n}(\cdot|\overline{\mathbf{x}^{1:D}})$ 来推导 $p_{n+\Delta n|n}(\mathbf{y}^d|\mathbf{x}^{1:D})$ 的分布形式。

$$\overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})} := \Delta n \beta(t) \left[\left(2 - \frac{\bar{\alpha}_{t|0} \langle f_t^{\theta,d}(\mathbf{x}^{1:D}), \mathbf{x}^d \rangle}{\bar{\alpha}_{t|0} + (1 - \bar{\alpha}_{t|0}) \langle \mathbf{x}^d, \mathbf{m}^d \rangle} \right) \mathbf{m}^d + \frac{\bar{\alpha}_{t|0}}{1 - \bar{\alpha}_{t|0}} f_t^{\theta,d}(\mathbf{x}^{1:D}) \right] \odot (\mathbf{1} - \mathbf{x}^d) \quad (97)$$

根据上述符号，采样概率可以进一步简化为

$$p_{n+\Delta n|n}(\mathbf{y}^d|\mathbf{x}^{1:D}) = \text{Cat}\left(\mathbf{y}^d; \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})} + (1 - \mathbf{1}^\top \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})}) \mathbf{x}^d\right) \quad (98)$$

请注意， Δn 应设置得足够小，使得 $\mathbf{1}^\top \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})} \leq 1$ ，这个条件可用于动态推导 Δn 。在实践中，我们还可以轻松地将 $\overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})}$ 的尺度裁剪到1，当 $\mathbf{1}^\top \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})} > 1$ ，以防止病态条件。直观上， $1 - \mathbf{1}^\top \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}^{1:D})}$ ，定义了修正步骤中保持第 d 个元素的比例，并且随着反向采样和修正期间 t 和 n 的增加而增大。

Algorithm 3 The MCMC correcting algorithm at time t

Input: The sample at time t from reverse sampling, $\mathbf{z}_t^{1:D}$; step size Δn ; learned f_t^θ ; total steps N .

Initialize $\mathbf{x}_{t,0}^{1:D} \leftarrow \mathbf{z}_t^{1:D}$

for i from 1 to N **do**

 Compute $\overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}_{t,i-1}^{1:D})}$ from Eq. (97);

$\forall d$, Draw $\mathbf{x}_{t,i}^d \sim \text{Cat}\left(\cdot; \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}_{t,i-1}^{1:D})} + (1 - \mathbf{1}^\top \overline{p_{\Delta n}^{\theta,d}(\cdot|\mathbf{x}_{t,i-1}^{1:D})}) \mathbf{x}_{t,i-1}^d\right)$

end for

Output: the improved (corrected) sample $\mathbf{x}_{t,N}^{1:D}$

A.17.2 训练中的噪声调度

请注意， $\beta(t) = -\frac{1}{\bar{\alpha}_t} \cdot \frac{d\bar{\alpha}_t}{dt}$ as $\bar{\alpha}_{t|s} = \exp(-\int_s^t \beta(a) da)$ 。我们现在首先介绍一种基于Chen [2023]的 α_t 调度的一般方法。对于任何连续函数 $h(t)$ ，我们定义 α_t 如下，使其满足 $\alpha_0 = 1$ 和 $\alpha_T = 0$ ：

$$\bar{\alpha}_t = \frac{h(T) - h(t)}{h(T) - h(0)} \quad (99)$$

我们可以很容易地推导出

$$\beta(t) = -\frac{1}{\bar{\alpha}_t} \cdot \frac{h'(t)}{h(0) - h(T)} = \frac{h'(t)}{h(T) - h(t)} \quad (100)$$

$$\bar{\alpha}_{t|s} = \exp\left(-\int_s^t \frac{dh(t)}{h(T) - h(t)}\right) = \exp\left(\int_s^t \frac{d(h(T) - h(t))}{h(T) - h(t)}\right) = \frac{h(T) - h(t)}{h(T) - h(s)} \quad (101)$$

基于上述通用公式，我们现在介绍一些广泛使用的噪声调度方案

表3：具有最大时间T的广泛使用的噪声调度方案

Type	参数。	参考	$h(t)$	$\bar{\alpha}_t$	$\beta(t)$	$\bar{\alpha}_{t s}$
余弦	$a = 0.008$	[Hoogeboom 等人, 2021]	$\cos(\frac{t/T+a}{1+a} \frac{\pi}{2})$	$\frac{h(t)}{h(0)}$	$\frac{\pi \tan(\frac{t/T+a}{1+a} \frac{\pi}{2})}{2T(1+a)}$	$\frac{h(t)}{h(s)}$
线性	-	[Ho 等人, 2020a]	t	$1 - \frac{t}{T}$	$\frac{1}{T-t}$	$\frac{T-t}{T-s}$
Exp.	a, b	[Campbell 等人, 2022]	-	$\exp(Ta(1 - b^{\frac{t}{T}}))$	$ab^{\frac{t}{T}} \log b$	$\exp(Ta(b^{\frac{s}{T}} - b^{\frac{t}{T}}))$

A.18 实验详情

A.18.1 Lakh 钢琴数据集详情

Lakh pianoroll 数据集包含 6,000 个训练和 973 个评估的钢琴序列，每个音乐序列总长度为 256。序列中的每个音乐音符可以取 128 个音乐音符中的任一值

of the 128 music notes plus 1 additional class meaning an empty note. The music note orderings are scrambled into the same random order as described in [Campbell et al. \[2022\]](#) such that the ordinal structure of the music notes are destroyed.

For evaluation, the first 32 notes of the 967 evaluation sequences are given to the model, while the model is asked to generate the resting 224 notes. Upon an analysis of the training and evaluation music sequences, we find that a total of 124 evaluation samples can be found with at least one matching training samples that have the same 32 dimensions. We separate out these samples and call the set PIANO-P. Among PIANO-P, 20 samples can be found to contain the same first 32 notes with 2 samples from the training sequences, three of them contain the same first 32 notes with 4 training samples each, one of them shares the same with 6 training samples, and one shares the same with 8 training samples. If the same 32 notes appear both in the training samples and as quests for the model to provide the inferences, the model is likely to directly memorizing the remaining 224 notes from the training set, and "parroting" music sequences according to the training samples. The rest 843 evaluation samples that do not have matching training samples are constituent of PIANO.

A.18.2 Pre-training VQGAN

To generate images in a categorical discrete latent space, we follow the implementation of VQGAN [Esser et al. \[2020\]](#) in MaskGIT [Chang et al. \[2022\]](#). Specifically, we use the same VQGAN setting as mentioned in [Sun et al. \[2023\]](#). VQGAN is a variant of Vector Quantized Variational Autoencoder (VQ-VAE) by [van den Oord et al. \[2018\]](#). In our setup for CIFAR10, a VQGAN encodes an image of shape $H \times W \times 3$ to $(\frac{H}{4} \times \frac{W}{4})$ tokens with vocabulary size of 512. For the encoder, we use three convolutional blocks, with filter sizes of 64, 128 and 256, and an average pooling between each blocks. For each block, it consists of two residual blocks. After an image is encoded, the output is mapped to a token index with a codebook of 512×256 . For VQGAN loss objective, the additional GAN loss and perceptual loss are added with weight 0.1. To train a general VQGAN model that allows us to embed CIFAR10 images without overfitting, we apply data augmentation (random flipping and cropping) to the 64×64 version of ImageNet dataset [[Deng et al. \[2009\]](#)], and train for 90 epochs. The VQGAN is trained with Adam optimizer ($\beta_1 = 0, \beta_2 = 0.99$), and the learning rate is linearly warmup to the peak of $1e^{-4}$ and then drops with cosine decay.

After VQGAN is trained, we freeze the VQGAN and apply encoder to the CIFAR10 images to create 8×8 latent codes. The latent codes then flattened to a vector of size 64 as the input of the diffusion model. Before we evaluate our diffusion methods, we will feed the generated latent codes back to the VQGAN decoder to reconstruct a sample in the image spaces. We test the effectiveness of VQGAN by trying to reconstruct the CIFAR10 dataset. The reconstruction gives a FID of 7.68 and IS of 10.42, using the Inception V3 Model⁴.

A.18.3 Music Generation Eval Metrics

In evaluation of the conditional music generation task, we apply the following metrics to measure generation quality:

- 1-gram Hellinger Distance ($\downarrow$) and 1-gram Proportion of Outliers ($\downarrow$): for these two metrics, we following the same evaluation as described in [Campbell et al. \[2022\]](#) and [Chen \[2023\]](#).
- $\{2, 3\}$ -gram Hellinger Distance ($\downarrow$) and $\{2, 3\}$ -gram Proportion of Outliers ($\downarrow$): similar to n-gram models, we first convert the music sequences into tuples of neighboring nodes. Then for Hellinger Distance, we compute the distance of the empirical probabilities of conditional generated samples to the ground truth samples. The empirical probabilities are constructed based on the histograms of the neighboring tuples, with bins being all possible $\{2, 3\}$ -gram nodes. Similarly, for the Proportion of Outliers, we count the fractions of newly appeared tuples that are not seen in the training samples. With these metrics, we are able to capture the sequence information instead of just measuring the single node distributions.
- Diverse Edit Distance ($\uparrow$): which accounts for the creativity/novelty of generated samples across multiple generation runs. For the conditionally generated music samples given the same first 32 notes, we calculate the edit distance, which is the minimum number of single-character edits (insertions, deletions, or substitutions) required to change one music sequence into the other, between each two of the generation samples. The mean and standard deviation are obtained for all edit distances between pairs. The higher the diverse edit distance, the further apart the music

加上 1 个额外的类别，表示空音符。音乐音符的顺序被打乱，与 Campbell 等人描述的随机顺序相同 [\[2022\]](#)，从而破坏了音乐音符的序数结构。

为了评估，前32个音符被提供给模型用于评估的967个序列，同时要求模型生成剩余的224个音符。在对训练和评估音乐序列进行分析后，我们发现共有124个评估样本至少与一个具有相同32维度的训练样本匹配。我们将这些样本分离出来，并将该集合称为PIANO-P。在PIANO-P中，有20个样本包含与训练序列中的2个样本相同的最初32个音符，其中3个样本包含与训练序列中的4个样本相同的最初32个音符，1个样本与6个训练样本共享相同的最初32个音符，还有1个样本与8个训练样本共享相同的最初32个音符。如果相同的32个音符既出现在训练样本中，又作为模型需要提供推理的任务出现，模型可能会直接从训练集中记忆剩余的224个音符，并根据训练样本“鹦鹉学舌”般地生成音乐序列。其余843个没有匹配训练样本的评估样本构成了PIANO集合。

A.18.2 预训练 VQGAN

在分类离散潜在空间中生成图像时，我们遵循 MaskGIT Chang 等人 [\[2022\]](#) 中 VQGAN Esser 等人 [\[2020\]](#) 的实现。具体来说，我们使用 Sun 等人 [\[2023\]](#) 中提到的相同 VQGAN 设置。VQGAN 是 van den Oord 等人 [\[2018\]](#) 提出的向量量化变分自动编码器 (VQ-VAE) 的一个变体。在我们的 CIFAR10 设置中，一个 VQGAN 将形状为 $H \times W \times 3$ 的图像编码为 $(\frac{H}{4} \times \frac{W}{4})$ 个词汇量为 512 的标记。对于编码器，我们使用三个卷积块，其滤波器大小分别为 64、128 和 256，并且每个块之间进行平均池化。

对于每个块，它由两个残差块组成。图像编码后，输出被映射到具有 512×256 代码簿的 token 索引。对于 VQGAN 损失目标，额外的 GAN 损失和感知损失以 0.1 的权重被添加。为了训练一个通用的 VQGAN 模型，使我们能够嵌入 CIFAR10 图像而不过拟合，我们将数据增强（随机翻转和裁剪）应用于 64×64 版本的 ImageNet 数据集 [[Deng et al. \[2009\]](#)]，并训练 90 个 epoch。VQGAN 使用 Adam 优化器 ($\beta_1 = 0, \beta_2 = 0.99$) 进行训练，学习率线性预热到 $1e^{-4}$ 的峰值，然后以余弦衰减下降。

VQGAN 训练完成后，我们冻结 VQGAN 并将编码器应用于 CIFAR10 图像以创建 8×8 潜在代码。这些潜在代码随后被展平为一个大小为 64 的向量，作为扩散模型的输入。在评估我们的扩散方法之前，我们将生成的潜在代码反馈给 VQGAN 解码器，以在图像空间中重建样本。我们通过尝试重建 CIFAR10 数据集来测试 VQGAN 的有效性。重建结果使用 Inception V3 模型⁴ 得到 FID 为 7.68，IS 为 10.42。

A.18.3 音乐生成评估指标

在条件音乐生成任务的评估中，我们应用以下指标来衡量生成质量：

- 1-gram Hellinger 距离 ($\downarrow$) 和 1-gram 异常值比例 ($\downarrow$): 对于这两个指标，我们遵循 Campbell 等人 [\[2022\]](#) 和 Chen [\[2023\]](#) 中描述的评估方法。
- $\{2, 3\}$ -gram Hellinger 距离 ($\downarrow$) 和 $\{2, 3\}$ -gram 异常值比例 ($\downarrow$): 与 n-gram 模型类似，我们首先将音乐序列转换为相邻节点的元组。然后对于 Hellinger 距离，我们计算条件生成样本的经验概率与真实样本的经验概率之间的距离。经验概率基于相邻元组的直方图构建，其中 bin 包含所有可能的 $\{2, 3\}$ -gram 节点。类似地，对于异常值比例，我们统计新出现且在训练样本中未出现的元组的比例。通过这些指标，我们能够捕捉序列信息，而不仅仅是测量单个节点的分布。
- 多样性编辑距离 ($\uparrow$): 用于衡量在不同生成运行中生成样本的创造性/新颖性。对于给定相同前32个音符的条件生成音乐样本，我们计算编辑距离，即改变一个音乐序列为另一个所需的最少单字符编辑次数（插入、删除或替换），在每两个生成样本之间。对所有成对编辑距离的均值和标准差进行计算。多样性编辑距离越高，音乐就越远

⁴https://github.com/openai/consistency_models/tree/main/evaluations

⁴https://github.com/openai/consistency_models/tree/main/evaluations

sequences are and more creativity are enforced in the generation process. However, there is also a trade-off between diverse edit distance and other accuracy measurements when the model processes large uncertainty about the underlying distribution.

- Train-to-Test Ratios ($\uparrow$) for $\{1, 2, 3\}$ -gram Hellinger as well as Proportion of Outliers, which compare the weighted distance of a generated sample to its evaluation (test) vs. training sequence that share the same first 32 notes. We evaluate the ratios only for PIANO-P. Denote the evaluation ground truth set in PIANO-P as tr , the corresponding set of training samples as ts , and conditionally generated samples as gs . For the selected distance metrics $dist()$ (from n-gram Hellinger or n-gram Proportion of Outliers), we calculate the ratio as: $\frac{1}{dist(tr,gs)+dist(ts,gs)} * \frac{dist(tr,gs)}{dist(ts,gs)}$. Such ratio measures quantify the extent of “parroting” in PIANO-P. The larger the ratio, the more equally distant the generated examples is from its training and evaluating set, and the less "parroting" occurs by simply memorizing all training sequences. We apply an additional coefficient to the ratio such that it will also penalize the models that provide unrealistic examples that do not conform both training and evaluation distributions.

A.18.4 Training Details

USD3 Lakh Pianoroll Training Details.

For the backbone sequence transformer structure, we adopt a similar transformer model as utilized by in τ -LDR-0 Campbell et al. [2022]. The sequence transformer is composed of several encoder blocks, where for each internal block, time is fed into the block through a FiLM layer and added to the input. The result is then fed into a multi-headed self-attention layer and fully connected layer. We use RELU for activation. At the output of self-attention layer and fully-connected layer, a dropout of 0.1 is applied. After obtaining the final embedding of each token, we feed that into a 1-block ResNet to obtain the predicted logits. In comparison with other baseline metrics, the transformer contains 6 encoder blocks, each containing 16 attention heads, input dimension of 1024 and MLP dimension of 4096. In ablation study, we also test our methods on a smaller architecture that contains 6 encoder blocks, each containing 8 attention heads, with input dimension of 128 and MLP dimension of 1024.

For training the pianoroll dataset, we use a batch size of 64, a learning rate of $5e^{-4}$, with a warmup of first 25 epochs. We adopt a constant learning rate decay scheduler, decaying the learning rate by half after every 500 epochs. The final result is given over 3000 epochs. We run our results with 2 A6000 GPUs. In discrete-time diffusion, we sample 1000 number of timesteps, fixing a cosine scheduler with $\alpha = 0.008$, In continuous-time diffusion, we sample time t between $[0, 1]$ and apply a constant scheduler with rate equals 0.007. We maintain an exponential moving average of parameters with decay factor 0.9999. We clip the gradient norm at a norm value of 1.0.

Baseline Training Details.

For **D3PM** and τ -LDR-0, for fair comparison, we use the same architecture, diffusion scheme and training scheme. For calculating the loss of both methods, we follow the previous literature and give 0.001 to CE loss, and 1 to the VLB loss. For **RDM**, we use the official implementation of loss function in reparam multinomial diffusion⁵. We keep the same neural network architecture, and apply reweighting = 1.0, with no label smoothing, and let sampling strategy be cmlm-dep. Similar to D3PM and USD3, we apply cosine scheduler with uniform noise. We remove the padding functions and sampling temperatures when adapting RDM from language tasks to music and image generation.

For **SEDD**, we follow the official code implementation with uniform noise and loss functions⁶. While keeping the same neural architectures as D3PM and USD3, we add an additional masking to the output (the same masking as in SEDD’s diffusion transformer), since SEDD models the ratio of two probabilities instead of direct log probabilities. We use log-linear noise schedules with $1e^{-3}$.

For **SDDM**, since it adopts a different architecture, directly utilizing SDDM with our experiment configurations is not feasible. Instead, we use the experiment configurations as given in the paper: the backbone structure is a hollow transformer Sun et al. [2023]. Each transformer block has 6 layers with embedding size of 256, 8 attention heads and hidden dimension of 2048. The batch size is 64 and number of training steps is 2 million. The weight decay is set to $1e^{-6}$. The learning rate is at constant $1e^{-3}$. For diffusion, SDDM adopts the constant noise scheduler with a uniform rate constant of 0.04.

⁵<https://github.com/HKUNLP/reparam-discrete-diffusion>

⁶<https://github.com/louaaron/Score-Entropy-Discrete-Diffusion>

生成过程中强制要求序列多样性和更多创造力。然而，当模型处理关于底层分布的大不确定性时，多样编辑距离与其他准确度测量之间也存在权衡。

- $\uparrow$ 的 $\{1, 2, 3\}$ -gram Hellinger以及异常值比例，用于比较生成的样本与其评估（测试）序列和共享前32个音符的训练序列之间的加权距离。我们仅对PIANO-P评估这些比率。将PIANO-P中的评估真实值集记为tr，相应的训练样本集记为ts，条件生成的样本记为gs。对于选定的距离度量dist()（来自n-gram Hellinger或n-gram异常值比例），我们计算比率为： $\frac{1}{dist(tr,gs)+dist(ts,gs)} * \frac{dist(tr,gs)}{dist(ts,gs)}$ 。该比率量化了PIANO-P中“鹦鹉学舌”的程度。比率越大，生成的示例与其训练和评估集的距离越相等，通过简单记忆所有训练序列就减少发生“鹦鹉学舌”。我们对比率应用了一个附加系数，使其也能惩罚那些提供不符合训练和评估分布的不切实际样本的模型。

A.18.4 训练详情

USD3 万美元 Pianoroll 训练详情。

对于骨干序列Transformer结构，我们采用与 τ -LDR-0 Campbell等人 [2022]使用的类似Transformer模型。序列Transformer由多个编码器块组成，其中每个内部块通过FiLM层将时间输入到块中，并添加到输入中。结果然后输入到多头自注意力层和全连接层。我们使用RELU作为激活函数。在自注意力层和全连接层的输出处，应用了0.1的dropout。在获得每个标记的最终嵌入后，我们将其输入到1块ResNet中以获得预测的logits。与其他基线指标相比，Transformer包含6个编码器块，每个块包含16个注意力头，输入维度为1024，MLP维度为4096。在消融研究中，我们还测试了我们的方法在一个更小的架构上，该架构包含6个编码器块，每个块包含8个注意力头，输入维度为128，MLP维度为1024。

在预训练钢琴卷数据集中，我们使用批大小为64，学习率为 $5e^{-4}$ ，并设置预热阶段为前25个epoch。我们采用恒定学习率衰减调度器，每500个epoch将学习率衰减一半。最终结果在3000个epoch后给出。我们使用2个A6000 GPU运行结果。在离散时间扩散中，我们采样1000个时间步，固定余弦调度器为 $\alpha = 0.008$ 。在连续时间扩散中，我们采样时间 t ，范围在 $[0, 1]$ 之间，并应用恒定调度器，速率等于0.007。我们保持参数的指数移动平均值，衰减因子为0.9999。我们将梯度范数裁剪在1.0的范数值上。

基线训练详情。

对于D3PM和 τ -LDR-0，为了公平比较，我们使用相同的架构、扩散方案和训练方案。在计算两种方法的损失时，我们遵循先前文献，将CE损失设为0.001，将VLB损失设为1。对于RDM，我们使用重参数多项式扩散⁵中的官方损失函数实现。我们保持相同的神经网络架构，并应用重加权 = 1。0，不进行标签平滑，并让采样策略为cmlm-dep。类似于D3PM和USD3，我们应用余弦调度器，并添加均匀噪声。在将RDM从语言任务适配到音乐和图像生成时，我们移除了填充函数和采样温度。

对于SEDD，我们遵循官方代码实现，使用均匀噪声和损失函数⁶。在保持与D3PM和USD3相同的神经架构的同时，我们向输出添加了额外的掩码（与SEDD的扩散Transformer中相同的掩码），因为SEDD建模的是两个概率的比值，而不是直接的对数概率。我们使用对数线性噪声调度，配合 $1e^{-3}$ 。

对于SDDM，由于它采用不同的架构，直接使用我们的实验配置与SDDM结合并不可行。相反，我们使用论文中给出的实验配置：骨干结构是一个空心Transformer，由Sun等人 [2023]提出。每个Transformer模块包含6层，嵌入维度为256，8个注意力头和2048的隐藏维度。批大小为64，训练步数为200万。权重衰减设置为 $1e^{-6}$ 。学习率为常数 $1e^{-3}$ 。对于扩散过程，SDDM采用恒定噪声调度器，其均匀率常数为0.04。

⁵<https://github.com/HKUNLP/reparam-discrete-diffusion>

⁶<https://github.com/louaaron/Score-Entropy-Discrete-Diffusion>

USD3 VQCIFAR10 Training Details.

For image generation task, we parameterize $f_t^\theta(\mathbf{x}_t)$ with the same sequence transformer as mentioned before. The model is a 12 layer transformer, where each layer has 16 attention heads, input embedding of 768 and a hidden layer size of 3072 for the MLPs. We use ReLU for activation. At the output of each internal block, a dropout of 0.1 is applied. The time is input into the network through FiLM layers before the self attention block. We use a learning rate of $5e^{-4}$, with warmup of 50000 steps, and a cosine learning rate decay scheduler, to train 2 million steps. In discrete-time diffusion, all the USD3 and its variants use the same cosine noise scheduler with $\alpha = 0.008$. For continuous-time diffusion, USD3-CE and USD3* apply the cosine noise scheduler with $\alpha = 0.008$. We find that USD3 is extremely hard to optimize due to the scale differences of coefficient $\beta(t)$, so we provide USD3* which clips the $\beta(t) = \max(1, \beta(t))$. USD3 utilizes a constant noise scheduler with 0.007, to match the scheduler in τ -LDR-0 and SDDM. We maintain an exponential moving average of parameters with decay factor 0.9999. We clip the gradient norm at a norm value of 1.0.

Baseline Training Details.

For τ -LDR-0 and D3PM, we again use the same transformer architecture and same training scheme. We apply a hybrid loss with cross entropy as a directly supervision, added with 0.001 CE loss. The τ -LDR-0 uses a constant rate noise scheduler of 0.007, while D3PM uses cosine scheduler with $\alpha = 0.008$. We train all models in parallel on 2 A6000 GPUs. For D3PM and τ -LDR-0, for fair comparison, we use the same architecture, diffusion scheme and training scheme. For calculating the loss of both methods, we follow the previous literature and give 0.001 to CE loss, and 1 to the VLB loss. For RDM, we apply reweighting = 1.0, with no label smoothing, and let sampling strategy be cmlm-dep. We apply cosine scheduler with uniform noise. For SEDD, we use log-linear noise schedules with $1e^{-3}$.

For SDDM, the model uses a masked modeling, where backbone neural network is BERT-based, with 12 layers of transformers. Each layer has 12 attention heads, embedding size of 768 and hidden layer size of 3072 for MLPs. After obtaining the final embedding of each token, the output is fed that into a 2-block ResNet to acquire the predicted logits. For diffusion, SDDM uses a constant uniform rate of 0.007 as the noise scheduler in the forward process. In Sun et al. [2023], the number of training step is set to 700,000, where the learning rate is warmed up to $1e^{-4}$ during the first 3% steps, and then decays to 0 in a linear schedule. We extended the training to $2m$ steps, but did not observe improved performance.

A.19 Additional Experiment Results

A.19.1 Full Experiment Results on Music Generation

We present the full evaluation metrics and results, including mean and standard deviation, for Lakh Pianoroll music generation tasks, as shown in Table 4.

Table 4: Metrics comparing generated conditional samples and evaluating ground truths for Lakh Pianoroll. For each of n-gram Hellinger Distance (ng.-Hellinger) and Proportion of Outliers (ng.-Prop. Outlier) and Diverse Edit Distance metrics, we show mean $\pm$ std with respect to 3 generated samples. The top two are highlighted by **First**, **Second**.

	Method	1g.-Hellinger(↓)	2g.-Hellinger(↓)	3g.-Hellinger(↓)	1g.-Prop.Out.(↓)	2g.-Prop.Out.(↓)	3g.-Prop.Out.(↓)	Edit Distance(↑)
Discrete-time	D3PM	0.3982±0.0004	0.5303±0.0038	0.5918±0.0046	0.1209±0.0008	0.2532±0.0053	0.3790±0.0059	0.2950±0.0775
	RDM	0.4123±0.0012	0.5964±0.0021	0.6198±0.0039	0.1401±0.0011	0.2459±0.0023	0.3864±0.0030	0.2891±0.0690
	USD3-CE	0.3754±0.0007	0.4835±0.0009	0.5741±0.0008	0.1079±0.0002	0.2099±0.0005	0.3034±0.0007	0.0472±0.0465
	USD3-VLB	0.3790±0.0009	0.4640±0.0010	0.5427±0.0009	0.1174±0.0007	0.1845±0.0010	0.2734±0.0011	0.0828±0.0567
	USD3	0.3770±0.0011	0.4693±0.0015	0.5525±0.0015	0.1077±0.0006	0.1861±0.0011	0.2861±0.0013	0.0648±0.0459
	USD3*	0.3753±0.0013	0.4704±0.0010	0.5550±0.0012	0.1107±0.0005	0.1911±0.0005	0.2832±0.0011	0.0664±0.0471
	美元3*	0.3753±0.0013	0.4704±0.0010	0.5550±0.0012	0.1107±0.0005	0.1911±0.0005	0.2832±0.0011	0.0664±0.0471
Continuous-time	SDDM	0.3759±0.0020	0.4856±0.0015	0.5773±0.0009	0.1101±0.0015	0.2059±0.0005	0.3409±0.0017	0.0606±0.0249
	τ -LDR-0	0.3796±0.0009	0.4811±0.0008	0.5710±0.0007	0.1149±0.0008	0.2078±0.0014	0.3202±0.0014	0.0553±0.0637
	SEDD	0.3814±0.0012	0.4712±0.0020	0.5470±0.0013	0.1052±0.0010	0.1823±0.0015	0.2709±0.0013	0.0632±0.0532
	USD3-CE	0.3734±0.0002	0.4837±0.0003	0.5776±0.0013	0.1158±0.0004	0.2218±0.0003	0.3461±0.0005	0.0434±0.0357
	USD3-VLB	0.3764±0.0006	0.4620±0.0014	0.5487±0.0018	0.1198±0.0009	0.1866±0.0011	0.2952±0.0011	0.0661±0.0478
	USD3	0.3735±0.0005	0.4617±0.0012	0.5432±0.0018	0.1123±0.0009	0.1840±0.0011	0.2704±0.0009	0.0661±0.0401
	USD3*	0.3737±0.0004	0.4623±0.0009	0.5490±0.0017	0.1143±0.0008	0.1844±0.0008	0.2707±0.0011	0.0516±0.0365

A.19.2 Study of “Parroting” with Ratio Metrics

As shown in Table 5, we find that models with combined losses, USD3 and USD3*, achieve higher Train-to-Test ratio than pure USD3-CE or USD3-VLB, suggesting that the combination of two losses

USD3 VQCIFAR10 训练详情。

对于图像生成任务，我们使用与之前提到的相同序列Transformer参数化 $f_t^\theta(\mathbf{x}_t)$ 。模型是一个12层的Transformer，其中每层有16个注意力头，输入嵌入维度为768，MLP的隐藏层大小为3072。我们使用ReLU作为激活函数。在每个内部模块的输出处，应用0.1的dropout。时间通过FiLM层输入到自注意力模块之前。我们使用学习率 $5e^{-4}$ ，预热50000步，并采用余弦学习率衰减调度器，训练200万步。在离散时间扩散中，所有USD3及其变体都使用相同的余弦噪声调度器，参数为 $\alpha = 0.008$ 。对于连续时间扩散，USD3-CE和USD3* 采用余弦噪声调度器，参数为 $\alpha = 0.008$ 。我们发现USD3由于系数 $\beta(t)$ 的尺度差异极难优化，因此我们提供了USD3*，该模型会裁剪 $\beta(t) = \max(1, \beta(t))$ 。USD3使用0.007的恒定噪声调度器，以匹配 τ -LDR-0和SDDM中的调度器。我们使用衰减因子 0.9999对参数进行指数移动平均。梯度范数裁剪在范数值为1.0处进行。

基线训练详情。

对于 τ -LDR-0 和 D3PM，我们再次使用相同的 Transformer 架构和相同的训练方案。我们应用了混合损失，以交叉熵作为直接监督，并添加了 0.001 的 CE 损失。 τ -LDR-0 使用 0.007 的恒定速率噪声调度器，而 D3PM 使用余弦调度器并带有 $\alpha = 0.008$ 。我们在 2 个 A6000 GPU 上并行训练所有模型。对于 D3PM 和 τ -LDR-0，为了公平比较，我们使用相同的架构、扩散方案和训练方案。在计算两种方法的损失时，我们遵循先前文献，将 CE 损失设为 0.001，将 VLB 损失设为 1。对于 RDM，我们应用 reweighting = 1.0，不进行标签平滑，并让采样策略为 cmlm-dep。我们应用余弦调度器并带有均匀噪声。对于 SEDD，我们使用对数线性噪声调度器并带有 $1e^{-3}$ 。

对于SDDM，该模型采用掩码建模，其骨干神经网络基于BERT，包含12层Transformer。每层有12个注意力头，嵌入维度为768，MLP的隐藏层维度为3072。在获取每个token的最终嵌入后，输出被输入到2块ResNet中以获取预测的logits。对于扩散过程，SDDM在正向过程中使用0.007的恒定均匀速率作为噪声调度器。在Sun等人 [2023] 的实验中，训练步数设置为700,000，其中学习率在前3%的步骤中预热至 $1e^{-4}$ ，然后在线性调度中衰减至0。我们将训练扩展至 $2m$ 步，但未观察到性能提升。

A.19 补充实验结果

A.19.1 音乐生成任务的完整实验结果

我们展示了Lakh钢琴卷音乐生成任务的完整评估指标和结果，包括均值和标准差，如表4所示。

表4：比较生成的条件样本和评估Lakh Pianoroll的地面真实数据。对于n-gram Hellinger距离（ng.-Hellinger）和异常值比例（ng.-Prop. Outlier）以及多样性编辑距离指标，我们展示了针对3个生成样本的均值 $\pm$ 标准差。前两个被标记为第一、第二。

	方法	1g.-Hellinger(↓)	2g.-Hellinger(↓)	3g.-Hellinger(↓)	1g.-Prop.Out.(↓)	2g.-属性.输出(↓)	3g.-属性.输出(↓)	编辑距离(↑)
Discrete-time	D3PM	0.3982±0.0004	0.5303±0.0038	0.5918±0.0046	0.1209±0.0008	0.2532±0.0053	0.3790±0.0059	0.2950±0.0775
	RDM	0.4123±0.0012	0.5964±0.0021	0.6198±0.0039	0.1401±0.0011	0.2459±0.0023	0.3864±0.0030	0.2891±0.0690
	USD3-CE	0.3754±0.0007	0.4835±0.0009	0.5741±0.0008	0.1079±0.0002	0.2099±0.0005	0.3034±0.0007	0.0472±0.0465
	USD3-VLB	0.3790±0.0009	0.4640±0.0010	0.5427±0.0009	0.1174±0.0007	0.1845±0.0010	0.2734±0.0011	0.0828±0.0567
	USD3	0.3770±0.0011	0.4693±0.0015	0.5525±0.0015	0.1077±0.0006	0.1861±0.0011	0.2861±0.0013	0.0648±0.0459
	USD3*	0.3753±0.0013	0.4704±0.0010	0.5550±0.0012	0.1107±0.0005	0.1911±0.0005	0.2832±0.0011	0.0664±0.0471
	美元3*	0.3753±0.0013	0.4704±0.0010	0.5550±0.0012	0.1107±0.0005	0.1911±0.0005	0.2832±0.0011	0.0664±0.0471
Continuous-time	SDDM	0.3759±0.0020	0.4856±0.0015	0.5773±0.0009	0.1101±0.0015	0.2059±0.0005	0.3409±0.0017	0.0606±0.0249
	τ -LDR-0	0.3796±0.0009	0.4811±0.0008	0.5710±0.0007	0.1149±0.0008	0.2078±0.0014	0.3202±0.0014	0.0553±0.0637
	SEDD	0.3814±0.0012	0.4712±0.0020	0.5470±0.0013	0.1052±0.0010	0.1823±0.0015	0.2709±0.0013	0.0632±0.0532
	USD3-CE	0.3734±0.0002	0.4837±0.0003	0.5776±0.0013	0.1158±0.0004	0.2218±0.0003	0.3461±0.0005	0.0434±0.0357
	USD3-VLB	0.3764±0.0006	0.4620±0.0014	0.5487±0.0018	0.1198±0.0009	0.1866±0.0011	0.2952±0.0011	0.0661±0.0478
	USD3	0.3735±0.0005	0.4617±0.0012	0.5432±0.0018	0.1123±0.0009	0.1840±0.0011	0.2704±0.0009	0.0661±0.0401
	USD3*	0.3737±0.0004	0.4623±0.0009	0.5490±0.0017	0.1143±0.0008	0.1844±0.0008	0.2707±0.0011	0.0516±0.0365

A.19.2 研究“模仿”与比率指标

如表5所示，我们发现，结合损失函数USD3和USD3*的模型，其训练集到测试集的比率高于纯USD3-CE或USD3-VLB模型，这表明两种损失函数的组合

can alleviate overfitting, i.e. “parroting” the training data. Overall, τ -LDR-0 is a strong competitor in n-gram Hellinger Train-to-Test ratios, suggesting that its VLB focused loss optimization can also alleviate parroting and generate diverse samples. Table 5: Train-to-test Ratio Metrics ($\uparrow$) for Lakh Pianoroll, using evaluation sequences from PIANO-P and the training sequences. The higher ratios indicate less "parroting" phenomenon and better generalization ability. Mean and std are calculated across 3 generated samples.The top two are highlighted by **First, Second**.

	Method	1g.-Hellinger Ratio	2g.-Hellinger Ratio	3g.-Hellinger Ratio	1g.-Prop.Out. Ratio	2g.-Prop.Out. Ratio	3g.-Prop.Out.r Ratio
Discrete-time	D3PM	1.7182±0.0500	1.3624±0.0411	1.1307±0.0304	5.6237±0.1463	3.7051±0.1162	2.2216±0.0616
	RDM	1.7534±0.0928	1.4023±0.00701	1.2023±0.0392	5.9981±0.0497	3.6053±0.1532	2.5211±0.0256
	USD3-CE	1.7922±0.0186	1.4028±0.0074	1.1906±0.0090	6.3964±0.4669	3.9206±0.1082	2.8887±0.0399
	USD3-VLB	1.7758±0.0097	1.4123±0.0118	1.2011±0.0101	6.2024±0.2584	3.9093± 0.0663	2.8635±0.0479
	USD3	1.8073±0.0880	1.4749±0.0708	1.2078±0.0518	6.8742±0.7366	4.2448± 0.4161	3.0830±0.2520
	USD3*	1.7705±0.0235	1.4959±0.0176	1.1928±0.0125	6.6452±0.2702	3.9867±0.1188	2.9591±0.0653
Continuous-time	SDDM	1.7872±0.0131	1.4516±0.0097	1.1812±0.0103	6.3019±0.2522	4.0652±0.0591	2.9014±0.0782
	τ -LDR-0	1.8648±0.0306	1.4726±0.0233	1.2497±0.0185	6.4586±0.3798	4.0863±0.2001	2.9651±0.1089
	SEDD	1.8230±0.0298	1.4902±0.0090	1.2152±0.0150	6.4361±0.2001	4.0598±0.0621	3.0421±0.0922
	USD3-CE	1.7951±0.0290	1.4023±0.0264	1.1474±0.0206	6.3712±0.4502	3.6232±0.1734	2.6379±0.0823
	USD3-VLB	1.7379±0.0220	1.4117±0.0250	1.1810±0.0232	6.2752±0.1232	3.9321±0.0949	2.9042±0.0853
	USD3	1.8623±0.0340	1.4516±0.0077	1.2017±0.0096	6.4218±0.1739	4.0877±0.0553	3.0828±0.0502
	USD3*	1.7970±0.0106	1.4898±0.0026	1.2377±0.0015	6.4361±0.2023	3.9340±0.0284	2.9234±0.0110

A.19.3 Model Sizes and Loss Analysis on Piano Dataset

Table 6 shows the evaluation metrics of different model sizes.This ablation study shows that CE loss is also preferred in combination with VLB, especially for smaller network structures, since VLB is harder to optimize alone.

Table 6: Metrics comparing different loss combinations and different model sizes for Lakh Pianoroll. For each of n-gram Hellinger Distance (ng.-Hellinger) and Proportion of Outliers (ng.-Prop. Outlier) metrics, we show mean $\pm$ std with respect to 3 generated samples. We use USD3-VLB to denote an additional variant of our model that only uses the exact VLB loss in training. "Small" refers to the backbone transformer model that has 6 Layers, 8 Attention Heads, Input Dimension of 128 and MLP dimension of 1024. The top two are highlighted by **First, Second**.

	Method	1g.-Hellinger($\downarrow$)	2g.-Hellinger($\downarrow$)	3g.-Hellinger($\downarrow$)	1g.-Prop.Out.($\downarrow$)	2g.-Prop.Out.($\downarrow$)	3g.-Prop.Out.($\downarrow$)	Edit Distance($\uparrow$)
Discrete-time	USD3-CE-S	0.3984±0.0006	0.4902 ±0.0004	0.5785±0.0004	0.1158±0.0002	0.1899±0.0006	0.3142±0.0005	0.1301±0.0613
	USD3-S	0.4011±0.0014	0.4902±0.0009	0.5707±0.0008	0.1215± 0.0007	0.1866±0.0006	0.3006±0.0013	0.1292±0.0623
	USD3-VLB-S	0.4115±0.0001	0.4954± 0.0001	0.5738±0.0005	0.1203±0.0006	0.1958±0.0009	0.3036±0.0010	0.2137±0.0843
	USD3-CE	0.3754±0.0007	0.4835±0.0009	0.5741±0.0008	0.1079±0.0002	0.2099±0.0005	0.3034±0.0007	0.0472±0.0465
	USD3	0.3770±0.0011	0.4693±0.0015	0.5525±0.0015	0.1077±0.0006	0.1861±0.0011	0.2861±0.0013	0.0648±0.0459
	USD3-VLB	0.3790±0.0009	0.4640±0.0010	0.5427±0.0009	0.1174±0.0007	0.1845±0.0010	0.2734±0.0011	0.0828±0.0567
Continuous-time	USD3-CE-S	0.4208±0.0170	0.5196±0.0078	0.6072±0.0005	0.1355±0.0147	0.2276±0.0007	0.3431±0.0181	0.2358±0.0619
	USD3-S	0.4239±0.0012	0.5083±0.0011	0.5852±0.0011	0.1403±0.0007	0.2070±0.0008	0.2943±0.0016	0.2468±0.0907
	USD3-VLB-S	0.4435±0.0012	0.5295±0.0011	0.6070±0.0009	0.1562±0.0014	0.2269±0.0013	0.3168±0.0013	0.2809±0.0866
	USD3-CE	0.3734±0.0002	0.4837±0.0003	0.5776±0.0013	0.1158±0.0004	0.2218±0.0003	0.3461±0.0005	0.0434±0.0357
	USD3	0.3735±0.0005	0.4617±0.0012	0.5432±0.0018	0.1123±0.0009	0.1840±0.0011	0.2704±0.0009	0.0661±0.0401
	USD3-VLB	0.3764±0.0006	0.4620±0.0014	0.5487±0.0018	0.1198±0.0009	0.1866±0.0011	0.2952±0.0011	0.0661±0.0478

A.19.4 Memory and Running-time Comparison

Table 7 provides the running time, number of parameters (in the transformer architecture) and GPU-memory. While D3PM, and τ -LDR-0 use the same backbone transformer structure as USD3 and contain the same number of parameters in the transformer, they require a passing of transition matrices, which incur additional memory and slow down the training process. RDM is similar to USD3-CE and USD3* in memory calculation and runtime. SDDM requires a special architecture and considerably larger memory during training. SEDD requires significantly less memory than USD3 (continuous version), and training time is almost equivalent to the discrete-version USD3-CE and USD3*. SEDD’s loss calculation removes the redundant additional forward sampling, yet the loss calculation is less exact than USD3 in continuous-time.

A.19.5 Qualitative Examples for VQCIFAR10 Image Generation

We provide 80 reconstructed images VQGAN decoder in Fig. 2. Most examples are easy to recognize from one of the 10 classes in the original CIFAR10: airplanes, cars, birds, cats, deer, dogs, frogs, horses, ships, and trucks.

可以缓解过拟合，即“鹦鹉学舌”式地复制训练数据。总体而言， τ -LDR-0在n-gram Hellinger训练集到测试集比率方面具有很强的竞争力，这表明其专注于VLB的损失优化也能缓解“鹦鹉学舌”现象并生成多样化的样本。T

Table 5: Train-to-test Ratio Metrics ($\uparrow$) for Lakh Pianoroll, using evaluation sequences from PIANO-P and the training sequences. The higher ratios indicate less "parroting" phenomenon and better generalization ability. Mean and std are calculated across 3 generated samples. The top two are highlighted by **First, Second**.

	方法	1g.-Hellinger比率	2g.-Hellinger比率	3g.-Hellinger比率	1g.-属性输出比率	2g.-属性输出比率	3g.-属性输出.r 比率
Discrete-time	D3PM	1.7182±0.0500	1.3624±0.0411	1.1307±0.0304	5.6237±0.1463	3.7051±0.1162	2.2216±0.0616
	RDM	1.7534±0.0928	1.4023±0.00701	1.2023±0.0392	5.9981±0.0497	3.6053±0.1532	2.5211±0.0256
	USD3-CE	1.7922±0.0186	1.4028±0.0074	1.1906±0.0090	6.3964±0.4669	3.9206±0.1082	2.8887±0.0399
	USD3-VLB	1.7758±0.0097	1.4123±0.0118	1.2011±0.0101	6.2024±0.2584	3.9093± 0.0663	2.8635±0.0479
	USD3	1.8073±0.0880	1.4749±0.0708	1.2078±0.0518	6.8742±0.7366	4.2448± 0.4161	3.0830±0.2520
	USD3*	1.7705±0.0235	1.4959±0.0176	1.1928±0.0125	6.6452±0.2702	3.9867±0.1188	2.9591±0.0653
Continuous-time	SDDM	1.7872±0.0131	1.4516±0.0097	1.1812±0.0103	6.3019±0.2522	4.0652±0.0591	2.9014±0.0782
	τ -LDR-0	1.8648±0.0306	1.4726±0.0233	1.2497±0.0185	6.4586±0.3798	4.0863±0.2001	2.9651±0.1089
	SEDD	1.8230±0.0298	1.4902±0.0090	1.2152±0.0150	6.4361±0.2001	4.0598±0.0621	3.0421±0.0922
	USD3-CE	1.7951±0.0290	1.4023±0.0264	1.1474±0.0206	6.3712±0.4502	3.6232±0.1734	2.6379±0.0823
	USD3-VLB	1.7379±0.0220	1.4117±0.0250	1.1810±0.0232	6.2752±0.1232	3.9321±0.0949	2.9042±0.0853
	USD3	1.8623±0.0340	1.4516±0.0077	1.2017±0.0096	6.4218±0.1739	4.0877±0.0553	3.0828±0.0502
	USD3*	1.7970±0.0106	1.4898±0.0026	1.2377±0.0015	6.4361±0.2023	3.9340±0.0284	2.9234±0.0110

A.19.3 模型尺寸与钢琴数据集上的损失分析

表6展示了不同模型尺寸的评估指标。这项消融研究表明，CE损失在与VLB结合时也更受青睐，特别是对于较小的网络结构，因为VLB单独优化更困难。

表6：比较不同损失组合和不同模型尺寸的Lakh钢琴卷积。对于n-gram Hellinger距离（ng.-Hellinger）和异常值比例（ng.-Prop. Outlier）指标中的每一个，我们展示了相对于3个生成样本的平均值 $\pm$ 标准差。我们用USD3-VLB表示我们模型的一个额外变体，该变体在训练中仅使用精确的VLB损失。“小”指的是具有6层、8个注意力头、输入维度为128和MLP维度为1024的主干Transformer模型。前两个用First、Second突出显示。

	方法	1g.-Hellinger($\downarrow$)	2g.-Hellinger($\downarrow$)	3g.-Hellinger($\downarrow$)	1g.-Prop.Out.($\downarrow$)	2g.-Prop.Out.($\downarrow$)	3g.-属性输出($\downarrow$)	编辑距离($\uparrow$)
Discrete-time	USD3-CE-S	0.3984±0.0006	0.4902 ±0.0004	0.5785±0.0004	0.1158±0.0002	0.1899±0.0006	0.3142±0.0005	0.1301±0.0613
	USD3-S	0.4011±0.0014	0.4902±0.0009	0.5707±0.0008	0.1215± 0.0007	0.1866±0.0006	0.3006±0.0013	0.1292±0.0623
	USD3-VLB-S	0.4115±0.0001	0.4954± 0.0001	0.5738±0.0005	0.1203±0.0006	0.1958±0.0009	0.3036±0.0010	0.2137±0.0843
	USD3-CE	0.3754±0.0007	0.4835±0.0009	0.5741±0.0008	0.1079±0.0002	0.2099±0.0005	0.3034±0.0007	0.0472±0.0465
	USD3	0.3770±0.0011	0.4693±0.0015	0.5525±0.0015	0.1077±0.0006	0.1861±0.0011	0.2861±0.0013	0.0648±0.0459
	USD3-VLB	0.3790±0.0009	0.4640±0.0010	0.5427±0.0009	0.1174±0.0007	0.1845±0.0010	0.2734±0.0011	0.0828±0.0567
Continuous-time	USD3-CE-S	0.4208±0.0170	0.5196±0.0078	0.6072±0.0005	0.1355±0.0147	0.2276±0.0007	0.3431±0.0181	0.2358±0.0619
	USD3-S	0.4239±0.0012	0.5083±0.0011	0.5852±0.0011	0.1403±0.0007	0.2070±0.0008	0.2943±0.0016	0.2468±0.0907
	USD3-VLB-S	0.4435±0.0012	0.5295±0.0011	0.6070±0.0009	0.1562±0.0014	0.2269±0.0013	0.3168±0.0013	0.2809±0.0866
	USD3-CE	0.3734±0.0002	0.4837±0.0003	0.5776±0.0013	0.1158±0.0004	0.2218±0.0003	0.3461±0.0005	0.0434±0.0357
	USD3	0.3735±0.0005	0.4617±0.0012	0.5432±0.0018	0.1123±0.0009	0.1840±0.0011	0.2704±0.0009	0.0661±0.0401
	USD3-VLB	0.3764±0.0006	0.4620±0.0014	0.5487±0.0018	0.1198±0.0009	0.1866±0.0011	0.2952±0.0011	0.0661±0.0478

A.19.4 内存与运行时比较

表7提供了运行时间、参数数量（在Transformer架构中）和GPU内存。虽然D3PM和 τ -LDR-0与USD3使用相同的骨干Transformer结构，并且Transformer中的参数数量相同，但它们需要传递过渡矩阵，这会带来额外的内存消耗并减慢训练过程。RDM在内存计算和运行时与USD3-CE和USD3* 相似。SDDM需要特殊的架构，并且在训练期间需要显著更大的内存。SEDD比USD3（连续版本）需要显著更少的内存，训练时间几乎与离散版本的USD3-CE和USD3*相当。SEDD的损失计算移除了冗余的前向采样，但连续时间下的损失计算不如USD3精确。

A.19.5 VQCIFAR10图像生成的定性示例

我们在图2中提供了80张由VQGAN解码器重建的图像。大多数示例都很容易从原始CIFAR10的10个类别之一中识别出来：飞机、汽车、鸟、猫、鹿、狗、青蛙、马、船和卡车。

Table 7: The GPU-memory, running time and number of network parameters in all methods.USD3 is easier to train and incurs the least GPU memory in both discrete- and continuous- time diffusions.

	Method	Num. Parameters	Memory	Runtime		Method	Num. Parameters	Memory	Runtime
Discrete-time	D3PM	~ 102,700,000	15669MiB	93 hrs	Continuous-time	SDDM	~ 12,350,000	85528MiB	96 hrs
	RDM	~ 102,700,000	9570MiB	86 hrs		SEDD	~ 102,700,000	4620MiB	85 hrs
	USD3-CE	~ 102,700,000	9735MiB	83 hrs		τ -LDR-0	~ 102,700,000	17669 MiB	129 hrs
	USD3*	~ 102,700,000	9735MiB	82 hrs		USD3-CE	~ 102,700,000	15703MiB	93 hrs
	USD3	~ 102,700,000	9735MiB	90 hrs		USD3*	~ 102,700,000	17620MiB	91 hrs
						USD3	~ 102,700,000	17620MiB	101 hrs

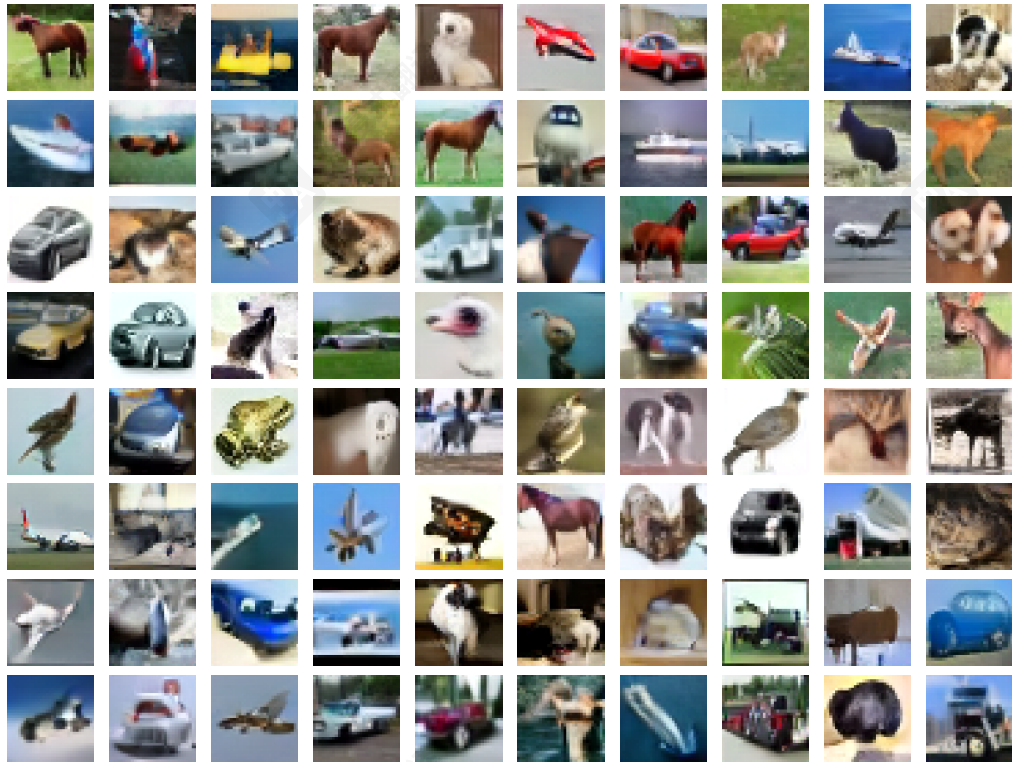


Figure 2: Example image samples generated by USD3* as trained on VQCIFAR10. Most images are easy to recognize as being from one of the 10 classes in CIFAR10.

A.19.6 MCMC for Discrete-time and Continuous-time Discrete Diffusion

To demonstrate the effectiveness of MCMC corrector steps, we take the top performing methods in discrete and continuous time diffusion models (for discrete time, USD3*, and for continuous time USD3-CE) and show improved quality metrics over generated images after MCMC corrector is applied. Due to extensive time required for MCMC corrector step, we could only conduct evaluation over 5,000 images, and thus the results are not comparable to the main VQCIFAR10 result in Table 2. We set the number of generation steps to be 100 and use MCMC corrector on the last 10, 20 generation timesteps, respectively.

From the results shown in Table 8 and Table 9, we can see that MCMC corrector can significantly improve the quality of the generated samples. Specifically, for discrete-time generation, IS is showing a significant improvement than the sampling process without the MCMC corrector. For continuous-time case, both IS and FID scores are improving from the baseline (without MCMC).

表7: 所有方法的GPU内存、运行时间和网络参数数量。USD3更易于训练, 并且在离散时间和连续时间扩散中均消耗最少的GPU内存。

	方法	参数数量	内存	运行时间		方法	参数数量	内存	运行时间
Discrete-time	D3PM	~ 102,700,000	15669MiB	93小时	Continuous-time	SDDM	~ 12,350,000	85528MiB	96 小时
	RDM	~ 102,700,000	9570MiB	86 小时		SEDD	~ 102,700,000	4620MiB	85 小时
	USD3-CE	~ 102,700,000	9735MiB	83 小时		τ -LDR-0	~ 102,700,000	17669 MiB	129 小时
	USD3*	~ 102,700,000	9735MiB	82 小时		USD3-CE	~ 102,700,000	15703MiB	93 小时
	USD3	~ 102,700,000	9735MiB	90 小时		USD3*	~ 102,700,000	17620MiB	91小时
						USD3	~ 102,700,000	17620MiB	101小时

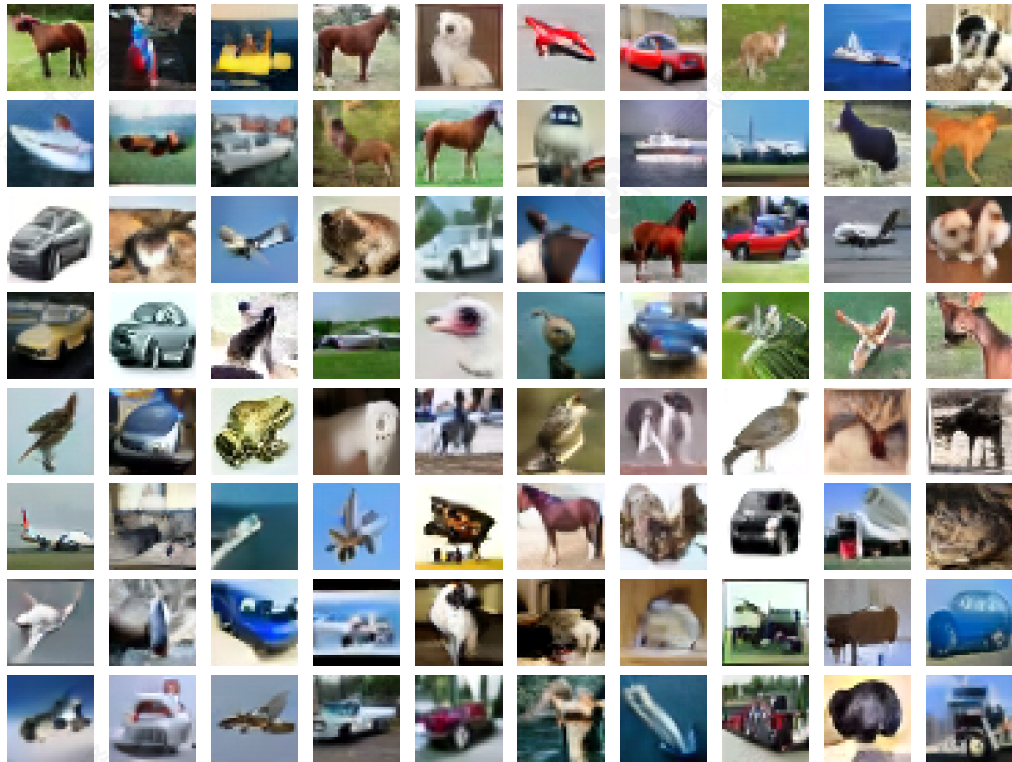


图2: 由USD3*在VQCIFAR10上训练生成的示例图像样本。大多数图像都很容易被识别为来自CIFAR10中的10个类别之一。

A.19.6 MCMC离散时间和连续时间离散扩散

为了展示MCMC校正步骤的有效性, 我们选取了离散时间和连续时间扩散模型中的表现最佳的方法 (对于离散时间, 是USD3*, 对于连续时间, 是USD3-CE), 并展示了在应用MCMC校正器后生成的图像质量指标有所提升。由于MCMC校正步骤需要大量时间, 我们只能对5,000张图像进行评估, 因此结果与表2中的主要VQCIFAR10结果不可比。我们将生成步数设置为100, 并在最后10个和20个生成时间步长上分别使用MCMC校正器。

从表8和表9的结果中可以看出, MCMC校正器可以显著提高生成样本的质量。具体来说, 对于离散时间生成, IS比没有MCMC校正器的采样过程显示出显著的改进。对于连续时间情况, IS和FID分数都从基线 (没有MCMC) 得到了提升。

Table 8: Image gen. quality w.r.t. Inception Score (IS) and the Frechet Inception Dist. (FID) over 5,000 samples unconditionally generated by USD3* in discrete-time case. MCMC corrector is conducted for the last 10,20 timesteps over 100 sampling steps.

MCMC Configuration	IS (↑)	FID (↓)
Without MCMC	9.01	19.79
$\Delta n = 0.001, N = 2$, Start Steps:10	9.28	18.46
$\Delta n = 0.001, N = 2$, Start Steps:20	9.59	18.36
$\Delta n = 0.005, N = 2$, Start Steps:10	9.43	18.26
$\Delta n = 0.005, N = 2$, Start Steps:20	9.43	20.37
$\Delta n = 0.001, N = 5$, Start Steps:10	9.29	18.02
$\Delta n = 0.001, N = 5$, Start Steps:20	9.47	18.56
$\Delta n = 0.002, N = 5$, Start Steps:10	9.35	18.18
$\Delta n = 0.002, N = 5$, Start Steps:20	9.48	20.37

A.19.7 Sampling Time Discrete-time and Continuous-time Discrete Diffusion

In Table 10, we have calculated the time required for baselines and our models to sample 50,000 VQ-encoded images with 1000 timesteps on a single A6000 GPU. In addition, we calculate the time required when USD3 and τ -LDR need MCMC corrector steps (=5) in sampling. We compare our results in the following tables. In terms of sampling time, RDM yields the most sampling time due to its re-routing sampling scheme. SEDD and τ -LDR are similar in sampling time, due to the tau-leaping scheme. USD3 falls short of SDDM, which requires a special NN architecture and different code base(jaxlib) than other methods. But when considering MCMC sampling, USD3 brings faster sampling time than tau-leaping in τ -LDR, while allowing sampling for both continuous-time and discrete-time models.

Table 10: Sample times and MCMC sampling steps for different models.

Model	Sample Time	MCMC Sampling Step=5
RDM	~23 hours	-
D3PM	~8 hours	-
τ -LDR	~9 hours	~2 days, 20 hours
SDDM	~7 hours	-
SEDD	~8 hours	-
USD3 (continuous version)	~8 hours	~2 days, 12 hours
USD3 (discrete version)	~8 hours	~2 days, 12 hours

Table 9: Image gen. quality w.r.t. Inception Score (IS) and the Frechet Inception Dist. (FID) over 5,000 samples generated by USD3-CE in continuous-time case. MCMC corrector is conducted for the last 10,20 timesteps over 100 sampling steps.

MCMC Configuration	IS (↑)	FID (↓)
Without MCMC	8.98	19.19
$\Delta n = 0.001, N = 2$, Start Steps:10	9.12	17.62
$\Delta n = 0.001, N = 2$, Start Steps:20	9.23	17.35
$\Delta n = 0.005, N = 2$, Start Steps:10	9.01	17.71
$\Delta n = 0.005, N = 2$, Start Steps:20	9.23	17.58
$\Delta n = 0.001, N = 5$, Start Steps:10	9.03	17.26
$\Delta n = 0.001, N = 5$, Start Steps:20	9.00	17.42
$\Delta n = 0.002, N = 5$, Start Steps:10	9.16	17.98
$\Delta n = 0.002, N = 5$, Start Steps:20	8.97	17.83

表 8：在离散时间情况下，由 USD3* 无条件生成的 5,000 个样本的图像生成质量相对于 Inception 分数 (IS) 和 Frechet Inception 距离 (FID)。在 100 个采样步骤中，对最后 10、20 个时间步长进行 MCMC 校正。

MCMC 配置	IS (↑)	FID (↓)
无 MCMC	9.01	19.79
$\Delta n = 0.001, N = 2$, 开始步数:10	9.28	18.46
$\Delta n = 0.001, N = 2$, 开始步数:20	9.59	18.36
$\Delta n = 0.005, N = 2$, 开始步骤: 10	9.43	18.26
$\Delta n = 0.005, N = 2$, 开始步骤: 20	9.43	20.37
$\Delta n = 0.001, N = 5$, 开始步骤: 10	9.29	18.02
$\Delta n = 0.001, N = 5$, 开始步骤: 20	9.47	18.56
$\Delta n = 0.002, N = 5$, 开始步骤: 10	9.35	18.18
$\Delta n = 0.002, N = 5$, 开始步骤: 20	9.48	20.37

A.19.7 采样时间 离散时间与连续时间离散扩散

在表10中，我们计算了基线我们的模型在单个A6000 GPU上对50,000张VQ编码图像进行采样所需的时间，采样步数为1000。此外，我们还计算了USD3和 τ -LDR在采样时需要MCMC校正步骤(=5)所需的时间。我们在以下表格中比较了我们的结果。在采样时间方面，RDM由于其重路由由采样方案产生了最长的采样时间。SEDD和 τ -LDR在采样时间上相似，因为它们使用了tau-跳跃方案。USD3不及SDDM，后者需要特殊的NN架构和与其他方法不同的代码库（jaxlib）。但当考虑MCMC采样时，USD3在 τ -LDR中比tau-跳跃方案带来更快的采样时间，同时允许对连续时间和离散时间模型进行采样。

表10：不同模型的样本时间和MCMC采样步数。

模型	样本时间	MCMC采样步数=5
RDM	~23 小时	-
D3PM	~8 小时	-
τ -LDR	~9 小时	~2 天, 20小时
SDDM	~7 小时	-
SEDD	~8 小时	-
美元3（连续版本）	~8 小时	~2 天, 12小时
美元3（离散版本）	~8 小时	~2 天, 12小时